

Student management System

A

Project Report

*Submitted in partial fulfillment of the requirement for the
award of Degree of*

Bachelor of Science in Information Technology

Submitted to

**444 - Swatantrata Senani Shankarlal Agarwal Prabandhan
Avum Takniki Mahavidyalay, Patna**



Submitted By
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Session-2023-2026

Under Joint Guidance of
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Computer Science &
Applications



Swatantrata Senani Shankarlal Agarwal Prabandhan
Avum Takniki Mahavidyalay, Patna

Department of Computer Science & Applications
Session 2023-2026

DATE:.....

CERTIFICATE

This is to certify that the work embodies in this project entitled, **Student Management System**, being submitted by **Sumit Swastik** (202344400198) in partial fulfillment of the requirement for the award of “**Bachelor of Science in Information Technology**” to **PATLIPUTRA UNIVERSITY, PATNA** during the academic year 2022-25 is a record of bonafide piece of work, carried out by him/her under our/my supervision and guidance in the “**Department of Computer Science & Applications**”, **SSSAPTM, Patna**.



Dr. Neeraj Agrawal
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Swatantrata Senani Shankarlal Agarwal Prabandhan
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Department of Computer Science & Applications
Session 2023-2026

Date:.....

APPROVAL CERTIFICATE

The project report entitled Student **management System** being submitted by **Sumit Swastik** (202344400198) has been examined by us and is hereby approved for the award of degree “**Bachelor of Science in Information Technology**”, for which it has been submitted. It is understood that by this approval the undersigned do not necessarily endorse or approve any statement made, the opinion expressed or conclusion drawn therein, but approve the project only for the purpose for which it has been submitted.



(Internal Examiner)

(External Examiner)

Date:

Date:



Swatantrata Senani Shankarlal Agarwal Prabandhan
Avum Takniki Mahavidyalay, Patna

Department of Computer Science & Applications
Session 2023-2026

Date:.....

DECLARATION

I, <your name> hereby declare that the work, which is being presented in the project report entitled **Student Management System**, partial fulfillment of the requirements for the award of degree of “**Bachelor of Science in Information Technology**” submitted in the department of **Computer Science & Applications (SSSAPTM)** is an authentic record of my own work carried under the joint guidance of **Prof. Niraj Kumar Singh Sir and Prof. Anjesh Kumar Sir**. I have not submitted the matter embodied in this report for the award of any other degree.

Sumit Swastik

Registration No: 202344400198

B. Sc.IT 3rd Year

Session-2023-2026

ACKNOWLEDGEMENT

“A journey is easier when you travel together. Interdependence is certainly more valuable than independence.”

I would like to thanks, **Prof. Niraj Kumar Singh Sir** and **Prof. Anjesh Kumar Sir**, for providing regular guidance and insight into my project work. I also thank them for all advice he has given me in the past year, and for always having time for me, whenever I needed.

I give special thanks to **Dr. Amit Kumar Shukla Sir**, **Prof. Raju Upadhyay Sir**, and **Prof. Ravi Kumar Soni Sir** for always being willing to help find solutions to any problems I had with my work.

“The completion of any project depends upon the cooperation, coordination, and combined efforts of several resources of knowledge, inspiration, and energy”.

I also extend my deepest gratitude to **Director**, CIMAGE GROUP OF INSTITUTIONS, PATNA, **Dr. Neeraj Agrawal Sir**, **Dean** CIMAGE GROUP OF INSTITUTIONS, PATNA, **Dr. Neeraj Kumar Sir** and **Centre Head Mrs. Megha Agrawal Ma’am** for providing all the necessary facilities and true encouraging environment to bring out the best in my endeavors.

I express my gratitude and thanks to all the staff members of Computer Science departments for their sincere cooperation in furnishing relevant information to complete this Project well in time successfully.

I extend a special word to my friends, who have been a constant source of inspiration throughout my project work.

Lastly but not least I must express my cordial thank to my parent and family members who gave me the moral support without that it is impossible to complete my project work. With this note, I thank everyone for the support.

Sumit Swastik

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Project Report

INTRODUCTION

The **Student Management System (SMS)** is a web-based application designed to manage all student-related information in a simple, secure, and efficient way. It helps educational institutions store and handle data such as student details, courses, attendance, and marks in a centralized digital system. In traditional methods, student data is maintained manually using paper records or spreadsheets, which can lead to errors, data loss, and difficulty in searching information. This project solves these problems by providing an automated system where data can be stored, updated, and accessed easily anytime.

The system provides different user roles:

- **Admin** – Manages students, teachers, courses, and system data
- **Teacher** – Updates attendance and student marks
- **Student** – Views personal details, attendance, and results

This project is developed using **Java (Servlets and JSP)** for backend processing, **MySQL** for database management, and runs on the **Apache Tomcat server**. It follows the **MVC (Model-View-Controller)** architecture, which separates the application into different layers for better organization and maintenance.

The main aim of this system is to reduce manual work, improve accuracy, save time, and make data management faster and more reliable. It also ensures data security and provides a user-friendly interface for smooth operation.

Overall, the Student Management System is a useful solution for modern educational institutions to manage their academic data efficiently.

OBJECTIVE

The main objectives of the **Student Management System (SMS)** are:

- **To automate student data management**

Reduce manual work by digitizing student records like personal details, attendance, and marks.

- **To maintain a centralized database**

Store all information in one place for easy access, update, and management.

- **To improve accuracy and reduce errors**

Minimize human mistakes that occur in manual record-keeping.

- **To provide role-based access**

Allow Admin, Teachers, and Students to access only relevant information securely.

- **To simplify administrative tasks**

Make it easy to manage students, teachers, courses, and subjects.

- **To enable quick report generation**

Generate student reports, attendance reports, and result summaries efficiently.

- **To enhance data security**

Protect sensitive student information using authentication and controlled access.

- **To save time and increase efficiency**

Speed up daily operations and reduce workload for staff.

PROJECT CATEGORY JAVA

The Java is a programming language originally developed by James Gosling at Sun Microsystems (which is now a subsidiary of Oracle Corporation) and released in 1995 as a core component of Sun Microsystems' Java platform. The language derives much of its syntax from C and C++ but has a simpler object model and fewer low-level facilities. Java applications are typically compiled to bytecode (class file) that can run on any Java Virtual Machine (JVM) regardless of computer architecture. Java is general-purpose, concurrent, class-based, and object-oriented, and is specifically designed to have as few implementation dependencies as possible. It is intended to let application developers "write once, run anywhere". Java is considered by many as one of the most influential programming languages of the 20th century, and widely used from application software to web application.[9][10]

The original and reference implementation Java compilers, virtual machines, and class libraries were developed by Sun from 1995. As of May 2007, in compliance with the specifications of the Java Community Process, Sun relicensed most of their Java technologies under the GNU General Public License. Others have also developed alternative implementations of these Sun technologies, such as the GNU Compiler for Java and GNUClasspath.

ADVANTAGES OF JAVA

1. **Java is simple:** Java was designed to be easy to use and is therefore easy to write, compile, debug, and learn than other programming languages. The reason that why Java is much simpler than C++ is because Java uses automatic memory allocation and garbage collection where else C++ requires the programmer to allocate memory and to collect garbage.
2. **Java is object-oriented:** Java is object-oriented because programming in Java is centered on creating objects, manipulating objects, and making objects work together. This allows you to create modular programs and reusable code.
3. **Java is platform-independent:** One of the most significant advantages of Java is its ability to move easily from one computer system to another.
4. **Java is distributed:** Distributed computing involves several computers on a network working together. Java is designed to make distributed computing easy with the networking capability that is inherently integrated into it.

5. **Java is interpreted:** An interpreter is needed in order to run Java programs. The programs are compiled into Java Virtual Machine code called bytecode.
6. **Java is secure:** Java is one of the first programming languages to consider security as part of its design. The Java language, compiler, interpreter, and runtime environment were each developed with security in mind.
7. **Java is robust:** Robust means reliable and no programming language can really assure reliability. Java puts a lot of emphasis on early checking for possible errors, as Java compilers are able to detect many problems that would first show up during execution time in other languages.
8. **Java is multithreaded:** Multithreaded is the capability for a program to perform several tasks simultaneously within a program. In Java, multithreaded programming has been smoothly integrated into it, while in other languages, operating system-specific procedures have to be called in order to enable multithreading.

TOOLS/PLATFORM REQUIRED

Operating System

- Windows / Linux / macOS

HARDWARE & SOFTWARE REQUIREMENTS

Hardware Requirements:

- Intel i3 or above
- Minimum 4GB RAM
- 20GB Free Disk Space

Software Requirements:

- **Programming Language:** Java (Servlets & JSP)
- **Database:** MySQL
- **Web Server:** Apache Tomcat 9.0
- **Database Connectivity:** JDBC (Java Database Connectivity)
- **Frontend Technologies:** HTML, CSS, JavaScript

PROBLEM DEFINITION

In many educational institutions, student information is still managed using manual methods such as paper records or basic spreadsheets. This traditional approach creates several challenges in handling and maintaining large amounts of data.

The main problem is the inefficient and error-prone management of student data, including personal details, attendance, courses, and examination results.

REQUIREMENT SPECIFICATIONS

1. Functional Requirements

These are the features and operations the system must perform:

- Admin Module
 - Add, update, delete student records
 - Manage teachers, courses, and subjects
 - View reports of students, attendance, and results
- Teacher Module
 - Login securely
 - Mark student attendance
 - Upload/update student marks
- Student Module
 - Login to the system
 - View personal details
 - Check attendance and results
- Authentication System
 - Secure login/logout functionality
 - Role-based access control
- Report Generation
 - Generate student reports
 - View attendance and performance reports

2. Non-Functional Requirements

These define how the system should perform:

- Performance
 - Fast response time for data access and updates
- Security
 - Secure login with authentication
 - Protection of sensitive data
- Usability
 - User-friendly interface
 - Easy navigation for all users
- Reliability
 - System should work without failure
 - Accurate data processing
- Scalability
 - Can handle increasing number of users and data

PROJECT PLANNING & SCHEDULING

1. Project Planning

The development of the Student Management System (SMS) is divided into different phases:

- Requirement Analysis
 - Understanding project needs, features, and system requirements.
- System Design
 - Designing system architecture, database structure, and user interface.
- Development (Coding)

Implementing the system using Java (Servlets & JSP), MySQL, and frontend technologies.

- **Testing**

Checking for errors, bugs, and system performance.

- **Implementation**

Deploying the system on the server (Apache Tomcat).

- **Maintenance**

Fixing issues and updating the system after deployment

2. Project Scheduling

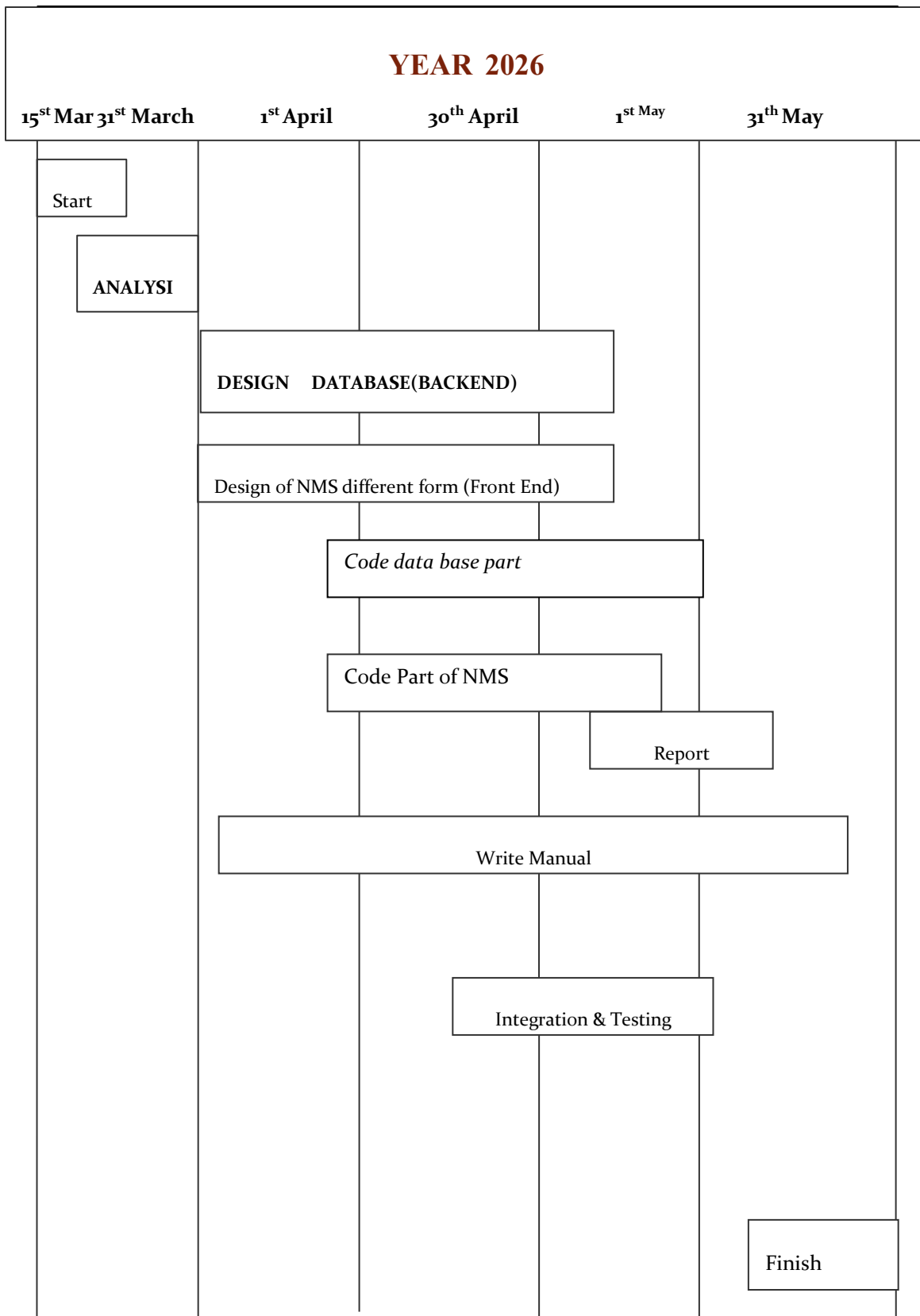
The project is scheduled over a fixed time period as follows:

Phase	Duration
Requirement Analysis	2Week
System Design	2 Week
Development	3 Weeks
Testing	2Week
Implementation	3 Days
Documentation	2 Week

GANTT CHART

Gantt charts (developed by Henry L. Gantt) are a project control technique that can be used for several purposes including scheduling, budgeting and resource planning. A Gantt chart is a bar chart with each bar representing an activity.

The bars are drawn against a time line. The length of each bar is proportional to the length of time planned for the activity. Gantt chart can take different phase depending on their intended use. The Gantt chart of **Student Management System** is drawn for the time management. The Gantt chart is drawn below:

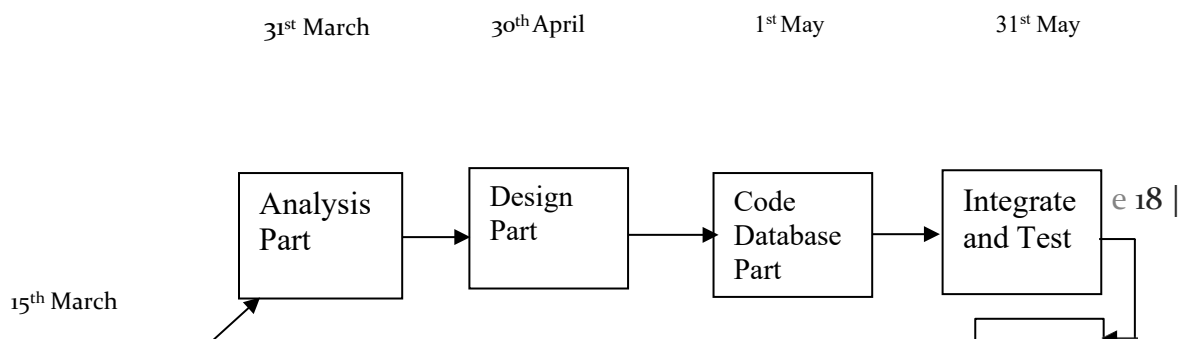


PERT CHART

PERT (Project Evaluation and Review Technique) charts consist of a network of boxes and arrows. The boxes represent activities and the arrows represent task dependencies.

PERT is organized by events and activities or tasks. PERT have more advantages and they are likely to be used for more complex projects. Through PERT chart the various task paths are defined. PERT enables the calculation of critical path. Each path consists of combination of tasks, which must be completed. The time and the cost associated with each task along a path are calculated, and the path that requires the greatest amount of elapsed time is the critical path. Calculation of the critical path enables project manager to monitor this series of task more closely than others to shift resources to it, if it begins to fall behind schedule.

PERT controls time and cost during the project and also facilitate finding the right balance between completing a project on time and completing it within the budget. There are thus not one but many critical paths, depending on the permutations of the estimates for each task. This makes analysis of critical path in PERT charts very complex. The PERT chart representation of the **Student Management System** is shown below.



SYSTEM ANALYSIS

IDENTIFICATION OF NEED

To Important Indication for software, we use the Principles of Requirement Engineering. Requirement engineering provides the appropriate mechanism for understanding what the customer wants, analyzing need, assessing feasibility, negotiating a reasonable solution, specifying the solution unambiguously, validating the specification and managing the requirement as they are transformed into an operational system. The requirement engineering process can be described in five distinct steps:

Requirement elicitation.

Requirement analysis & negotiation.

Requirement specification.

System Modeling.

Requirement validation.

Requirement Management.

In other words we can say that requirement analysis is a software task that bridges the gap between system level requirement engineering and software design.

Requirement analysis allows the software engineering to refine the software allocation and build models of the data, functional and behavioral domains that will be treated by software. Requirement analysis provides the software designer with a representation of information, function and behavior that can be translated into data, architectural, interface and component level design; finally, the requirement specification provides the developer and the customer with the means to assess quality once software is built.

The most commonly used requirement elicitation technique is to conduct a meeting or interview. The first meeting between a software engineer and customer can be likened to the awareness of a first date between two adolescents. Neither person knows what to say or ask; Both are worried that what they do say will be misinterpreted; both are thinking about where it might lead (Both likely have radically different expectations here); Both want to get the thing over with, but at the same time, both want it to be a success.

Here according to this principle, the analyst starts by asking context-free questions. That is a set of questions that will lead to a basic understanding of the problem, the people who want a solution, the nature of solution that is desired, and the effectiveness of the first

encounter itself. The first set of Context-free question focuses on the customer, the overall goals, and the benefits. For example, the analyst might ask: -

Who is behind the request for this Work?

Who will use the solution?

What will be economic benefit of a successful solution?

Is there another source for the solution that you need?

The next set of questions enables the software engineer to gain a better Understanding of the problem and the customer to voice his or her perceptions about a solution: -

How would you characterize “good” output that would be generated by a successful solution?

What problem(s) will this solution address?

Can you show me (or describe) the environment in which the solution will be used?

Will special performance issues or constraints affect the way the solution is approached?

The final set of question focuses on the effectiveness of the meeting:

Are you the right person to answer these questions? Are your answers? Official”?

Are my questions relevant to the problem that you have?

Am I asking too many questions?

Can anyone else provide additional information?

Should I be asking you anything else?

According to the above concepts I went to institution and met its staffs, management, and some persons related with the organizational work, which advised me related with my project such as details of agent, policy related information will be cater this system.

Preliminary Investigation: -

The first step in the system development life cycle is the preliminary investigation to determine the feasibility of the System. The purpose of the preliminary investigation is to evaluate project requests. It is not a design study nor does it include the collection of details to describe the business system in all respect. Rather, it is collecting of information that helps committee members to evaluate the merits of the project request and make an informed judgment about the feasibility of the proposed project.

Here for the Ngo Management System, I have worked on the preliminary investigation that accomplished the following objectives: -

Clarify and understand the project request.

Determine the size of the project.

Assess costs and benefits of alternative approaches.

Determine the technical and operational; feasibility of alternative approaches.

Report the findings to management, with recommendation outlining the acceptances or rejection of the proposal.

Considering above criteria, I also keep in mind that the requirements are clearly understandable when the clarification of project request is enquired. The data of the Ngo Management System, which are collected by me during preliminary investigation, are through Reviewing organization Documents Onsite observation and

Conducting interviews.

FEASIBILITY STUDY

Not everything imaginable is feasible, not even in software, even as it may appear to outsiders. Feasibility is the determination of whether or not a project is worth doing. On the contrary, software feasibility has seven solid dimensions as below: -

- **Technical feasibility.**
- **Operational feasibility.**
- **Economic feasibility.**
- **Social feasibility.**
- **Management feasibility.**
- **Legal feasibility.**
- **Time feasibility.**

The process followed in making this determination is called a feasibility study. The type of study determines if a project can and should be taken. Once it has been determined that a project is feasible, that analyst can go ahead and prepare the project specification which finalizes project requirements. Generally, feasibility studies are undertaken within tight time constraints and normally culminate in a written and oral feasibility report. The contents and recommendation of such a study will be used as a sound basis for deciding whether to proceed, postpone or cancel the project. Thus, since the feasibility study may lead to the commitment of large resources, it becomes

necessary that it should be conducted competently and that no fundamental errors of judgment are made.

In other words, we can say that a feasibility study is conducted to select the best system that meets performance requirement. This contains and identification description, an evaluation of candidate system and the selection of best system for the job. The system required a statement of constraints; the identification of specific system objective and a description of outputs define performance.

TECHNICAL FEASIBILITY: -

- The system is developed using **Java (Servlets & JSP), MySQL, and Apache Tomcat**, which are widely available and easy to use.
- The required tools (IDE, server, database) are easily accessible.
- The project can be implemented using standard computers and basic technical knowledge.

ECONOMICAL FEASIBILITY: -

- The software used (Java, MySQL, Tomcat) is **free and open-source**.
- No expensive hardware is required.
- Development cost is low compared to benefits like time-saving and efficiency.

OPERATIONAL FEASIBILITY: -

My project is also fit in Operational Feasibility Study. As I have explained before this about Technical Feasibility and Economical Feasibility, this Software is very easy and user-friendly.

I have used a very easy menu system and also used some control buttons in the easy way by which one can choose the options on his desire.

Though this is fully protected with the security by word (Password and user name), if both are matched only then one can get access of this software.

Various Reports, Forms and Queries can be generated on the fingertips for the user.

In nutshell we can say that it has following operational features:

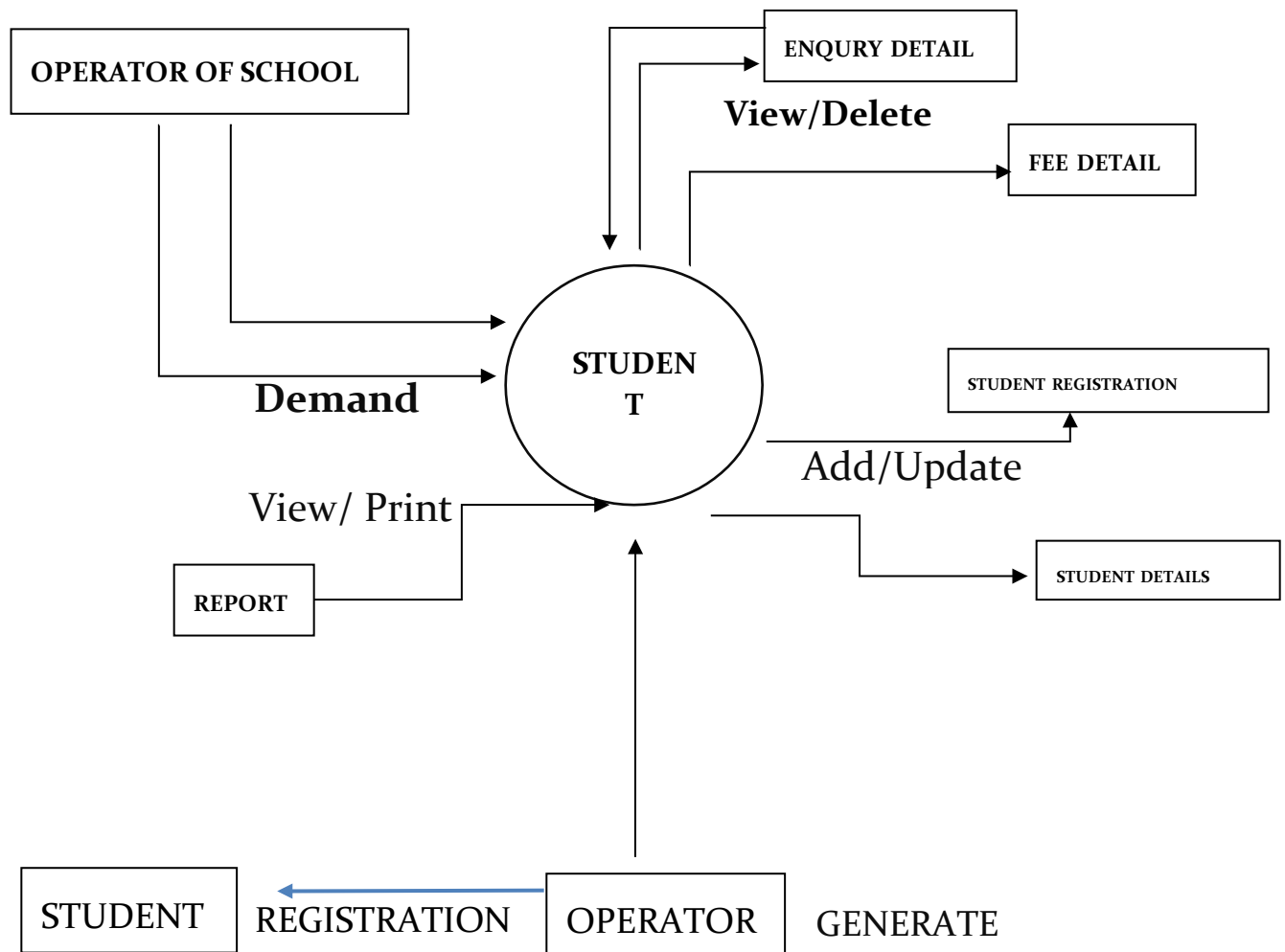
1. It is User-Friendly.
2. It is having less paperwork.
3. Efficient tractability.
4. Query can be generated.
5. Various Reports and Forms can be generated.
6. Fully protected by Password and User name for unauthorized access.

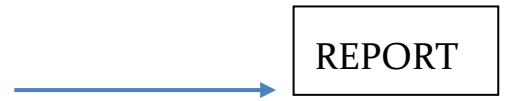
ANALYSIS

System Analysis refers into the process of examining a situation with the intent of improving it through better procedures and methods. System Analysis is the process of planning a new System to either replace or complement an existing system.

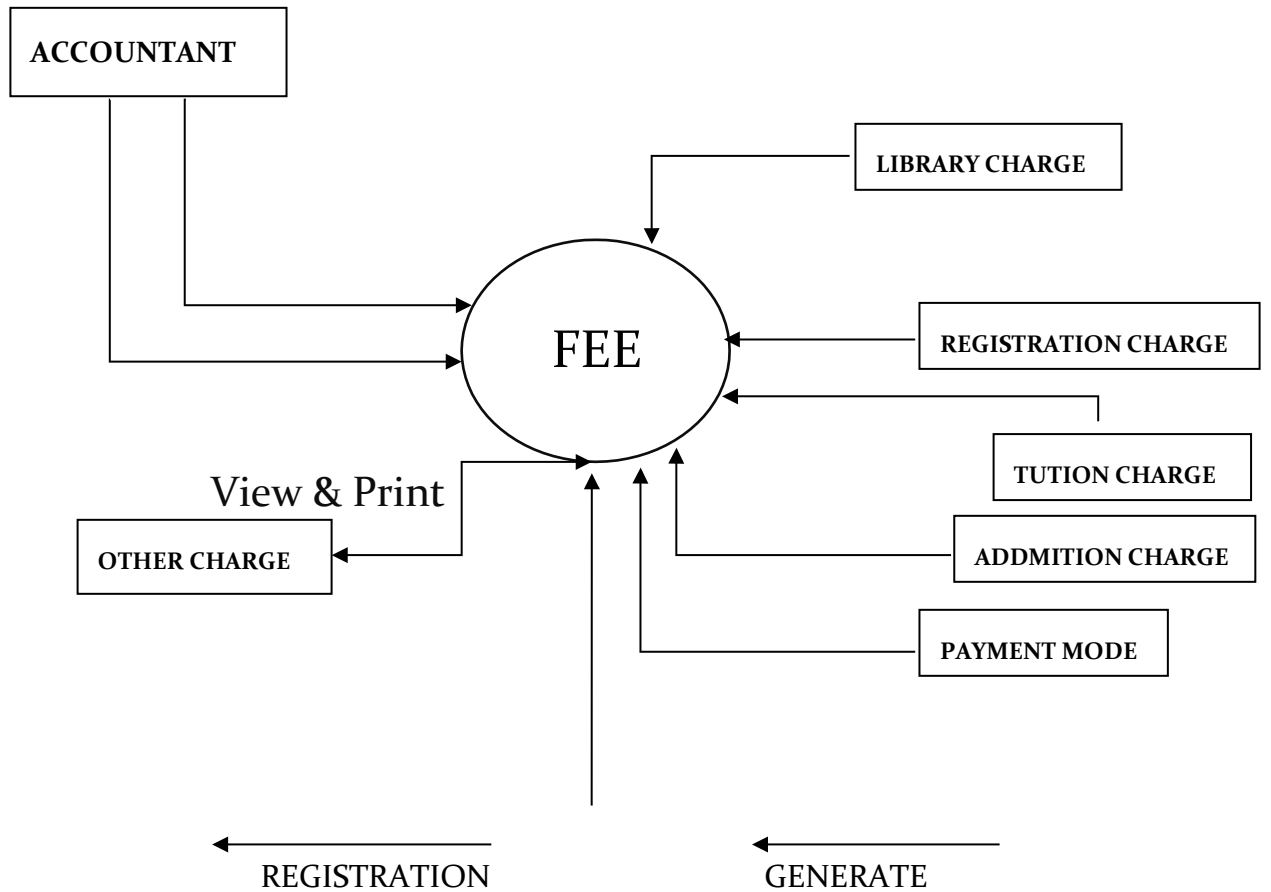
DFD

The DFD is an excellent communication tool for analysts to model processes and functional requirements. Used effectively, it is a useful and easy to understand modeling tool. It has broad application

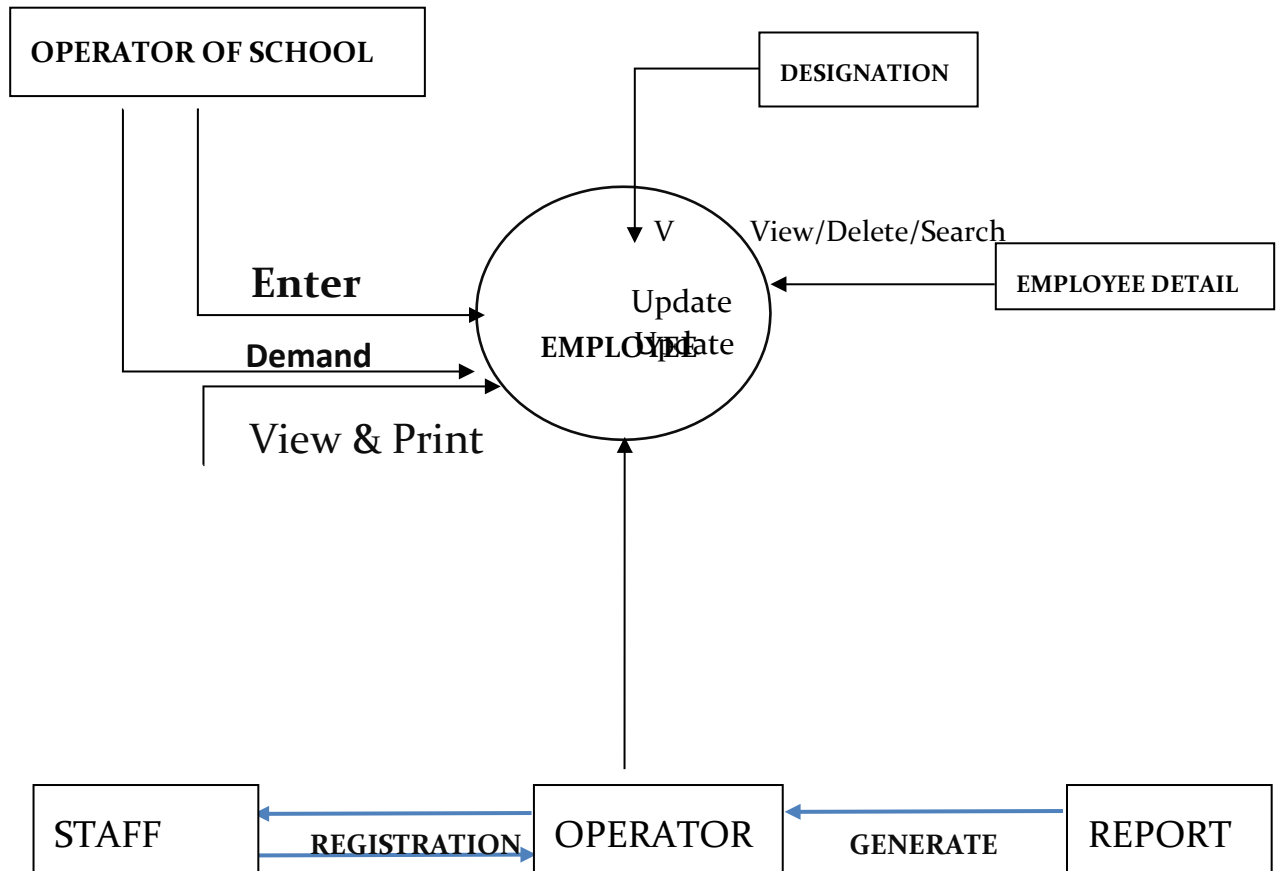




DFD 2: MAINTAIN STUDENT RECORD



DFD 3: MAINTAIN FEE RECORD



DFD 4: MAINTAIN EMPLOYEE RECORD

MODULE DESCRIPTION

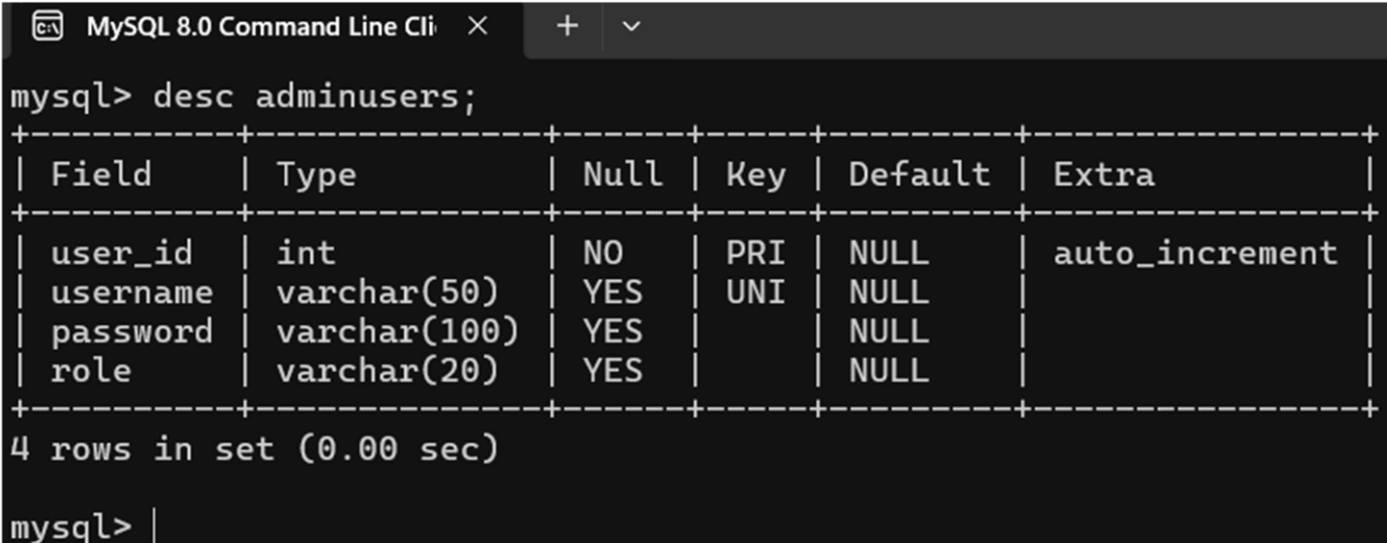
These are the following Modules in the project:-

1. **STUDENT** :- This module is used for the information about the students through which it can be considered for particular student information as roll_no, address, course, admission status etc. In this module admission, student record updation , deletion etc happen.
2. **STAFF/EMPLOYEE** :- This module consists, the information about employees (working members) in the college for their different posts as faculties, coordinator, peon and other teaching & non-teaching staffs.
3. **FEE** : - This module maintains the information of student's admission fee , monthly fee and different type of fee Dues etc.
4. **SETTING** :- It is necessary to maintain the security level by implementing the concept of user authentication checking. The security module deals with user authentication and security measures. In which the user login id and password is used to secure & maintain the protection feature.
5. **REPORT** :- This module is used to view or print different type of Report like admission details, fee details etc.

DATA STRUCTURE SNAP SHOT

DATA DICTIONARY: - The data dictionary stores description of data items and structures, as well as systems processes. Data dictionary used for database design in this project is mentioned. Care was taken during the database design for future expansion of school's need.

Admin (Login)



```
mysql> desc adminusers;
```

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	auto_increment
username	varchar(50)	YES	UNI	NULL	
password	varchar(100)	YES		NULL	
role	varchar(20)	YES		NULL	

```
4 rows in set (0.00 sec)

mysql> |
```

Teacher (Login):

```
MySQL 8.0 Command Line Cli x + v
mysql> desc techusers;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| user_id    | int           | NO   | PRI | NULL    | auto_increment |
| username   | varchar(50)   | YES  | UNI | NULL    |                |
| password   | varchar(100)  | YES  |     | NULL    |                |
| role       | varchar(20)   | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```

Student (Login):

```
MySQL 8.0 Command Line Cli x + v
mysql> desc stusers;
+-----+-----+-----+-----+-----+-----+
| Field      | Type          | Null | Key | Default | Extra          |
+-----+-----+-----+-----+-----+-----+
| user_id    | int           | NO   | PRI | NULL    | auto_increment |
| username   | varchar(50)   | YES  | UNI | NULL    |                |
| password   | varchar(100)  | YES  |     | NULL    |                |
| role       | varchar(20)   | YES  |     | NULL    |                |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> |
```

Student Details (mngstudent): -

```
mysql> desc mngstudent;
```

Field	Type	Null	Key	Default	Extra
rollno	int	NO	PRI	NULL	
studentname	varchar(100)	YES		NULL	
email	varchar(100)	YES		NULL	
contact	varchar(15)	YES		NULL	
course	varchar(100)	YES		NULL	

```
5 rows in set (0.00 sec)
```

```
mysql> |
```

Student Details (mngstudent): -

```
mysql> desc mngteachers;
```

Field	Type	Null	Key	Default	Extra
teacher_id	int	NO	PRI	NULL	auto_increment
teacher_name	varchar(50)	NO		NULL	
teacher_email	varchar(50)	NO	UNI	NULL	
teacher_contact	varchar(15)	NO		NULL	

```
4 rows in set (0.02 sec)
```

```
mysql> |
```

Attendance Details: -

```
mysql> desc attendance;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
roll_no	varchar(20)	YES		NULL	
attendance_date	date	YES		NULL	
status	varchar(10)	YES		NULL	
studentname	varchar(50)	YES		NULL	
sub_name	varchar(50)	YES		NULL	

```
6 rows in set (0.00 sec)
```

```
mysql> |
```

Marks Details: -

```
mysql> desc marks;
```

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI	NULL	auto_increment
roll_no	varchar(20)	YES		NULL	
student_name	varchar(100)	YES		NULL	
subject_name	varchar(100)	YES		NULL	
marks	int	YES		NULL	

```
5 rows in set (0.01 sec)
```

```
mysql> |
```

Course Details:

```
mysql> desc mngcourses;
```

Field	Type	Null	Key	Default	Extra
course_id	int	NO	PRI	NULL	auto_increment
course_name	varchar(50)	YES	UNI	NULL	
course_duration	varchar(20)	YES		NULL	
course_desc	varchar(100)	YES		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> |
```

Subject_assign:

```
mysql> desc subject_assign;
```

Field	Type	Null	Key	Default	Extra
assign_id	int	NO	PRI	NULL	auto_increment
teacher_name	varchar(100)	NO		NULL	
course_name	varchar(100)	NO		NULL	
subject_name	varchar(100)	NO		NULL	

```
4 rows in set (0.00 sec)
```

```
mysql> |
```

Subjects: -

```
mysql> desc subjects;
```

Field	Type	Null	Key	Default	Extra
sub_id	int	NO	PRI	NULL	auto_increment
sub_name	varchar(50)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> |
```

LIST OF REPORTS

The following reports will be generated: -

1. Student Reports

- List of all students
- Individual student details report
- Students by course report

2. Teacher Reports

- List of all teachers
- Teacher-wise subject report

3. Attendance Reports

- Daily attendance report
- Student-wise attendance report
- Monthly attendance summary

4. Marks/Result Reports

- Student result report
- Subject-wise marks report
- Overall performance report

5. Course & Subject Reports

- List of courses
- Subject details by course

6. Admin Reports

- Complete system report
- Summary of students, teachers, and courses

SCOPE AND FUTURE ENHANCEMENT

The Student Management System (SMS) has a wide scope in educational institutions such as schools, colleges, and training centers.

- Student Data Management
Store and manage student personal and academic information.
- Teacher & Course Management
Handle teacher details, subjects, and course structures.
- Attendance & Result Management
Track attendance and manage student marks efficiently.
- User-Based Access
Provide secure login for Admin, Teachers, and Students.
- Web-Based Access
Can be accessed from any system through a web browser.
- Report Generation
Generate various reports for analysis and decision-making.

Note: - The system improves efficiency, reduces manual work, and ensures better data management.

- Future Enhancements
- The system can be further improved by adding advanced features:

- **Mobile Application**
Develop an Android/iOS app for easy access.
- **Cloud Integration**
Store data on cloud for better scalability and remote access.
- **Notification System**
Send SMS/Email alerts for attendance, results, and updates.
- **Online Fee Management**
Add fee payment and billing system.
- **Advanced Analytics**
Provide graphical reports and performance analysis.
- **Biometric Attendance**
Integrate fingerprint or face recognition for attendance.
- **Multi-User & Multi-Institution Support**
Expand system for multiple schools/colleges.

IMPLEMENTATION METHODOLOGY

1. Requirement Gathering

- Collected system requirements from users (Admin, Teacher, Student)
- Identified features like student management, attendance, and result handling

2. System Design

- Designed system architecture using MVC (Model-View-Controller)
- Created database structure (tables like students, teachers, marks, etc.)
- Designed user interface (JSP pages, forms, dashboards)

3. Development (Coding Phase)

- Backend developed using Java Servlets
- Frontend created using JSP, HTML, CSS, JavaScript
- Database connectivity implemented using JDBC with MySQL
- Implemented modules:
 - Admin Module

- Teacher Module
- Student Module

4. Testing

- Performed unit testing for individual modules
- Conducted system testing for overall functionality
- Fixed bugs and ensured smooth performance

5. Deployment

- Deployed the application on Apache Tomcat Server
- Connected system with MySQL database
- Tested the system in real environment

6. Maintenance

- Regular monitoring of system performance
- Fixing errors and updating features when needed

Development Approach Used

- Incremental Model

The project was developed step-by-step, adding features gradually and testing each module.

TESTING

Software testing consists of the following stages.

a) TEST PLAN DEVELOPMENT:-

The importance of software testing and its implications need not be emphasized. Software testing is a critical element of software quality assurance and represents the ultimate review of specifications, design and coding. The main objective of testing is to uncover a host of errors,

systematically, with minimum effort and time. It improves the reliability of the software.

b) TESTING STRATEGY:-

A testing strategy explains how to conduct a test to uncover all the errors with minimum possible time and effort. It should be flexible enough to promote customization that may be necessary in due course of development process.

c) UNIT TESTING:-

Unit testing focuses verification effort on the smallest unit of software module using the detailed design criteria. Process specifications testing are done to uncover errors within the boundaries of the module. All modules must be successful in the unit test before start of the integration testing. The software developed for e-banking is tested module wise by giving normal values and extreme values as input. All the bugs have been identified and removed. The software designed is tested module wise. Before using the software, an employee has to give organization name and password.

In the next stage he has to provide division name, employee ID and password. If any of this information is wrong, the system gives an error message- “invalid user and password”

- Each module (Student, Teacher, Attendance, Marks) is tested individually

- Ensures every function works correctly

INTEGRATION TESTING:-

It is a systematic technique for constructing different program module in to an integrated software structure. This test uncovers the errors during integration process. The integration aspect is tested after integrating all the modules and validated.

- Checks interaction between modules
- Example: Student data connected with attendance and marks

VALIDATION TESTING:-

After the culmination of integration testing the software is ready as per the specification. But it has to be validated as per the specification and to uncover the unexpected future errors and to improve its reliability. The random value and boundary values as input is fed to the software developed and studied the software performance and validated

TEST CASE DESIGN

The techniques that are used in deriving the test cases are explained below.

CONDITION TESTING:-

Condition testing is a test case design method that exercises the logical conditions contained in the program. The possible components in a condition statement are a Boolean operator, Boolean variable,

relational operator, arithmetic expression and parenthesis.

BOUNDARY VALUE ANALYSIS:-

Boundary value analysis leads to the selection of test cases that exercise the boundary conditions or boundary values. It is observed that a large number of errors tend to appear at the boundaries of the input domain than in the center. The boundary condition is tested by inserting values beyond specification/availability of project allocations. E.g. the software is tested by giving more amount of money in the indent form than available in the budget. Under this conciliation the software gives an error message stating that insufficient fund.

EQUIVALENCE PARTITIONING:-

Equivalence portioning is a black box testing method that divides the input domain of a program into classes of data from which test cases can be derived. A typical test uncovers a class errors might otherwise require many more test cases before the error is observed. Equivalence classes for input determine the valid and invalid inputs for the program

System Testing:

- Entire system is tested as a whole
- Ensures all features work together properly

User Acceptance Testing (UAT):

- Tested by users (Admin/Teacher/Student)
- Confirms system meets user requirements

Test Case	Input	Expected Output	Result
Login	Valid username/password	Login successful	Pass
Add Student	Valid student details	Student added successfully	Pass
Attendance Entry	Select student & status	Attendance recorded	Pass
View Result	Student ID	Display marks and grade	Pass
Invalid Login	Wrong credentials	Error message	Pass

LIMITATIONS AND FURTHER ENHANCEMENTS

LIMITATIONS:

1. Limited Accessibility

- The system works mainly on a local server (Apache Tomcat)
- Requires proper setup to access from different locations

2. Basic Security Features

- Provides login authentication, but advanced security (encryption, multi-factor authentication) may not be fully implemented
-

3. No Mobile Application

- The system is web-based and not optimized as a dedicated mobile app
-

4. Internet/Server Dependency

- Requires server availability and network connection for proper functioning
-

5. Limited Scalability

- Suitable for small to medium institutions
 - May need upgrades to handle large-scale data and users
-

6. Manual Data Entry

- Data like student details, attendance, and marks must be entered manually
-

7. Basic User Interface

- UI may be simple and may not include advanced design or user experience features

RECOMMENDED AREAS FOR FURTHER RESEARCH

Considering the fact that has already gone ahead the following areas needs to be further probed for improving the system to meet the changing scenario.

- 1) There is a lot of scope for future research in the areas like automatic component/material planning, automatic advanced indenting, micro transactions, and customer relationship management.
- 2) Components like electronic payment system, digital certificates, electronic tokens, electronic cheques etc. are still in an embryonic stage of development. This area opens up a lot of scope for further research.

In the absence of universal standards for electronic transactions, the organization should develop a standard internal to it and that translate between its formats and those supported by its trading partners. This approach reduces dependence on industry schedules for adopting standards, and avoids the pitfalls of an organization-imposed solution.

1. User Authentication

- Users must log in using username and password
 - Only authorized users (Admin, Teacher, Student) can access the system
-

2. Role-Based Access Control

- Different users have different permissions:
 - Admin: Full access
 - Teacher: Limited access (attendance, marks)
 - Student: View-only access

Note: Prevents unauthorized actions and data misuse

3. Password Protection

- Passwords are stored securely (preferably encrypted)
 - Prevents unauthorized access to user accounts
-

4. Session Management

- Sessions are used to track logged-in users
 - Automatic logout after session ends
 - Prevents unauthorized access after logout
-

5. Database Security

- Controlled database access using JDBC
 - Only authorized queries are executed
 - Prevents unauthorized data modification
-

6. Input Validation

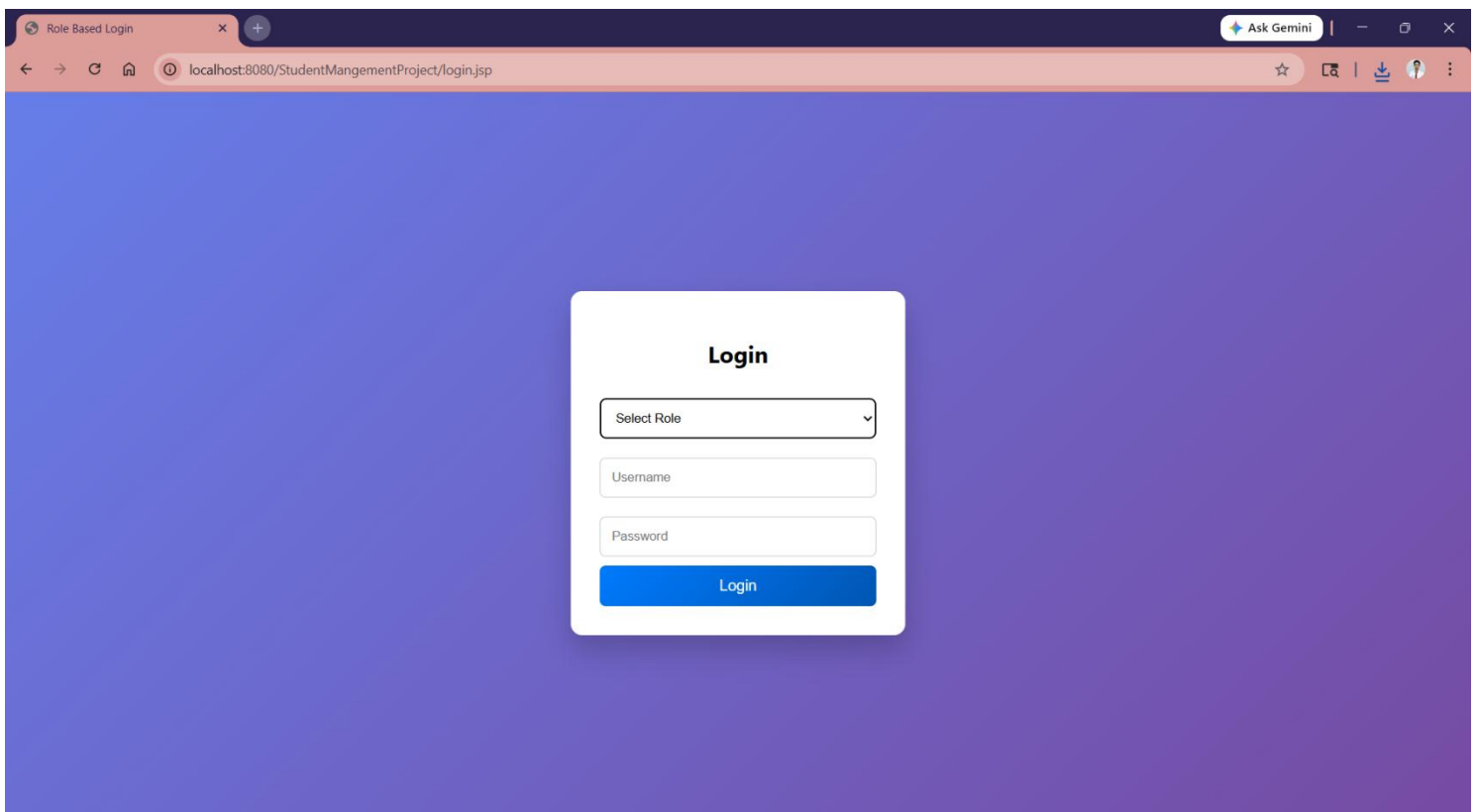
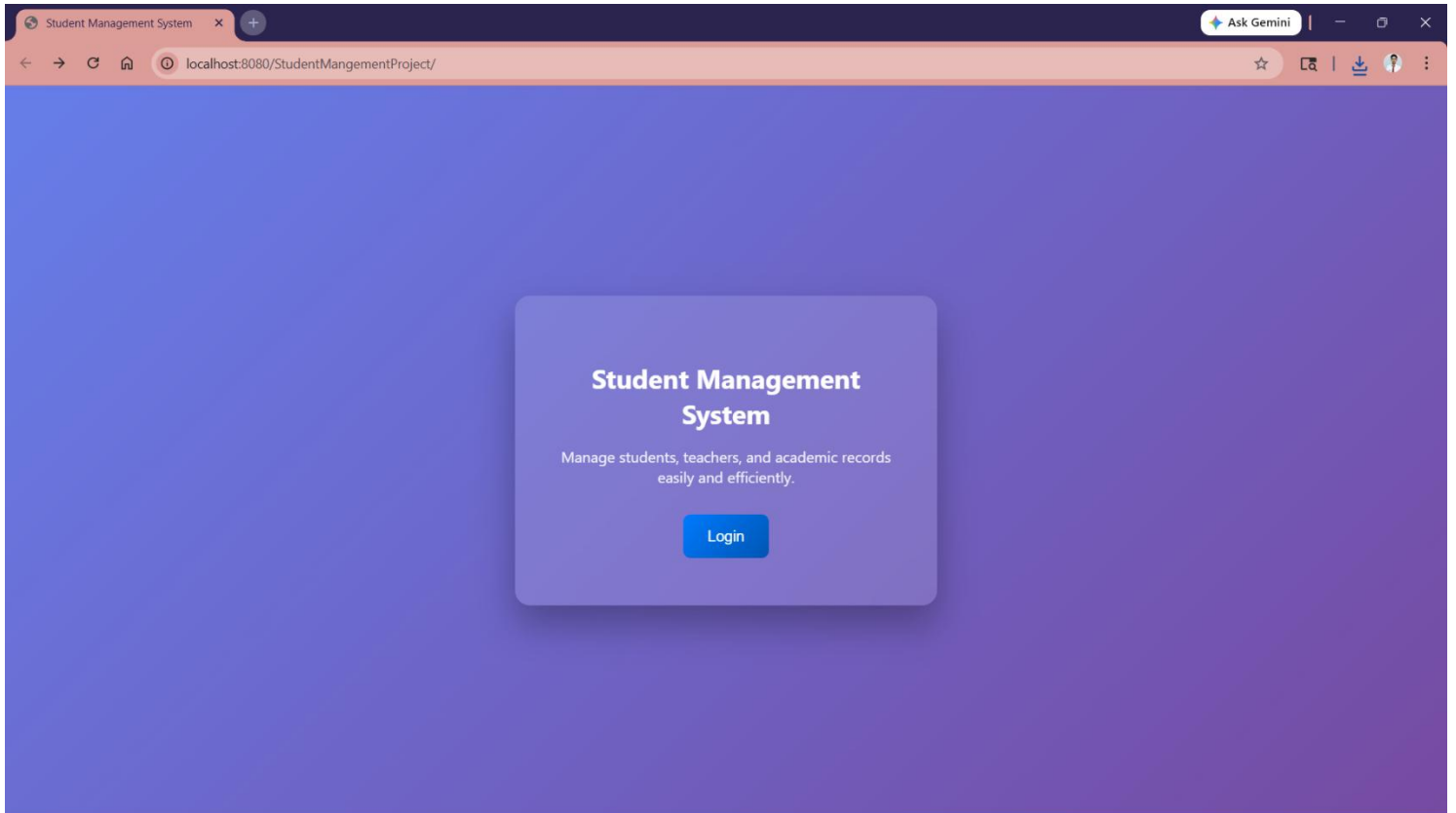
- User inputs are validated before processing
 - Prevents invalid data entry and basic attacks (like SQL Injection)
-

7. Error Handling

- System shows controlled error messages
- Sensitive information is not exposed to users

SNAPSHOTS

LOGIN Page:



Login.jsp Code:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Role Based Login</title>

<style>
*{ box-sizing:border-box; }
body{
    font-family:"Segoe UI",Tahoma,sans-serif;
    background:linear-gradient(135deg,#667eea,#764ba2);
    display:flex;
    justify-content:center;
    align-items:center;
    height:100vh;
    margin:0;
}
.login-box{
    background:#fff;
    padding:30px;
    width:100%;
    max-width:350px;
    border-radius:12px;
    box-shadow:0 15px 30px rgba(0,0,0,0.2);
}
h2{
    text-align:center;
    margin-bottom:20px;
}
input, select{
    width:100%;
    padding:12px;
    margin:10px 0;
    border-radius:6px;
    border:1px solid #ccc;
}
button{
    width:100%;
    padding:12px;
    background:linear-gradient(135deg,#007bff,#0056b3);
    color:white;
    border:none;
    border-radius:6px;
```

```

    font-size:16px;
    cursor:pointer;
}
.error{
    color:#dc3545;
    text-align:center;
    font-size:14px;
}
.signup-link{
    text-align:center;
    margin-top:15px;
}
.signup-link a{
    color:#667eea;
    text-decoration:none;
    font-weight:600;
}
</style>
</head>

<body>

<div class="login-box">
    <h2>Login</h2>

    <form action="LoginServlet" method="post">

        <select name="role" required>
            <option value="">Select Role</option>
            <option value="admin">Admin</option>
            <option value="teacher">Teacher</option>
            <option value="student">Student</option>
        </select>

        <input type="text" name="username" placeholder="Username" required>
        <input type="password" name="password" placeholder="Password" required>

        <div class="error">
            <%= request.getAttribute("errorMsg") != null ? request.getAttribute("errorMsg") : "" %>
        </div>

        <button type="submit">Login</button>
    </form>

    <!-- <div class="signup-link">
        Don't have an account?
        <a href="signup.jsp">Signup</a>
    </div> -->

```

</div>

</body>

</html>

Authfilter.java:

```
package com.mangestudent.signup;
import java.io.IOException;

import javax.servlet.Filter;
import javax.servlet.FilterChain;
import javax.servlet.FilterConfig;
import javax.servlet.ServletException;
import javax.servlet.ServletRequest;
import javax.servlet.ServletResponse;
import javax.servlet.annotation.WebFilter;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebFilter(urlPatterns = {
    "/adminDashboard.jsp",
    "/teacherDashboard.jsp",
    "/studentDashboard.jsp"
})
public class AuthFilter implements Filter {

    public void init(FilterConfig filterConfig) throws ServletException {}

    public void doFilter(ServletRequest request, ServletResponse response, FilterChain chain)
        throws IOException, ServletException {

        HttpServletRequest req = (HttpServletRequest) request;
        HttpServletResponse res = (HttpServletResponse) response;

        HttpSession session = req.getSession(false);

        String uri = req.getRequestURI();

        if (session == null ||
            session.getAttribute("username") == null ||
            session.getAttribute("role") == null) {

            res.sendRedirect("login.jsp");
            return;
        }
    }
}
```

```

String role = session.getAttribute("role").toString();

// Role-based page access
if (uri.contains("adminDashboard.jsp") && !role.equals("admin")) {
    res.sendRedirect("login.jsp");
    return;
}

if (uri.contains("teacherDashboard.jsp") && !role.equals("teacher")) {
    res.sendRedirect("login.jsp");
    return;
}

if (uri.contains("studentDashboard.jsp") && !role.equals("student")) {
    res.sendRedirect("login.jsp");
    return;
}

// User is authorized
chain.doFilter(request, response);
}

public void destroy() {}
}

```

LoginServlet.java:

```

package com.mangestudent.signup;
import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/LoginServlet")
public class LoginServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    private static final String DB_URL =
        "jdbc:mysql://localhost:3306/diffusers?useSSL=false&allowPublicKeyRetrieval=true";

```

```

private static final String DB_USER = "root";
private static final String DB_PASS = "2005";

protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {

    String role = request.getParameter("role");
    String username = request.getParameter("username");
    String password = request.getParameter("password");

    String table = "";

    if ("admin".equals(role)) {
        table = "adminusers";
    } else if ("teacher".equals(role)) {
        table = "techusers";
    } else if ("student".equals(role)) {
        table = "stusers";
    } else {
        request.setAttribute("errorMsg", "Invalid role selected!");
        request.getRequestDispatcher("login.jsp").forward(request, response);
        return;
    }

    try {
        Class.forName("com.mysql.cj.jdbc.Driver");
        Connection con = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);

        String sql = "SELECT * FROM " + table +
            " WHERE username=? AND password=? AND role=?";
        PreparedStatement ps = con.prepareStatement(sql);

        ps.setString(1, username);
        ps.setString(2, password);
        ps.setString(3, role);

        ResultSet rs = ps.executeQuery();

        if (rs.next()) {
            // Create session
            HttpSession session = request.getSession();
            session.setAttribute("username", username);
            session.setAttribute("role", role);

            // Redirect based on role
            if (role.equals("admin")) {
                response.sendRedirect("adminDashboard.jsp");
            } else if (role.equals("teacher")) {

```

```

        response.sendRedirect("teacherDashboard.jsp");
    } else if (role.equals("student")) {
        response.sendRedirect("studentDashboard.jsp");
    }

    } else {
        request.setAttribute("errorMsg", "Invalid username or password!");
        request.getRequestDispatcher("login.jsp").forward(request, response);
    }

    rs.close();
    ps.close();
    con.close();

    } catch (Exception e) {
        e.printStackTrace();
        request.setAttribute("errorMsg", "Database error!");
        request.getRequestDispatcher("login.jsp").forward(request, response);
    }
}
}
}

```

LogoutServlet.java:

```

package com.mangestudent.signup;
import java.io.IOException;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

@WebServlet("/LogoutServlet")
public class LogoutServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);
        if (session != null) {
            session.invalidate(); // destroy session
        }
    }
}

```

```

// Prevent caching (important)
response.setHeader("Cache-Control", "no-cache, no-store, must-revalidate");
response.setHeader("Pragma", "no-cache");
response.setDateHeader("Expires", 0);

response.sendRedirect("index.jsp");
}

// Optional: allow GET logout too
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    doPost(request, response);
}
}

```

SignupServlet.java:

```

package com.mangestudent.signup;

import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/SignupServlet")
public class SignupServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    private static final String DB_URL =
        "jdbc:mysql://localhost:3306/diffusers?useSSL=false&allowPublicKeyRetrieval=true";
    private static final String DB_USER = "root";
    private static final String DB_PASS = "2005";

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String fullname = request.getParameter("fullname"); // username

```



```

String password = request.getParameter("password");
String role = request.getParameter("role");

String tableName = "";

// Decide table based on role
if ("admin".equalsIgnoreCase(role)) {
    tableName = "adminusers";
} else if ("teacher".equalsIgnoreCase(role)) {
    tableName = "techusers";
} else if ("student".equalsIgnoreCase(role)) {
    tableName = "stusers";
} else {
    request.setAttribute("errorMsg", "Invalid role selected!");
    request.getRequestDispatcher("signup.jsp").forward(request, response);
    return;
}

Connection con = null;
PreparedStatement ps = null;

try {
    // Load MySQL Driver
    Class.forName("com.mysql.cj.jdbc.Driver");

    // Create Connection
    con = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);

    // SQL Insert
    String sql = "INSERT INTO " + tableName +
        " (username, password, role) VALUES (?, ?, ?)";

    ps = con.prepareStatement(sql);
    ps.setString(1, fullname);
    ps.setString(2, password); // Plain text (hash later)
    ps.setString(3, role);

    int rows = ps.executeUpdate();

    if (rows > 0) {
        request.setAttribute("successMsg", "Signup successful! Please login.");
    } else {
        request.setAttribute("errorMsg", "Signup failed. Try again.");
    }

    request.getRequestDispatcher("signup.jsp").forward(request, response);
} catch (Exception e) {

```

```

        e.printStackTrace();
        request.setAttribute("errorMsg", "Database Error!");
        request.getRequestDispatcher("signup.jsp").forward(request, response);

    } finally {
        try {
            if (ps != null) ps.close();
            if (con != null) con.close();
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
}
}

```

DBConnection.java:

```

package com.sms.util;

import java.sql.Connection;
import java.sql.DriverManager;

public class DBConnection {

    private static final String URL =
        "jdbc:mysql://localhost:3306/admindb?useSSL=false&serverTimezone=UTC";
    private static final String USER = "root";
    private static final String PASS = "2005";

    public static Connection getConnection() {
        Connection con = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(URL, USER, PASS);
        } catch (Exception e) {
            e.printStackTrace();
        }
        return con;
    }
}

```

Index.Jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Student Management System</title>

<style>
    * {
        box-sizing: border-box;
    }

    body {
        margin: 0;
        font-family: "Segoe UI", Tahoma, sans-serif;
        height: 100vh;
        background: linear-gradient(135deg, #667eea, #764ba2);
        display: flex;
        justify-content: center;
        align-items: center;
        color: #fff;
    }

    .welcome-box {
        background: rgba(255, 255, 255, 0.12);
        padding: 50px 40px;
        border-radius: 16px;
        text-align: center;
        max-width: 450px;
        width: 100%;
        backdrop-filter: blur(8px);
        box-shadow: 0 20px 40px rgba(0,0,0,0.3);
        animation: fadeIn 0.8s ease;
    }

    @keyframes fadeIn {
        from {
            opacity: 0;
            transform: translateY(20px);
        }
        to {
            opacity: 1;
            transform: translateY(0);
        }
    }
}
```

```

h1 {
  font-size: 28px;
  margin-bottom: 10px;
}



p {
  font-size: 16px;
  margin-bottom: 30px;
  opacity: 0.9;
}



.btn-group {
  display: flex;
  gap: 15px;
  justify-content: center;
}

.btn {
  padding: 14px 26px;
  font-size: 16px;
  border-radius: 8px;
  border: none;
  cursor: pointer;
  transition: transform 0.2s, box-shadow 0.2s;
}

.btn-login {
  background: linear-gradient(135deg, #007bff, #0056b3);
  color: #fff;
}

.btn-signup {
  background: linear-gradient(135deg, #28a745, #1e7e34);
  color: #fff;
}

.btn:hover {
  transform: translateY(-3px);
  box-shadow: 0 8px 20px rgba(0,0,0,0.3);
}

@media (max-width: 480px) {
  .btn-group {
    flex-direction: column;
  }
}
</style>

```

```

</head>
<body>

<div class="welcome-box">
  <h1>Student Management System</h1>
  <p>Manage students, teachers, and academic records easily and efficiently.</p>

  <div class="btn-group">
    <button class="btn btn-login" onclick="location.href='login.jsp'">Login</button>
  </div>
</div>

</body>
</html>

```

Signup.jsp:

```

<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>

<%
String role = (String) session.getAttribute("role");
if (role == null || !role.equals("admin")) {
    response.sendRedirect("login.jsp");
    return;
}
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Create Account</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:linear-gradient(135deg,#1d2671,#c33764);
    min-height:100vh;

```

```

    display:flex;
    align-items:center;
    justify-content:center;
}

.card{
    border-radius:16px;
    box-shadow:0 15px 30px rgba(0,0,0,0.25);
    animation:fadeIn 0.6s ease-in-out;
}

@keyframes fadeIn{
    from{opacity:0; transform:translateY(15px);}
    to{opacity:1; transform:translateY(0);}
}

.form-control, .form-select{
    border-radius:10px;
    padding:12px;
}

.btn{
    border-radius:10px;
    padding:12px;
    font-weight:600;
}

.error{
    color:#dc3545;
    font-size:14px;
    text-align:center;
}

.success{
    color:#28a745;
    font-size:14px;
    text-align:center;
}
</style>
</head>

<body>

<div class="col-md-4 col-sm-10">

<div class="card">
    <div class="card-header bg-primary text-white text-center">
        <h4><i class="fa fa-user-plus"></i> Create Account</h4>

```

```

</div>

<div class="card-body">

    <form action="SignupServlet" method="post" onsubmit="return validateSignup()">

        <div class="mb-3">
            <label class="form-label">Full Name</label>
            <input type="text" id="fullname" name="fullname" class="form-control"
                placeholder="Enter full name">
        </div>

        <div class="mb-3">
            <label class="form-label">Email</label>
            <input type="email" id="email" name="email" class="form-control"
                placeholder="Enter email address">
        </div>

        <div class="mb-3">
            <label class="form-label">Role</label>
            <select id="role" name="role" class="form-select">
                <option value="">Select Role</option>
                <option value="admin">Admin</option>
                <option value="teacher">Teacher</option>
                <option value="student">Student</option>
            </select>
        </div>

        <div class="mb-3">
            <label class="form-label">Password</label>
            <input type="password" id="password" name="password" class="form-control"
                placeholder="Enter password">
        </div>

        <div class="mb-3">
            <label class="form-label">Confirm Password</label>
            <input type="password" id="confirmPassword" class="form-control"
                placeholder="Confirm password">
        </div>

        <!-- Messages -->
        <div class="error mb-2">
            <%= request.getAttribute("errorMsg") != null ? request.getAttribute("errorMsg") : "" %>
        </div>

        <div class="success mb-2">
            <%= request.getAttribute("successMsg") != null ? request.getAttribute("successMsg") : "" %>
        </div>
    </form>

```

```

        <button type="submit" class="btn btn-success w-100">
            <i class="fa fa-check-circle"></i> Create Account
        </button>
    </form>

    <div class="text-center mt-3">
        <small>
            Go back to
            <a href="adminDashboard.jsp" class="fw-bold text-primary text-decoration-none">
                Admin Dashboard
            </a>
        </small>
    </div>

</div>
</div>

</div>

<script>
function validateSignup() {
    const name = document.getElementById("fullname").value.trim();
    const email = document.getElementById("email").value.trim();
    const role = document.getElementById("role").value;
    const password = document.getElementById("password").value;
    const confirmPassword = document.getElementById("confirmPassword").value;

    if (name === "" || email === "" || role === "" || password === "" || confirmPassword === "") {
        alert("All fields are required!");
        return false;
    }

    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(email)) {
        alert("Enter a valid email address!");
        return false;
    }

    if (password.length < 6) {
        alert("Password must be at least 6 characters!");
        return false;
    }

    if (password !== confirmPassword) {
        alert("Passwords do not match!");
        return false;
    }
}

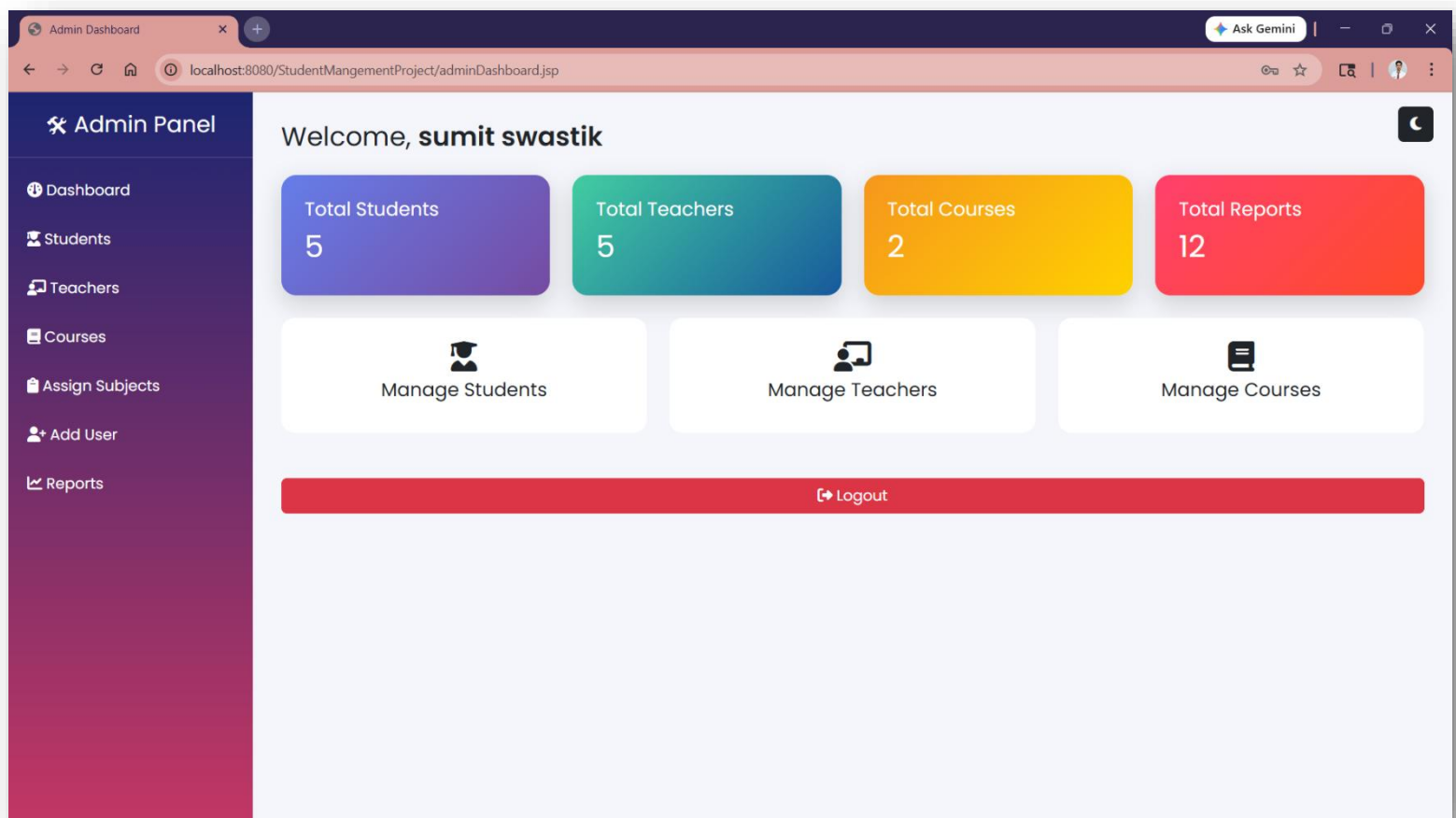
```



```
    return true;
}
</script>

</body>
</html>
```

Admin Dashboard



AdminDashboard.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>

<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>

<%
    String username = (String) session.getAttribute("username");
    String role = (String) session.getAttribute("role");

    if (username == null || role == null || !role.equals("admin")) {
        response.sendRedirect("login.jsp");
        return;
    }
%>
<%@ page import="java.sql.*" %>

<%
int studentCount = 0;
int teacherCount = 0;
int courseCount = 0;
int reportCount = 0;

try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/admindb",
        "root",
        "2005" // password (keep empty if none)
    );

    Statement st = con.createStatement();

    // Total Students
    ResultSet rs1 = st.executeQuery("SELECT COUNT(rollno) FROM mngstudent");
    if (rs1.next()) studentCount = rs1.getInt(1);

    // Total Teachers
    ResultSet rs2 = st.executeQuery("SELECT COUNT(teacher_id) FROM mngteachers");
    if (rs2.next()) teacherCount = rs2.getInt(1);

    // Total Courses
```

```

ResultSet rs3 = st.executeQuery("SELECT COUNT(course_id) FROM mngcourses");
if (rs3.next()) courseCount = rs3.getInt(1);

// Total Reports (Attendance + Marks)
ResultSet rs4 = st.executeQuery(
    "SELECT (SELECT COUNT(id) FROM attendance) + " +
    "(SELECT COUNT(id) FROM marks) AS totalReports"
);
if (rs4.next()) reportCount = rs4.getInt("totalReports");

con.close();

} catch (Exception e) {
    e.printStackTrace();
}
%>

```

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Admin Dashboard</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:#f4f6fb;
    transition:0.3s;
}
.dark-mode{
    background:#121212;
    color:#f5f5f5;
}

/* Sidebar */
.sidebar{
    width:260px;
    min-height:100vh;

```

```

    background:linear-gradient(180deg,#1d2671,#c33764);
    color:white;
    position:fixed;
}
.sidebar h4{
    text-align:center;
    padding:20px 0;
    border-bottom:1px solid rgba(255,255,255,0.2);
}
.sidebar a{
    display:block;
    padding:14px 20px;
    color:white;
    text-decoration:none;
    transition:0.3s;
}
.sidebar a:hover{
    background:rgba(255,255,255,0.15);
}

/* Main Content */
.main{
    margin-left:260px;
    padding:30px;
}

/* Cards */
.stat-card{
    border-radius:16px;
    padding:25px;
    color:white;
    box-shadow:0 10px 25px rgba(0,0,0,0.15);
}
.bg1{background:linear-gradient(135deg,#667eea,#764ba2);}
.bg2{background:linear-gradient(135deg,#43cea2,#185a9d);}
.bg3{background:linear-gradient(135deg,#f7971e,#ffd200);}
.bg4{background:linear-gradient(135deg,#ff416c,#ff4b2b);}

/* Dashboard Action Cards */
.action-card{
    background:white;
    border-radius:14px;
    padding:25px;
    text-align:center;
    cursor:pointer;
    transition:0.3s;
}
.action-card:hover{

```

```

    transform:translateY(-5px);
    box-shadow:0 15px 30px rgba(0,0,0,0.2);
}
.dark-mode .action-card{
    background:#1e1e1e;
}

/* Dark button */
.toggle-btn{
    position:absolute;
    top:15px;
    right:20px;
}
</style>
</head>

<body>

<!-- Sidebar -->
<div class="sidebar">
    <h4>✖ Admin Panel</h4>
    <a href="#"><i class="fa fa-gauge"></i> Dashboard</a>
    <a href="manageStudents.jsp"><i class="fa fa-user-graduate"></i> Students</a>
    <a href="manageTeachers.jsp"><i class="fa fa-chalkboard-teacher"></i> Teachers</a>
    <a href="manageCourses.jsp"><i class="fa fa-book"></i> Courses</a>
    <a href="assignSubjects.jsp"><i class="fa fa-clipboard"></i> Assign Subjects</a>
    <a href="signup.jsp"><i class="fa fa-user-plus"></i> Add User</a>
    <a href="AdminReportServlet"><i class="fa fa-chart-line"></i> Reports</a>
</div>

<!-- Main Content -->
<div class="main">

    <button class="btn btn-dark toggle-btn" onclick="toggleDark()">
        <i class="fa fa-moon"></i>
    </button>

    <h3 class="mb-4">Welcome, <b><%= username %></b></h3>

    <!-- Stats Cards -->
    <div class="row g-4 mb-4">

        <div class="col-md-3">
            <div class="stat-card bg1">
                <h5>Total Students</h5>
                <h2><%= studentCount %></h2>
            </div>
        </div>
    </div>

```

```

<div class="col-md-3">
  <div class="stat-card bg2">
    <h5>Total Teachers</h5>
    <h2><%= teacherCount %></h2>
  </div>
</div>

<div class="col-md-3">
  <div class="stat-card bg3">
    <h5>Total Courses</h5>
    <h2><%= courseCount %></h2>
  </div>
</div>

<div class="col-md-3">
  <div class="stat-card bg4">
    <h5>Total Reports</h5>
    <h2><%= reportCount %></h2>
  </div>
</div>

</div>

<!-- Action Cards -->
<div class="row g-4">
  <div class="col-md-4">
    <div class="action-card" onclick="go('manageStudents.jsp')">
      <i class="fa fa-user-graduate fa-2x mb-2"></i>
      <h5>Manage Students</h5>
    </div>
  </div>
  <div class="col-md-4">
    <div class="action-card" onclick="go('manageTeachers.jsp')">
      <i class="fa fa-chalkboard-teacher fa-2x mb-2"></i>
      <h5>Manage Teachers</h5>
    </div>
  </div>
  <div class="col-md-4">
    <div class="action-card" onclick="go('manageCourses.jsp')">
      <i class="fa fa-book fa-2x mb-2"></i>
      <h5>Manage Courses</h5>
    </div>
  </div>
</div>

<!-- Logout -->

```

```

<form action="LogoutServlet" method="post" class="mt-5">
  <button class="btn btn-danger w-100">
    <i class="fa fa-sign-out-alt"></i> Logout
  </button>
</form>

</div>

<script>
function go(page){
  window.location.href = page;
}
function toggleDark(){
  document.body.classList.toggle("dark-mode");
}
</script>

</body>
</html>

```

Admin Mange Student

The screenshot shows a web application titled "Manage Students" with a dark purple header. In the top right corner, there is a "Dashboard" button. Below the header, there is a section titled "Add New Student" with a blue background. This section contains five input fields: "Roll No", "Student Name", "Email", "Contact", and "Course". A green button labeled "+ Add Student" is positioned to the right of these fields. Below the "Add New Student" section is a "Student List" section with a dark header. It contains a table with six columns: "Roll", "Name", "Email", "Contact", "Course", and "Action". The table lists five students, each with a red trash icon in the "Action" column.

Roll	Name	Email	Contact	Course	Action
1	vishal kumar	vishal@gmail.com	7870315835	BSC.IT	
2	krrish soni	krrish@gmail.com	7079786840	BSC.IT	
3	vivek kumar	Vivek@gmail.com	7077786849	BSC.IT	
4	Nitish kumar	nitish@gmail.com	7079786878	BSC.IT	
5	div kumari	div@gmail.com	7079786876	BSC.IT	

Managestudent.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>

<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>

<%@ page import="java.util.*" %>
<%@ page import="com.sms.dao.StudentDAO" %>
<%@ page import="com.sms.model.Student" %>

<%
    StudentDAO dao = new StudentDAO();
    List<Student> students = dao.getAllStudents();
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Manage Students</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:linear-gradient(135deg,#1d2671,#c33764);
    min-height:100vh;
}
.card{
    border-radius:16px;
    box-shadow:0 15px 30px rgba(0,0,0,0.25);
}
.form-control{
    border-radius:10px;
```



```

}
.btn{
  border-radius:10px;
}
.table th{
  background:#1d2671;
  color:#fff;
}
.table tbody tr:hover{
  background:#fff3ff;
}
.error{
  border:2px solid red;
}
</style>

<script>
function validateForm(){
  let roll = document.getElementsByName("roll")[0];
  let name = document.getElementsByName("name")[0];
  let email = document.getElementsByName("email")[0];
  let contact = document.getElementsByName("contact")[0];
  let course = document.getElementsByName("course")[0];

  let valid = true;
  [roll,name,email,contact,course].forEach(f=>f.classList.remove("error"));

  if(roll.value <= 0){
    roll.classList.add("error");
    alert("Roll number must be positive");
    valid = false;
  }
  if(name.value.trim().length < 3){
    name.classList.add("error");
    alert("Student name must be at least 3 characters");
    valid = false;
  }
  if(!email.value.match(/^[^ ]+@[^ ]+\.[a-z]{2,}$/)){
    email.classList.add("error");
    alert("Invalid email format");
    valid = false;
  }
  if(!contact.value.match(/^[6-9]\d{9}$/)){
    contact.classList.add("error");
    alert("Enter valid 10-digit mobile number");
    valid = false;
  }
  if(course.value.trim() === ""){

```

```

        course.classList.add("error");
        alert("Course is required");
        valid = false;
    }
    return valid;
}

function confirmDelete(){
    return confirm("Are you sure you want to delete this student?");
}
</script>
</head>

<body>

<div class="container py-4">

    <!-- Header -->
    <div class="d-flex justify-content-between align-items-center mb-4 text-white">
        <h3><i class="fa fa-user-graduate"></i> Manage Students</h3>
        <button class="btn btn-light" onclick="goBack()">
            <i class="fa fa-arrow-left"></i> Dashboard
        </button>
    </div>

    <!-- Add Student Card -->
    <div class="card mb-4">
        <div class="card-header bg-primary text-white">
            <i class="fa fa-user-plus"></i> Add New Student
        </div>
        <div class="card-body">
            <form method="post" action="StudentServlet" onsubmit="return validateForm()">
                <input type="hidden" name="action" value="add">

                <div class="row g-3">
                    <div class="col-md-2">
                        <input type="number" name="roll" class="form-control" placeholder="Roll No" required>
                    </div>
                    <div class="col-md-3">
                        <input type="text" name="name" class="form-control" placeholder="Student Name" required>
                    </div>
                    <div class="col-md-3">
                        <input type="email" name="email" class="form-control" placeholder="Email" required>
                    </div>
                    <div class="col-md-2">
                        <input type="text" name="contact" class="form-control" placeholder="Contact" required>
                    </div>
                    <div class="col-md-2">

```

```

        <input type="text" name="course" class="form-control" placeholder="Course" required>
    </div>
</div>

<div class="text-end mt-3">
    <button class="btn btn-success">
        <i class="fa fa-plus"></i> Add Student
    </button>
</div>
</form>
</div>
</div>

<!-- Student Table -->
<div class="card">
    <div class="card-header bg-dark text-white">
        <i class="fa fa-table"></i> Student List
    </div>
    <div class="card-body table-responsive">
        <table class="table table-bordered align-middle text-center">
            <thead>
                <tr>
                    <th>Roll</th>
                    <th>Name</th>
                    <th>Email</th>
                    <th>Contact</th>
                    <th>Course</th>
                    <th>Action</th>
                </tr>
            </thead>
            <tbody>
                <%
                if(students.isEmpty()){
                %>
                <tr>
                    <td colspan="6">No students found</td>
                </tr>
                <%
                } else {
                for(Student s : students){
                %>
                <tr>
                    <td><%=s.getRollNo()%></td>
                    <td><%=s.getStudentName()%></td>
                    <td><%=s.getEmail()%></td>
                    <td><%=s.getContact()%></td>
                    <td><%=s.getCourse()%></td>
                    <td>

```

```

        <form method="post" action="StudentServlet" onsubmit="return confirmDelete()">
            <input type="hidden" name="action" value="delete">
            <input type="hidden" name="roll" value="<%=s.getRollNo()%>">
            <button class="btn btn-danger btn-sm">
                <i class="fa fa-trash"></i>
            </button>
        </form>
    </td>
</tr>
</tbody>
</table>
</div>
</div>

<script>
function goBack(){
    window.location.href="adminDashboard.jsp";
}
</script>

</body>
</html>

```

StudentServlet.java:

```

package com.sms.controller;

import com.sms.dao.StudentDAO;
import com.sms.model.Student;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;
import java.io.IOException;

@WebServlet("/StudentServlet")
public class StudentServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

```

```

private StudentDAO dao = new StudentDAO();

@Override
protected void doPost(HttpServletRequest req, HttpServletResponse resp)
    throws ServletException, IOException {

    String action = req.getParameter("action");

    // =====
    // ADD STUDENT
    // =====
    if ("add".equals(action)) {

        Student s = new Student();
        s.setRollNo(Integer.parseInt(req.getParameter("roll")));
        s.setStudentName(req.getParameter("name"));
        s.setEmail(req.getParameter("email"));
        s.setContact(req.getParameter("contact"));
        s.setCourse(req.getParameter("course"));

        dao.addStudent(s);
    }

    // =====
    // DELETE STUDENT
    // =====
    if ("delete".equals(action)) {
        int rollNo = Integer.parseInt(req.getParameter("roll"));
        dao.deleteStudent(rollNo);
    }

    // =====
    // REDIRECT
    // =====
    resp.sendRedirect("manageStudents.jsp");
}
}

```

StudentDao.java:

```

package com.sms.dao;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

import com.sms.model.Student;
import com.sms.util.DBConnection;

public class StudentDAO {

    // =====
    // ADD STUDENT
    // =====
    public void addStudent(Student s) {

        String sql = "INSERT INTO mngstudent (rollno, studentname, email, contact, course) VALUES (?,?,,?,?)";

        try (Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setInt(1, s.getRollNo());
            ps.setString(2, s.getStudentName());
            ps.setString(3, s.getEmail());
            ps.setString(4, s.getContact());
            ps.setString(5, s.getCourse());

            ps.executeUpdate();

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // =====
    // FETCH ALL STUDENTS
    // =====
    public List<Student> getAllStudents() {

        List<Student> list = new ArrayList<>();
        String sql = "SELECT * FROM mngstudent";

        try (Connection con = DBConnection.getConnection();
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery(sql)) {

            while (rs.next()) {
                Student s = new Student();

```

```

        s.setRollNo(rs.getInt("rollno"));
        s.setStudentName(rs.getString("studentname"));
        s.setEmail(rs.getString("email"));
        s.setContact(rs.getString("contact"));
        s.setCourse(rs.getString("course"));

        list.add(s);
    }

    } catch (Exception e) {
        e.printStackTrace();
    }

    return list;
}

// =====
// DELETE STUDENT
// =====
public void deleteStudent(int rollNo) {

    String sql = "DELETE FROM mngstudent WHERE rollno=?";

    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setInt(1, rollNo);
        ps.executeUpdate();

    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

Studentmodel.java:

```

package com.sms.model;

public class Student {

    private int rollNo;
    private String studentName;
    private String email;
    private String contact;
    private String course;

```

```

public int getRollNo() {
    return rollNo;
}
public void setRollNo(int rollNo) {
    this.rollNo = rollNo;
}

public String getStudentName() {
    return studentName;
}
public void setStudentName(String studentName) {
    this.studentName = studentName;
}

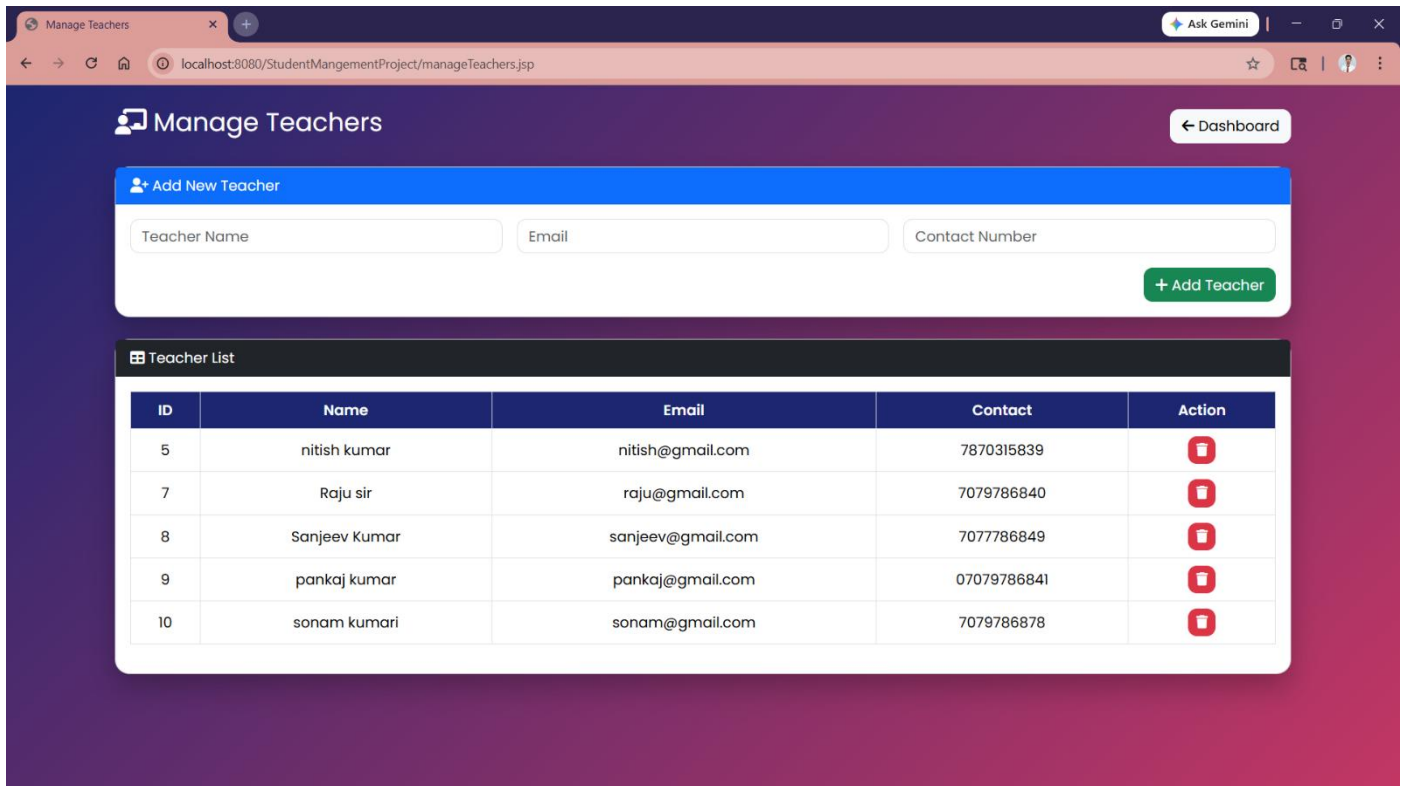
public String getEmail() {
    return email;
}
public void setEmail(String email) {
    this.email = email;
}

public String getContact() {
    return contact;
}
public void setContact(String contact) {
    this.contact = contact;
}

public String getCourse() {
    return course;
}
public void setCourse(String course) {
    this.course = course;
}
}

```


Admin Manage Teachers



ManageTeacher.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
```

```
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>
```

```
<%@ page import="java.util.*" %>
<%@ page import="com.sms.dao.TeacherDAO" %>
```

```

<%@ page import="com.sms.model.Teacher" %>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Manage Teachers</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:linear-gradient(135deg,#1d2671,#c33764);
    min-height:100vh;
}
.card{
    border-radius:16px;
    box-shadow:0 15px 30px rgba(0,0,0,0.25);
}
.form-control{
    border-radius:10px;
}
.btn{
    border-radius:10px;
}
.table th{
    background:#1d2671;
    color:#fff;
}
.table tbody tr:hover{
    background:#f1f3ff;
}
</style>
</head>

<body>

<div class="container py-4">

    <!-- Header -->

```

```

<div class="d-flex justify-content-between align-items-center mb-4 text-white">
  <h3><i class="fa fa-chalkboard-teacher"></i> Manage Teachers</h3>
  <button class="btn btn-light" onclick="goBack()">
    <i class="fa fa-arrow-left"></i> Dashboard
  </button>
</div>

<!-- Add Teacher Card -->
<div class="card mb-4">
  <div class="card-header bg-primary text-white">
    <i class="fa fa-user-plus"></i> Add New Teacher
  </div>
  <div class="card-body">
    <form method="post" action="<%=request.getContextPath()%>/TeacherServlet">
      <input type="hidden" name="action" value="add">

      <div class="row g-3">
        <div class="col-md-4">
          <input type="text" name="name" class="form-control" placeholder="Teacher Name" required>
        </div>
        <div class="col-md-4">
          <input type="email" name="email" class="form-control" placeholder="Email" required>
        </div>
        <div class="col-md-4">
          <input type="text" name="contact" class="form-control" placeholder="Contact Number" required>
        </div>
      </div>

      <div class="text-end mt-3">
        <button class="btn btn-success">
          <i class="fa fa-plus"></i> Add Teacher
        </button>
      </div>
    </form>
  </div>
</div>

<!-- Teacher Table -->
<div class="card">
  <div class="card-header bg-dark text-white">
    <i class="fa fa-table"></i> Teacher List
  </div>
  <div class="card-body table-responsive">
    <table class="table table-bordered align-middle text-center">
      <thead>
        <tr>
          <th>ID</th>
          <th>Name</th>

```

```

        <th>Email</th>
        <th>Contact</th>
        <th>Action</th>
    </tr>
</thead>
<tbody>
<%
TeacherDAO dao = new TeacherDAO();
List<Teacher> list = dao.getAllTeachers();

if(list.isEmpty()){
%>
    <tr>
        <td colspan="5">No teachers found</td>
    </tr>
<%
} else {
    for(Teacher t : list){
%>
        <tr>
            <td><%= t.getId() %></td>
            <td><%= t.getName() %></td>
            <td><%= t.getEmail() %></td>
            <td><%= t.getContact() %></td>
            <td>
                <form method="post" action="<%=request.getContextPath()%>/TeacherServlet">
                    <input type="hidden" name="action" value="delete">
                    <input type="hidden" name="id" value="<%= t.getId() %>">
                    <button class="btn btn-danger btn-sm">
                        <i class="fa fa-trash"></i>
                    </button>
                </form>
            </td>
        </tr>
<%
    }
}
%>
</tbody>
</table>
</div>
</div>

<script>
function goBack(){
    window.location.href="<%=request.getContextPath()%>/adminDashboard.jsp";

```

```
}  
</script>  
  
</body>  
</html>
```

TeacherServlet.java:

```
package com.sms.controller;  
  
import com.sms.dao.TeacherDAO;  
import com.sms.model.Teacher;  
  
import javax.servlet.ServletException;  
import javax.servlet.annotation.WebServlet;  
import javax.servlet.http.*;  
import java.io.IOException;  
  
@WebServlet("/TeacherServlet")  
public class TeacherServlet extends HttpServlet {  
  
    private static final long serialVersionUID = 1L;  
    private TeacherDAO dao = new TeacherDAO();  
  
    protected void doPost(HttpServletRequest req, HttpServletResponse resp)  
        throws ServletException, IOException {  
  
        String action = req.getParameter("action");  
  
        if ("add".equals(action)) {  
            Teacher t = new Teacher();  
            t.setTeacherName(req.getParameter("name"));  
            t.setTeacherEmail(req.getParameter("email"));  
            t.setTeacherContact(req.getParameter("contact"));  
            dao.addTeacher(t);  
        }  
  
        if ("delete".equals(action)) {  
            int id = Integer.parseInt(req.getParameter("id"));  
            dao.deleteTeacher(id);  
        }  
  
        resp.sendRedirect("manageTeachers.jsp");  
    }  
}
```

TeacherDao.java:

```
package com.sms.dao;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

import com.sms.model.Teacher;
import com.sms.util.DBConnection;

public class TeacherDAO {

    // ADD TEACHER
    public void addTeacher(Teacher t) {
        String sql = "INSERT INTO mngteachers(teacher_name, teacher_email, teacher_contact) VALUES (?, ?, ?)";
        try (Connection con = DBConnection.getConnection();
            PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, t.getTeacherName());
            ps.setString(2, t.getTeacherEmail());
            ps.setString(3, t.getTeacherContact());
            ps.executeUpdate();

        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // FETCH ALL TEACHERS
    public List<Teacher> getAllTeachers() {
        List<Teacher> list = new ArrayList<>();
        String sql = "SELECT * FROM mngteachers";

        try (Connection con = DBConnection.getConnection();
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery(sql)) {

            while (rs.next()) {
                Teacher t = new Teacher();
                t.setTeacherId(rs.getInt("teacher_id"));
                t.setTeacherName(rs.getString("teacher_name"));
                t.setTeacherEmail(rs.getString("teacher_email"));
                t.setTeacherContact(rs.getString("teacher_contact"));
                list.add(t);
            }
        }
    }
}
```

```

    }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

// DELETE TEACHER
public void deleteTeacher(int id) {
    String sql = "DELETE FROM mngteachers WHERE teacher_id=?";
    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setInt(1, id);
        ps.executeUpdate();
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

Teachermodel.java:

```

package com.sms.model;

public class Teacher {

    private int teacherId;
    private String teacherName;
    private String teacherEmail;
    private String teacherContact;

    public int getTeacherId() {
        return teacherId;
    }

    public void setTeacherId(int teacherId) {
        this.teacherId = teacherId;
    }

    public String getTeacherName() {
        return teacherName;
    }

    public void setTeacherName(String teacherName) {
        this.teacherName = teacherName;
    }
}

```

```

}

public String getTeacherEmail() {
    return teacherEmail;
}

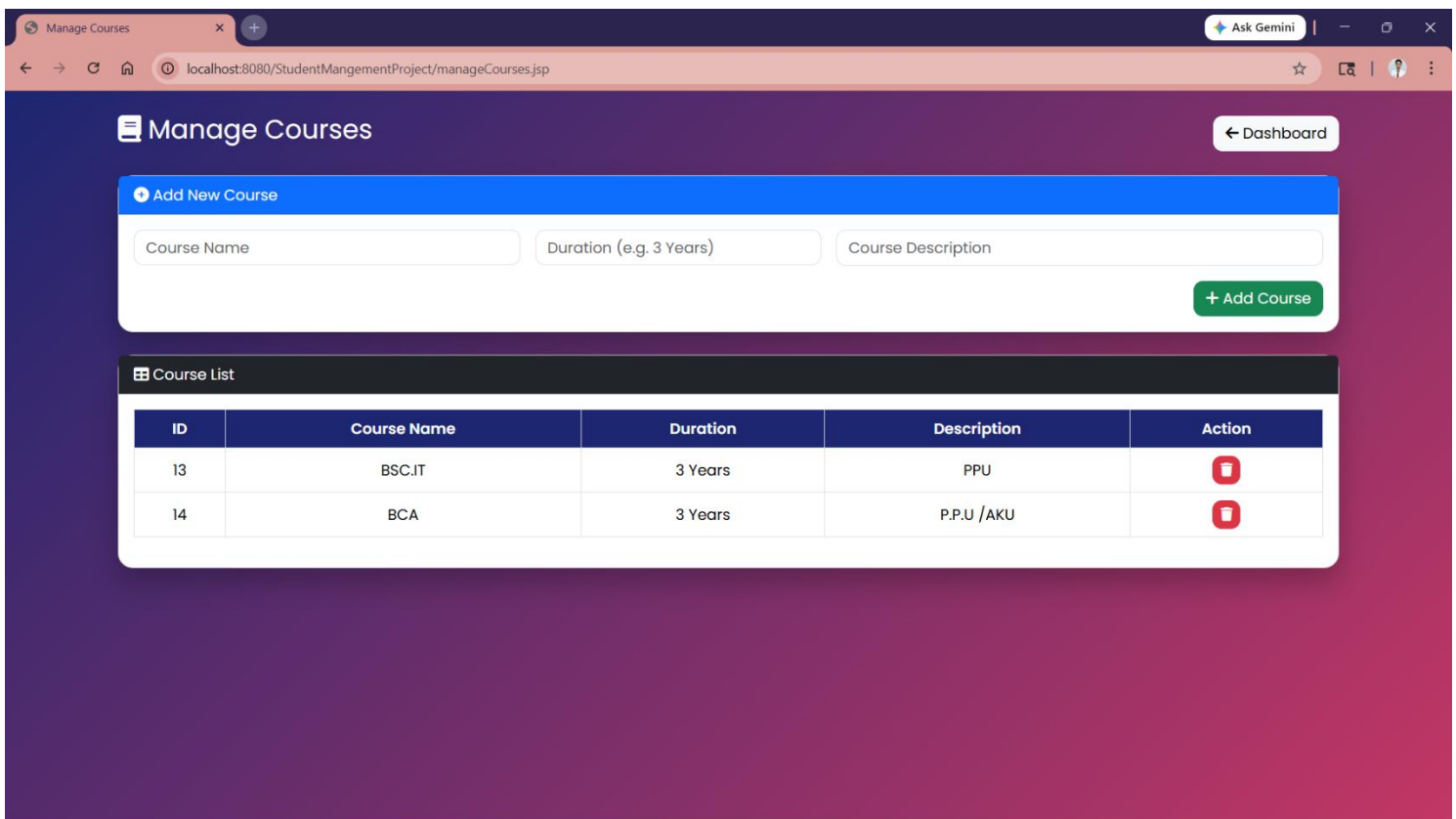
public void setTeacherEmail(String teacherEmail) {
    this.teacherEmail = teacherEmail;
}

public String getTeacherContact() {
    return teacherContact;
}

public void setTeacherContact(String teacherContact) {
    this.teacherContact = teacherContact;
}
}

```

Admin Manage Course



The screenshot shows a web application titled "Manage Courses" running on a browser at the URL `localhost:8080/StudentMangementProject/manageCourses.jsp`. The interface has a dark purple header with a "Manage Courses" title and a "Dashboard" link. Below the header is a blue "Add New Course" section with three input fields: "Course Name", "Duration (e.g. 3 Years)", and "Course Description". A green "+ Add Course" button is at the bottom right of this section. Below the add section is a "Course List" section containing a table with the following data:

ID	Course Name	Duration	Description	Action
13	BSC.IT	3 Years	PPU	
14	BCA	3 Years	P.P.U /AKU	

ManageCourse.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>

<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>

<%@ page import="java.util.*" %>
<%@ page import="com.sms.dao.CourseDAO" %>
<%@ page import="com.sms.model.Course" %>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Manage Courses</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:linear-gradient(135deg,#1d2671,#c33764);
    min-height:100vh;
}
.card{
    border-radius:16px;
    box-shadow:0 15px 30px rgba(0,0,0,0.25);
}
.form-control{
    border-radius:10px;
}
textarea.form-control{
    resize:none;
```

```

}
.btn{
  border-radius:10px;
}
.table th{
  background:#1d2671;
  color:#fff;
}
.table tbody tr:hover{
  background:#f1f3ff;
}
</style>
</head>

<body>

<div class="container py-4">

  <!-- Header -->
  <div class="d-flex justify-content-between align-items-center mb-4 text-white">
    <h3><i class="fa fa-book"></i> Manage Courses</h3>
    <button class="btn btn-light" onclick="goBack()">
      <i class="fa fa-arrow-left"></i> Dashboard
    </button>
  </div>

  <!-- Add Course Card -->
  <div class="card mb-4">
    <div class="card-header bg-primary text-white">
      <i class="fa fa-plus-circle"></i> Add New Course
    </div>
    <div class="card-body">
      <form method="post" action="<%=request.getContextPath()%>/CourseServlet">
        <input type="hidden" name="action" value="add">

        <div class="row g-3">
          <div class="col-md-4">
            <input type="text" name="name" class="form-control"
              placeholder="Course Name" required>
          </div>
          <div class="col-md-3">
            <input type="text" name="duration" class="form-control"
              placeholder="Duration (e.g. 3 Years)" required>
          </div>
          <div class="col-md-5">
            <textarea name="desc" class="form-control"
              placeholder="Course Description" rows="1" required></textarea>
          </div>
        </div>
      </form>
    </div>
  </div>

```

```

</div>

<div class="text-end mt-3">
  <button class="btn btn-success">
    <i class="fa fa-plus"></i> Add Course
  </button>
</div>
</form>
</div>
</div>

<!-- Course Table -->
<div class="card">
  <div class="card-header bg-dark text-white">
    <i class="fa fa-table"></i> Course List
  </div>
  <div class="card-body table-responsive">
    <table class="table table-bordered align-middle text-center">
      <thead>
        <tr>
          <th>ID</th>
          <th>Course Name</th>
          <th>Duration</th>
          <th>Description</th>
          <th>Action</th>
        </tr>
      </thead>
      <tbody>
        <%
CourseDAO dao = new CourseDAO();
List<Course> list = dao.getAllCourses();

if(list.isEmpty()){
%>
  <tr>
    <td colspan="5">No courses available</td>
  </tr>
  <%
} else {
  for(Course c : list){
%>
    <tr>
      <td><%= c.getCourseId() %></td>
      <td><%= c.getCourseName() %></td>
      <td><%= c.getCourseDuration() %></td>
      <td><%= c.getCourseDesc() %></td>
      <td>
        <form method="post" action="<%=request.getContextPath()%>/CourseServlet">

```

```

        <input type="hidden" name="action" value="delete">
        <input type="hidden" name="id" value="<%= c.getCourseId() %>">
        <button class="btn btn-danger btn-sm">
            <i class="fa fa-trash"></i>
        </button>
    </form>
</td>
</tr>
</tbody>
</table>
</div>
</div>

<script>
function goBack(){
    window.location.href="<%=request.getContextPath()%>/adminDashboard.jsp";
}
</script>

</body>
</html>

```

CourseServlet.java:

```

package com.sms.controller;

import com.sms.dao.CourseDAO;
import com.sms.model.Course;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

import java.io.IOException;

```

```

@WebServlet("/CourseServlet")
public class CourseServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;
    private CourseDAO dao = new CourseDAO();

    @Override
    protected void doPost(HttpServletRequest req, HttpServletResponse resp)
        throws ServletException, IOException {

        String action = req.getParameter("action");

        if ("add".equals(action)) {
            Course c = new Course();
            c.setCourseName(req.getParameter("name"));
            c.setCourseDuration(req.getParameter("duration"));
            c.setCourseDesc(req.getParameter("desc"));

            dao.addCourse(c);
        }
        else if ("delete".equals(action)) {
            int id = Integer.parseInt(req.getParameter("id"));
            dao.deleteCourse(id);
        }

        resp.sendRedirect("manageCourses.jsp");
    }
}

```

CourseDao.java:

```

package com.sms.dao;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

import com.sms.model.Course;
import com.sms.util.DBConnection;

public class CourseDAO {

```

```

// INSERT
public boolean addCourse(Course c) {
    String sql = "INSERT INTO mngcourses(course_name, course_duration, course_desc) VALUES (?, ?, ?)";
    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setString(1, c.getCourseName());
        ps.setString(2, c.getCourseDuration());
        ps.setString(3, c.getCourseDesc());
        return ps.executeUpdate() > 0;

    } catch (Exception e) {
        e.printStackTrace();
    }
    return false;
}

// FETCH ALL
public List<Course> getAllCourses() {
    List<Course> list = new ArrayList<>();
    String sql = "SELECT * FROM mngcourses";

    try (Connection con = DBConnection.getConnection();
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery(sql)) {

        while (rs.next()) {
            Course c = new Course();
            c.setCourseId(rs.getInt("course_id"));
            c.setCourseName(rs.getString("course_name"));
            c.setCourseDuration(rs.getString("course_duration"));
            c.setCourseDesc(rs.getString("course_desc"));
            list.add(c);
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

// DELETE
public void deleteCourse(int id) {
    String sql = "DELETE FROM mngcourses WHERE course_id=?";
    try (Connection con = DBConnection.getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {
        ps.setInt(1, id);
        ps.executeUpdate();
    } catch (Exception e) {

```

```

        e.printStackTrace();
    }
}
}

```

Coursemodel.java:

```

package com.sms.model;

public class Course {
    private int courseId;
    private String courseName;
    private String courseDuration;
    private String courseDesc;

    // Getters & Setters
    public int getCourseId() {
        return courseId;
    }
    public void setCourseId(int courseId) {
        this.courseId = courseId;
    }
    public String getCourseName() {
        return courseName;
    }
    public void setCourseName(String courseName) {
        this.courseName = courseName;
    }
    public String getCourseDuration() {
        return courseDuration;
    }
    public void setCourseDuration(String courseDuration) {
        this.courseDuration = courseDuration;
    }
    public String getCourseDesc() {
        return courseDesc;
    }
    public void setCourseDesc(String courseDesc) {
        this.courseDesc = courseDesc;
    }
}

```

Assign Subject

The screenshot shows a web browser window with the URL `localhost:8080/StudentMangementProject/assignSubjects.jsp`. The page has a dark purple header with the title "Assign Subjects" and a "Dashboard" link. Below the header is a blue bar with the text "Assign Subject to Teacher". Underneath this bar are three dropdown menus labeled "Select Teacher", "Select Course", and "Select Subject". To the right of these dropdowns is a green button labeled "+ Assign Subject". Below the dropdowns is a table titled "Assigned Subjects". The table has four columns: "Teacher", "Course", "Subject", and "Action". The table contains six rows of data, each with a red trash icon in the "Action" column.

Teacher	Course	Subject	Action
nitish kumar	BCA	HTML	
Raju sir	BSC.IT	java	
Sanjeev Kumar	BSC.IT	HTML	
Sanjeev Kumar	BSC.IT	css	
sonam kumari	BSC.IT	PHP	
Raju sir	BSC.IT	SQL Server	

AssignSubject.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
```

```
<%  
if(session.getAttribute("username")==null){  
    response.sendRedirect("login.jsp");  
}  
%>
```

```
<%@ page import="java.util.*" %>  
<%@ page import="com.sms.dao.SubjectAssignDAO" %>
```



```

<%@ page import="com.sms.model.SubjectAssign" %>

<%
SubjectAssignDAO dao = new SubjectAssignDAO();
List<String> teachers = dao.getTeachers();
List<String> courses = dao.getCourses();
List<String> subjects = dao.getSubjects();
List<SubjectAssign> list = dao.getAllAssignments();
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Assign Subjects</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:linear-gradient(135deg,#1d2671,#c33764);
    min-height:100vh;
}
.card{
    border-radius:16px;
    box-shadow:0 15px 30px rgba(0,0,0,0.25);
}
.form-select{
    border-radius:10px;
}
.btn{
    border-radius:10px;
}
.table th{
    background:#1d2671;
    color:#fff;
}
.table tbody tr:hover{
    background:#f1f3ff;
}

```

```

</style>
</head>

<body>

<div class="container py-4">

    <!-- Header -->
    <div class="d-flex justify-content-between align-items-center mb-4 text-white">
        <h3><i class="fa fa-clipboard-list"></i> Assign Subjects</h3>
        <button class="btn btn-light" onclick="goBack()">
            <i class="fa fa-arrow-left"></i> Dashboard
        </button>
    </div>

    <!-- Assign Subject Card -->
    <div class="card mb-4">
        <div class="card-header bg-primary text-white">
            <i class="fa fa-plus-circle"></i> Assign Subject to Teacher
        </div>
        <div class="card-body">
            <form method="post" action="<%=request.getContextPath()%>/SubjectAssignServlet">
                <input type="hidden" name="action" value="assign">

                <div class="row g-3">
                    <div class="col-md-4">
                        <select name="teacher" class="form-select" required>
                            <option value="">Select Teacher</option>
                            <% for(String t : teachers){ %>
                                <option value="<%=t%>"><%=t%></option>
                            <% } %>
                        </select>
                    </div>

                    <div class="col-md-4">
                        <select name="course" class="form-select" required>
                            <option value="">Select Course</option>
                            <% for(String c : courses){ %>
                                <option value="<%=c%>"><%=c%></option>
                            <% } %>
                        </select>
                    </div>

                    <div class="col-md-4">
                        <select name="subject" class="form-select" required>
                            <option value="">Select Subject</option>
                            <% for(String s : subjects){ %>
                                <option value="<%=s%>"><%=s%></option>
                            <% } %>
                        </select>
                    </div>
                </div>
            </form>
        </div>
    </div>

```

```

        <% } %>
    </select>
</div>
</div>

<div class="text-end mt-3">
    <button class="btn btn-success">
        <i class="fa fa-plus"></i> Assign Subject
    </button>
</div>
</form>
</div>
</div>

<!-- Assignment Table -->
<div class="card">
    <div class="card-header bg-dark text-white">
        <i class="fa fa-table"></i> Assigned Subjects
    </div>
    <div class="card-body table-responsive">
        <table class="table table-bordered align-middle text-center">
            <thead>
                <tr>
                    <th>Teacher</th>
                    <th>Course</th>
                    <th>Subject</th>
                    <th>Action</th>
                </tr>
            </thead>
            <tbody>
                <% if(list.isEmpty()){ %>
                    <tr>
                        <td colspan="4">No assignments found</td>
                    </tr>
                <% } else {
                    for(SubjectAssign s : list){ %>
                        <tr>
                            <td><%= s.getTeacherName() %></td>
                            <td><%= s.getCourseName() %></td>
                            <td><%= s.getSubjectName() %></td>
                            <td>
                                <form method="post" action="<%=request.getContextPath()%>/SubjectAssignServlet">
                                    <input type="hidden" name="action" value="delete">
                                    <input type="hidden" name="id" value="<%= s.getAssignId() %>">
                                    <button class="btn btn-danger btn-sm">
                                        <i class="fa fa-trash"></i>
                                    </button>
                                </form>
                            </td>
                        </tr>
                    <% } %>
                </tbody>
            </table>
        </div>
    </div>

```

```

        </td>
    </tr>
<% } } %>
</tbody>
</table>
</div>
</div>

</div>

<script>
function goBack(){
    window.location.href="<%=request.getContextPath()%>/adminDashboard.jsp";
}
</script>

</body>
</html>

```

SubjectAssignServlet.java:

```

package com.sms.controller;

import java.io.IOException;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

import com.sms.dao.SubjectAssignDAO;
import com.sms.model.SubjectAssign;

@WebServlet("/SubjectAssignServlet")
public class SubjectAssignServlet extends HttpServlet {

    /**
     *
     */
    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest req, HttpServletResponse resp)
        throws ServletException, IOException {

        SubjectAssignDAO dao = new SubjectAssignDAO();
        String action = req.getParameter("action");
    }
}

```

```

if ("assign".equals(action)) {
    SubjectAssign s = new SubjectAssign();
    s.setTeacherName(req.getParameter("teacher"));
    s.setCourseName(req.getParameter("course"));
    s.setSubjectName(req.getParameter("subject"));
    dao.assignSubject(s);
}

if ("delete".equals(action)) {
    int id = Integer.parseInt(req.getParameter("id"));
    dao.deleteAssignment(id);
}

resp.sendRedirect("assignSubjects.jsp");
}
}

```

SubjectAssignDao.java:

```

package com.sms.dao;

import java.sql.*;
import java.util.*;
import com.sms.model.SubjectAssign;

public class SubjectAssignDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/admindb", "root", "2005");
    }

    /* Dropdown Data */
    public List<String> getTeachers() {
        List<String> list = new ArrayList<>();
        try (Connection con = getConnection();
            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT teacher_name FROM mngteachers")) {
            while (rs.next()) list.add(rs.getString(1));
        } catch (Exception e) { e.printStackTrace(); }
        return list;
    }
}

```

```

}

public List<String> getCourses() {
    List<String> list = new ArrayList<>();
    try (Connection con = getConnection();
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT course_name FROM mngcourses")) {
        while (rs.next()) list.add(rs.getString(1));
    } catch (Exception e) { e.printStackTrace(); }
    return list;
}

public List<String> getSubjects() {
    List<String> list = new ArrayList<>();
    try (Connection con = getConnection();
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT sub_name FROM subjects")) {
        while (rs.next()) list.add(rs.getString(1));
    } catch (Exception e) { e.printStackTrace(); }
    return list;
}

/* Insert */
public void assignSubject(SubjectAssign s) {
    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(
            "INSERT INTO subject_assign(teacher_name,course_name,subject_name) VALUES(?,?,?)")) {
        ps.setString(1, s.getTeacherName());
        ps.setString(2, s.getCourseName());
        ps.setString(3, s.getSubjectName());
        ps.executeUpdate();
    } catch (Exception e) { e.printStackTrace(); }
}

/* Fetch */
public List<SubjectAssign> getAllAssignments() {
    List<SubjectAssign> list = new ArrayList<>();
    try (Connection con = getConnection();
        Statement st = con.createStatement();
        ResultSet rs = st.executeQuery("SELECT * FROM subject_assign")) {

        while (rs.next()) {
            SubjectAssign s = new SubjectAssign();
            s.setAssignId(rs.getInt("assign_id"));
            s.setTeacherName(rs.getString("teacher_name"));
            s.setCourseName(rs.getString("course_name"));
            s.setSubjectName(rs.getString("subject_name"));
            list.add(s);
        }
    }
}

```

```

    }
    } catch (Exception e) { e.printStackTrace(); }
    return list;
}

/* Delete */
public void deleteAssignment(int id) {
    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(
            "DELETE FROM subject_assign WHERE assign_id=?")) {
        ps.setInt(1, id);
        ps.executeUpdate();
    } catch (Exception e) { e.printStackTrace(); }
}
}

```

SubjectAssignmodel.java:

```

package com.sms.model;

public class SubjectAssign {

    private int assignId;
    private String teacherName;
    private String courseName;
    private String subjectName;

    public int getAssignId() {
        return assignId;
    }

    public void setAssignId(int assignId) {
        this.assignId = assignId;
    }

    public String getTeacherName() {
        return teacherName;
    }

    public void setTeacherName(String teacherName) {
        this.teacherName = teacherName;
    }

    public String getCourseName() {
        return courseName;
    }
}

```

```

}
public void setCourseName(String courseName) {
    this.courseName = courseName;
}

public String getSubjectName() {
    return subjectName;
}
public void setSubjectName(String subjectName) {
    this.subjectName = subjectName;
}
}

```

Add User

The screenshot shows a web browser window with the address bar displaying 'localhost:8080/StudentMangementProject/signup.jsp'. The page features a 'Create Account' form with the following elements:

- Header:** A blue bar with a user icon and the text 'Create Account'.
- Full Name:** A text input field with the placeholder 'Enter full name'.
- Email:** A text input field with the placeholder 'Enter email address'.
- Role:** A dropdown menu with the text 'Select Role' and a downward arrow.
- Password:** A text input field with the placeholder 'Enter password'.
- Confirm Password:** A text input field with the placeholder 'Confirm password'.
- Buttons:** A green 'Create Account' button and a blue link 'Go back to Admin Dashboard'.

Signup.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>

<%
String role = (String) session.getAttribute("role");
if (role == null || !role.equals("admin")) {
    response.sendRedirect("login.jsp");
    return;
}
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Create Account</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/@fortawesome/fontawesome-free@6.5.1/css/all.min.css" rel="stylesheet">

<!-- Google Font -->
<link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;600&display=swap" rel="stylesheet">

<style>
body{
    font-family:'Poppins',sans-serif;
    background:linear-gradient(135deg,#1d2671,#c33764);
    min-height:100vh;
    display:flex;
    align-items:center;
    justify-content:center;
}

.card{
    border-radius:16px;
    box-shadow:0 15px 30px rgba(0,0,0,0.25);
    animation:fadeIn 0.6s ease-in-out;
}

@keyframes fadeIn{
    from{opacity:0; transform:translateY(15px);}
```

```

    to{opacity:1; transform:translateY(0);}
}

.form-control, .form-select{
    border-radius:10px;
    padding:12px;
}

.btn{
    border-radius:10px;
    padding:12px;
    font-weight:600;
}

.error{
    color:#dc3545;
    font-size:14px;
    text-align:center;
}

.success{
    color:#28a745;
    font-size:14px;
    text-align:center;
}
</style>
</head>

<body>

<div class="col-md-4 col-sm-10">

<div class="card">
    <div class="card-header bg-primary text-white text-center">
        <h4><i class="fa fa-user-plus"></i> Create Account</h4>
    </div>

    <div class="card-body">

        <form action="SignupServlet" method="post" onsubmit="return validateSignup()">

            <div class="mb-3">
                <label class="form-label">Full Name</label>
                <input type="text" id="fullname" name="fullname" class="form-control"
                    placeholder="Enter full name">
            </div>

            <div class="mb-3">

```

```

        <label class="form-label">Email</label>
        <input type="email" id="email" name="email" class="form-control"
            placeholder="Enter email address">
    </div>

    <div class="mb-3">
        <label class="form-label">Role</label>
        <select id="role" name="role" class="form-select">
            <option value="">Select Role</option>
            <option value="admin">Admin</option>
            <option value="teacher">Teacher</option>
            <option value="student">Student</option>
        </select>
    </div>

    <div class="mb-3">
        <label class="form-label">Password</label>
        <input type="password" id="password" name="password" class="form-control"
            placeholder="Enter password">
    </div>

    <div class="mb-3">
        <label class="form-label">Confirm Password</label>
        <input type="password" id="confirmPassword" class="form-control"
            placeholder="Confirm password">
    </div>

    <!-- Messages -->
    <div class="error mb-2">
        <%= request.getAttribute("errorMsg") != null ? request.getAttribute("errorMsg") : "" %>
    </div>

    <div class="success mb-2">
        <%= request.getAttribute("successMsg") != null ? request.getAttribute("successMsg") : "" %>
    </div>

    <button type="submit" class="btn btn-success w-100">
        <i class="fa fa-check-circle"></i> Create Account
    </button>
</form>

<div class="text-center mt-3">
    <small>
        Go back to
        <a href="adminDashboard.jsp" class="fw-bold text-primary text-decoration-none">
            Admin Dashboard
        </a>
    </small>
</div>

```

```

        </div>

    </div>
</div>

</div>

<script>
function validateSignup() {
    const name = document.getElementById("fullname").value.trim();
    const email = document.getElementById("email").value.trim();
    const role = document.getElementById("role").value;
    const password = document.getElementById("password").value;
    const confirmPassword = document.getElementById("confirmPassword").value;

    if (name === "" || email === "" || role === "" || password === "" || confirmPassword === "") {
        alert("All fields are required!");
        return false;
    }

    const emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(email)) {
        alert("Enter a valid email address!");
        return false;
    }

    if (password.length < 6) {
        alert("Password must be at least 6 characters!");
        return false;
    }

    if (password !== confirmPassword) {
        alert("Passwords do not match!");
        return false;
    }
    return true;
}
</script>

</body>
</html>

```

SignupServlet.java:

```
package com.mangestudent.signup;
```

```

import java.io.IOException;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

@WebServlet("/SignupServlet")
public class SignupServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    private static final String DB_URL =
        "jdbc:mysql://localhost:3306/diffusers?useSSL=false&allowPublicKeyRetrieval=true";
    private static final String DB_USER = "root";
    private static final String DB_PASS = "2005";

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String fullname = request.getParameter("fullname"); // username
        String password = request.getParameter("password");
        String role = request.getParameter("role");

        String tableName = "";

        // Decide table based on role
        if ("admin".equalsIgnoreCase(role)) {
            tableName = "adminusers";
        } else if ("teacher".equalsIgnoreCase(role)) {
            tableName = "techusers";
        } else if ("student".equalsIgnoreCase(role)) {
            tableName = "stusers";
        } else {
            request.setAttribute("errorMsg", "Invalid role selected!");
            request.getRequestDispatcher("signup.jsp").forward(request, response);
            return;
        }

        Connection con = null;
        PreparedStatement ps = null;

        try {

```

```

// Load MySQL Driver
Class.forName("com.mysql.cj.jdbc.Driver");

// Create Connection
con = DriverManager.getConnection(DB_URL, DB_USER, DB_PASS);

// SQL Insert
String sql = "INSERT INTO " + tableName +
    " (username, password, role) VALUES (?, ?, ?)";

ps = con.prepareStatement(sql);
ps.setString(1, fullname);
ps.setString(2, password); // Plain text (hash later)
ps.setString(3, role);

int rows = ps.executeUpdate();

if (rows > 0) {
    request.setAttribute("successMsg", "Signup successful! Please login.");
} else {
    request.setAttribute("errorMsg", "Signup failed. Try again.");
}

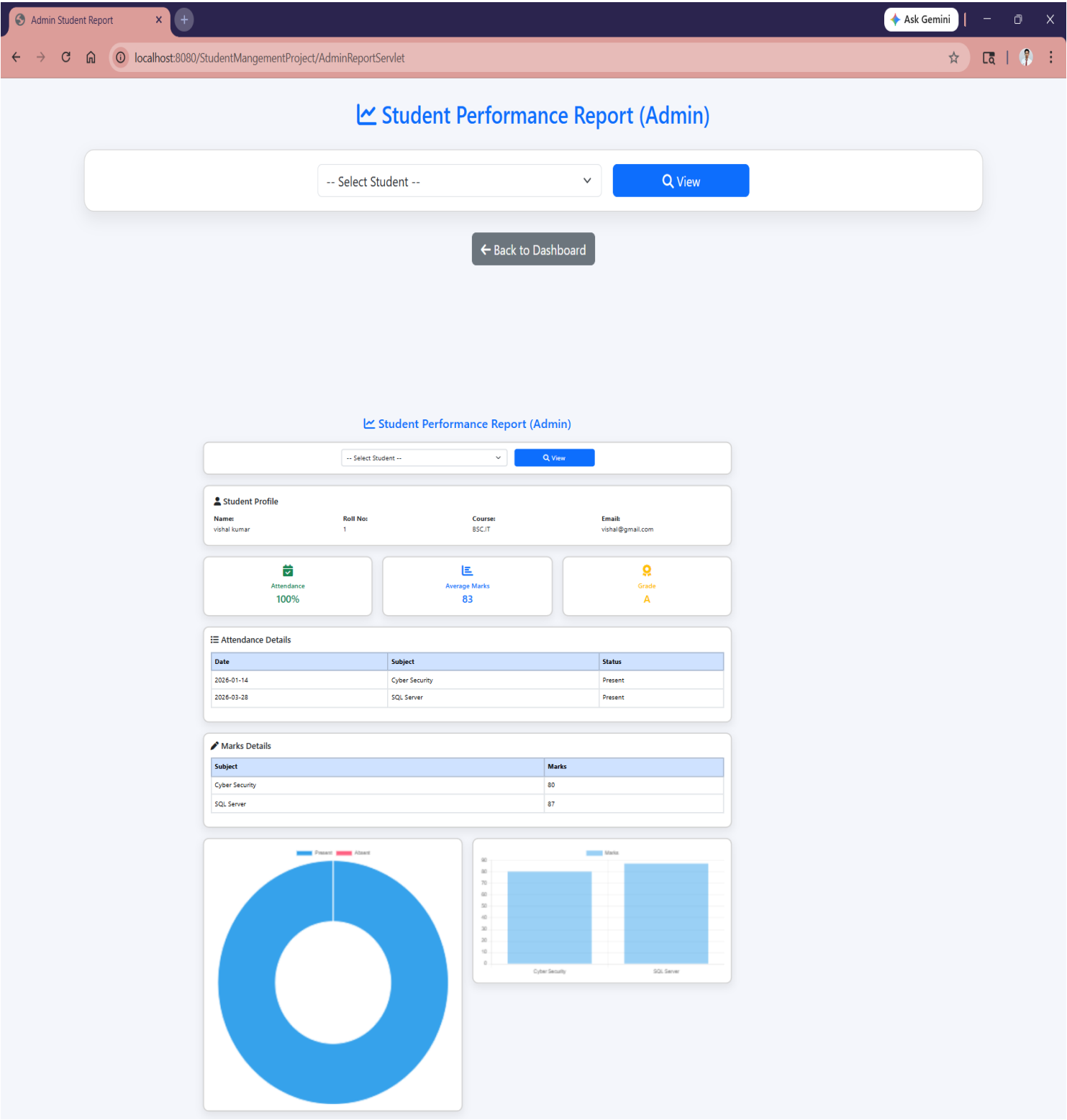
request.getRequestDispatcher("signup.jsp").forward(request, response);

} catch (Exception e) {
    e.printStackTrace();
    request.setAttribute("errorMsg", "Database Error!");
    request.getRequestDispatcher("signup.jsp").forward(request, response);

} finally {
    try {
        if (ps != null) ps.close();
        if (con != null) con.close();
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}
}

```

Student Performance Report (Admin)



AdminReport.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util. *" %>
<%@ page import="com.sms.model. *" %>

<%
AdminReportStudent student =
    (AdminReportStudent) request.getAttribute("student");

List<AdminReportAttendance> attendanceList =
    (List<AdminReportAttendance>) request.getAttribute("attendanceList");

List<AdminReportMarks> marksList =
    (List<AdminReportMarks>) request.getAttribute("marksList");

int total = 0, present = 0;
if (attendanceList != null) {
    for (AdminReportAttendance a : attendanceList) {
        total++;
        if ("Present".equalsIgnoreCase(a.getStatus())) present++;
    }
}
int attendancePercent = total == 0 ? 0 : (present * 100 / total);

int totalMarks = 0;
if (marksList != null) {
    for (AdminReportMarks m : marksList) totalMarks += m.getMarks();
}
int avg = (marksList == null || marksList.size() == 0) ? 0 : totalMarks / marksList.size();

String grade =
    avg >= 90 ? "A+" :
    avg >= 80 ? "A" :
    avg >= 70 ? "B" :
    avg >= 60 ? "C" : "Fail";
%>

<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Admin Student Report</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">
```



```

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/chart.js" rel="stylesheet">

<!-- Chart.js -->
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>

<style>
body{
    background:#f4f6fb;
    font-family:"Segoe UI",sans-serif;
}
.card{
    border-radius:14px;
    box-shadow:0 8px 20px rgba(0,0,0,0.08);
}
.stat-card{
    text-align:center;
    padding:20px;
    font-weight:600;
}
.stat-card i{
    font-size:28px;
    margin-bottom:8px;
}
</style>
</head>

<body>

<div class="container py-4">

<h3 class="text-center mb-4 text-primary">
    <i class="fa fa-chart-line"></i> Student Performance Report (Admin)
</h3>

<!-- Student Selector -->
<div class="card mb-4 p-3">
<form class="row g-3 justify-content-center" method="get" action="AdminReportServlet">
    <div class="col-md-4">
        <select name="username" class="form-select" required>
            <option value="">-- Select Student --</option>
            <%
                List<String> students = (List<String>) request.getAttribute("students");
                if (students != null) {
                    for (String s : students) {
                        <option value="<%= s %>"><%= s %></option>

```

```

        <% }} %>
    </select>
</div>
<div class="col-md-2">
    <button class="btn btn-primary w-100">
        <i class="fa fa-search"></i> View
    </button>
</div>
</form>
</div>

<% if (student != null) { %>

<!-- Profile -->
<div class="card mb-4 p-4">
<h5 class="mb-3"><i class="fa fa-user"></i> Student Profile</h5>
<div class="row">
    <div class="col-md-3"><b>Name:</b><br><%= student.getStudentname() %></div>
    <div class="col-md-3"><b>Roll No:</b><br><%= student.getRollno() %></div>
    <div class="col-md-3"><b>Course:</b><br><%= student.getCourse() %></div>
    <div class="col-md-3"><b>Email:</b><br><%= student.getEmail() %></div>
</div>
</div>

<!-- Summary Cards -->
<div class="row mb-4">
    <div class="col-md-4">
        <div class="card stat-card text-success">
            <i class="fa fa-calendar-check"></i>
            Attendance<br>
            <span class="fs-4"><%= attendancePercent %>%</span>
        </div>
    </div>
    <div class="col-md-4">
        <div class="card stat-card text-primary">
            <i class="fa fa-chart-bar"></i>
            Average Marks<br>
            <span class="fs-4"><%= avg %></span>
        </div>
    </div>
    <div class="col-md-4">
        <div class="card stat-card text-warning">
            <i class="fa fa-award"></i>
            Grade<br>
            <span class="fs-4"><%= grade %></span>
        </div>
    </div>
</div>
</div>

```

```

<!-- Attendance Table -->
<div class="card mb-4 p-3">
<h5><i class="fa fa-list"></i> Attendance Details</h5>
<table class="table table-bordered table-hover mt-2">
<tr class="table-primary">
<th>Date</th><th>Subject</th><th>Status</th>
</tr>
<% for (AdminReportAttendance a : attendanceList) { %>
<tr>
<td><%= a.getAttendanceDate() %></td>
<td><%= a.getSubName() %></td>
<td><%= a.getStatus() %></td>
</tr>
<% } %>
</table>
</div>

```

```

<!-- Marks Table -->
<div class="card mb-4 p-3">
<h5><i class="fa fa-pen"></i> Marks Details</h5>
<table class="table table-bordered table-hover mt-2">
<tr class="table-primary">
<th>Subject</th><th>Marks</th>
</tr>
<% for (AdminReportMarks m : marksList) { %>
<tr>
<td><%= m.getSubjectName() %></td>
<td><%= m.getMarks() %></td>
</tr>
<% } %>
</table>
</div>

```

```

<!-- Charts -->
<div class="row mb-4">
<div class="col-md-6">
<div class="card p-3">
<canvas id="attendanceChart"></canvas>
</div>
</div>
<div class="col-md-6">
<div class="card p-3">
<canvas id="marksChart"></canvas>
</div>
</div>
</div>

```

```

<script>
new Chart(document.getElementById("attendanceChart"),{
  type:'doughnut',
  data:{
    labels:['Present','Absent'],
    datasets:[{ data:[<%= present %>,<%= total-present %>] }]
  }
});

```

```

new Chart(document.getElementById("marksChart"),{
  type:'bar',
  data:{
    labels:[
      <% for(AdminReportMarks m:marksList){ %>
      "<%= m.getSubjectName() %>",
      <% } %>
    ],
    datasets:[{
      label:'Marks',
      data:[
        <% for(AdminReportMarks m:marksList){ %>
        <%= m.getMarks() %>,
        <% } %>
      ]
    }]
  },
  options:{ scales:{ y:{ beginAtZero:true } } }
});

```

```

</script>

```

```

<% } %>

```

```

<div class="text-center mt-3">
<button class="btn btn-secondary"
onclick="location.href='adminDashboard.jsp'">
<i class="fa fa-arrow-left"></i> Back to Dashboard
</button>
</div>

```

```

</div>
</body>
</html>

```

AdminReportServlet.java:

```
package com.sms.controller;

import java.io.IOException;
import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

import com.sms.dao.AdminReportDAO;

@WebServlet("/AdminReportServlet")
public class AdminReportServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        String username = request.getParameter("username");
        AdminReportDAO dao = new AdminReportDAO();

        // First load → only dropdown
        if (username == null || username.trim().isEmpty()) {
            request.setAttribute("students", dao.getAllStudentNames());
            request.getRequestDispatcher("adminReport.jsp")
                .forward(request, response);
            return;
        }

        // Report load
        request.setAttribute("students", dao.getAllStudentNames());
        request.setAttribute("student", dao.getStudent(username));
        request.setAttribute("attendanceList", dao.getAttendance(username));
        request.setAttribute("marksList", dao.getMarks(username));

        request.getRequestDispatcher("adminReport.jsp")
            .forward(request, response);
    }
}
```

AdminReportDao.java:

```
package com.sms.dao;

import java.sql.*;
import java.util.*;
import com.sms.model.*;

public class AdminReportDAO {

    Connection con;

    public AdminReportDAO() {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/admindb",
                "root",
                "2005"
            );
        } catch (Exception e) {
            e.printStackTrace();
        }
    }

    // ♦ Student List (for dropdown)
    public List<String> getAllStudentNames() {
        List<String> list = new ArrayList<>();
        try {
            PreparedStatement ps =
                con.prepareStatement("select studentname from mngstudent");
            ResultSet rs = ps.executeQuery();
            while (rs.next()) {
                list.add(rs.getString("studentname"));
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
        return list;
    }

    // ♦ Student Profile
    public AdminReportStudent getStudent(String username) {
        AdminReportStudent s = null;
```

```

try {
    PreparedStatement ps =
        con.prepareStatement(
            "select studentname, rollno, course, email from mngstudent where studentname=?"
        );
    ps.setString(1, username);
    ResultSet rs = ps.executeQuery();
    if (rs.next()) {
        s = new AdminReportStudent();
        s.setStudentname(rs.getString("studentname"));
        s.setRollno(rs.getString("rollno"));
        s.setCourse(rs.getString("course"));
        s.setEmail(rs.getString("email"));
    }
} catch (Exception e) {
    e.printStackTrace();
}
return s;
}

```

// ♦ Attendance

```

public List<AdminReportAttendance> getAttendance(String username) {
    List<AdminReportAttendance> list = new ArrayList<>();
    try {
        PreparedStatement ps =
            con.prepareStatement(
                "select attendance_date, sub_name, status from attendance where studentname=?"
            );
        ps.setString(1, username);
        ResultSet rs = ps.executeQuery();
        while (rs.next()) {
            AdminReportAttendance a = new AdminReportAttendance();
            a.setAttendanceDate(rs.getString("attendance_date"));
            a.setSubName(rs.getString("sub_name"));
            a.setStatus(rs.getString("status"));
            list.add(a);
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}

```

// ♦ Marks

```

public List<AdminReportMarks> getMarks(String username) {
    List<AdminReportMarks> list = new ArrayList<>();
    try {
        PreparedStatement ps =

```

```

        con.prepareStatement(
            "select subject_name, marks from marks where student_name=?"
        );
        ps.setString(1, username);
        ResultSet rs = ps.executeQuery();
        while (rs.next()) {
            AdminReportMarks m = new AdminReportMarks();
            m.setSubjectName(rs.getString("subject_name"));
            m.setMarks(rs.getInt("marks"));
            list.add(m);
        }
    } catch (Exception e) {
        e.printStackTrace();
    }
    return list;
}
}

```

AdminReportAttendanceModel.java:

```

package com.sms.model;

public class AdminReportAttendance {
    private String attendanceDate;
    private String subName;
    private String status;

    public String getAttendanceDate() {
        return attendanceDate;
    }

    public void setAttendanceDate(String attendanceDate) {
        this.attendanceDate = attendanceDate;
    }

    public String getSubName() {
        return subName;
    }

    public void setSubName(String subName) {
        this.subName = subName;
    }

    public String getStatus() {

```



```

        return status;
    }
    public void setStatus(String status) {
        this.status = status;
    }
}

```

AdminReportmarksmodel.java:

```

package com.sms.model;

public class AdminReportMarks {
    private String subjectName;
    private int marks;

    public String getSubjectName() {
        return subjectName;
    }
    public void setSubjectName(String subjectName) {
        this.subjectName = subjectName;
    }

    public int getMarks() {
        return marks;
    }
    public void setMarks(int marks) {
        this.marks = marks;
    }
}

```

AdminReportStudentmodel.java:

```

package com.sms.model;

public class AdminReportStudent {
    private String studentname;
    private String rollno;
    private String course;
}

```

```
private String email;

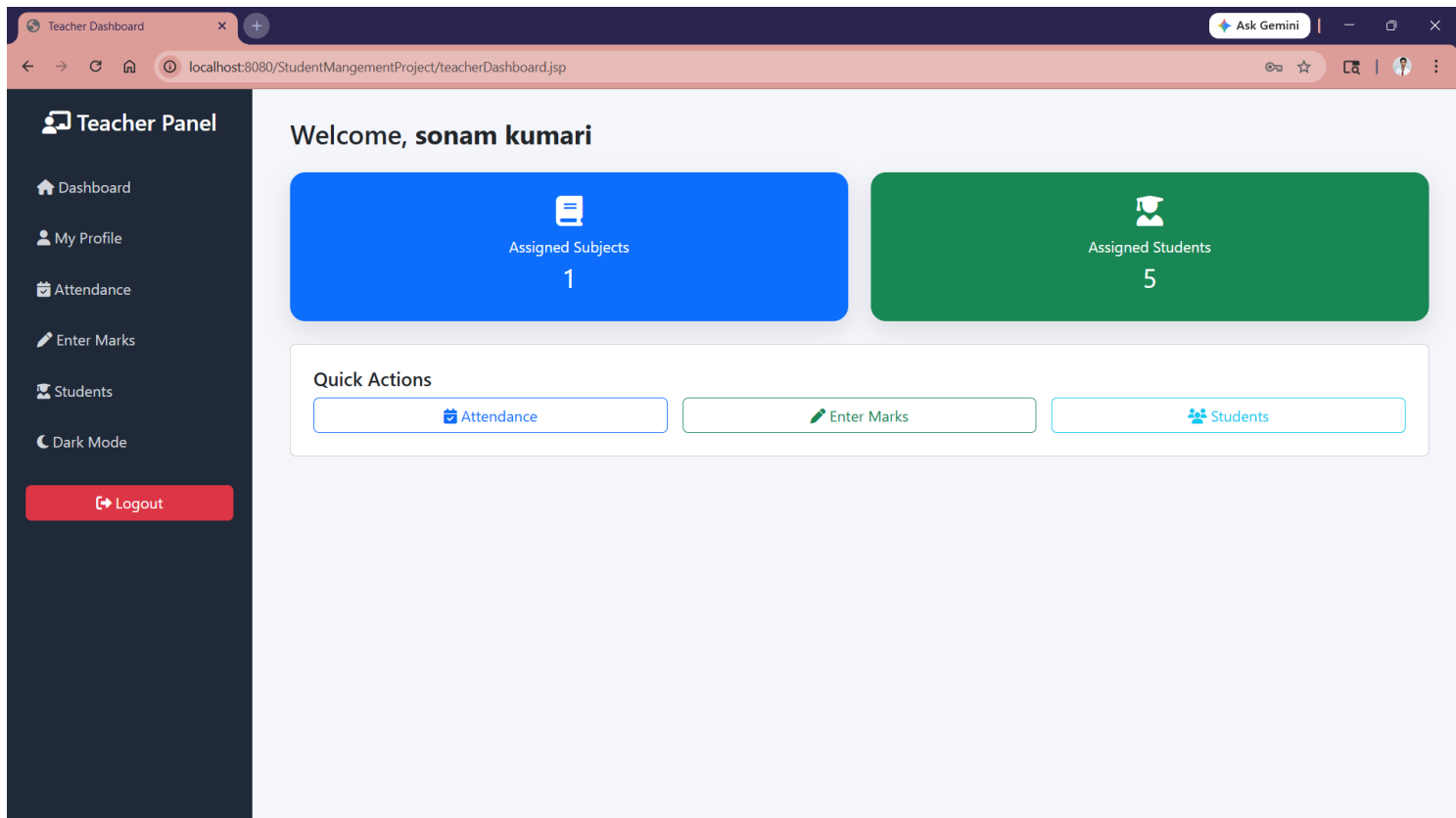
public String getStudentname() {
    return studentname;
}
public void setStudentname(String studentname) {
    this.studentname = studentname;
}

public String getRollno() {
    return rollno;
}
public void setRollno(String rollno) {
    this.rollno = rollno;
}

public String getCourse() {
    return course;
}
public void setCourse(String course) {
    this.course = course;
}

public String getEmail() {
    return email;
}
public void setEmail(String email) {
    this.email = email;
}
}
```

Teacher Dashboard



Teacher Dashboard.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
```

```
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
    return;
}
```

```
String username = (String) session.getAttribute("username");
String role = (String) session.getAttribute("role");
```

```
if (!"teacher".equals(role)) {
    response.sendRedirect("login.jsp");
```

```

    return;
}

/* ===== REAL DB STATS ===== */
int totalSubjects = 0;
int totalStudents = 0;

try{
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/admindb","root","2005");

    // Count assigned subjects
    PreparedStatement ps1 = con.prepareStatement(
        "SELECT COUNT(*) FROM subject_assign WHERE teacher_name=?");
    ps1.setString(1, username);
    ResultSet rs1 = ps1.executeQuery();
    if(rs1.next()) totalSubjects = rs1.getInt(1);

    // Count students from assigned courses
    PreparedStatement ps2 = con.prepareStatement(
        "SELECT COUNT(*) FROM mngstudent WHERE course IN " +
        "(SELECT DISTINCT course_name FROM subject_assign WHERE teacher_name=?)");
    ps2.setString(1, username);
    ResultSet rs2 = ps2.executeQuery();
    if(rs2.next()) totalStudents = rs2.getInt(1);

    con.close();
}catch(Exception e){
    e.printStackTrace();
}
%>

<!DOCTYPE html>
<html>
<head>
<title>Teacher Dashboard</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.1/css/all.min.css" rel="stylesheet">

<style>
body{
    font-family:"Segoe UI",sans-serif;
    background:#f4f6fb;

```

```

}

/* Sidebar */
.sidebar{
  width:260px;
  height:100vh;
  position:fixed;
  background:#1f2937;
  color:white;
  padding:20px;
}
.sidebar h4{
  text-align:center;
  margin-bottom:30px;
}
.sidebar a{
  display:block;
  color:#d1d5db;
  padding:12px;
  border-radius:8px;
  text-decoration:none;
  margin-bottom:6px;
}
.sidebar a:hover{
  background:#374151;
  color:white;
}

/* Main */
.main{
  margin-left:270px;
  padding:30px;
}

/* Cards */
.stat-card{
  border-radius:16px;
  padding:25px;
  text-align:center;
  box-shadow:0 10px 25px rgba(0,0,0,0.1);
}
.stat-card i{
  font-size:32px;
  margin-bottom:10px;
}

/* Dark Mode */
.dark{

```

```

    background:#020617;
    color:white;
}
.dark .stat-card,
.dark .card{
    background:#1f2937;
    color:white;
}
.dark .sidebar{
    background:#020617;
}
</style>
</head>

<body id="body">

<!-- SIDEBAR -->
<div class="sidebar">
    <h4><i class="fa fa-chalkboard-teacher"></i> Teacher Panel</h4>

    <a href="teacherDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
    <a href="teacherProfile.jsp"><i class="fa fa-user"></i> My Profile</a>
    <a href="markAttendance.jsp"><i class="fa fa-calendar-check"></i> Attendance</a>
    <a href="enterMarks.jsp"><i class="fa fa-pen"></i> Enter Marks</a>
    <a href="ViewStudents.jsp"><i class="fa fa-user-graduate"></i> Students</a>

    <a href="#" onclick="toggleDark()">
        <i class="fa fa-moon"></i> Dark Mode
    </a>

    <form action="LogoutServlet" method="post">
        <button class="btn btn-danger w-100 mt-3">
            <i class="fa fa-sign-out-alt"></i> Logout
        </button>
    </form>
</div>

<!-- MAIN CONTENT -->
<div class="main">

<h3 class="mb-4">
    Welcome, <b><%= username %></b>
</h3>

<!-- STATS -->
<div class="row g-4 mb-4">

    <div class="col-md-6">

```

```

    <div class="stat-card bg-primary text-white">
      <i class="fa fa-book"></i><br>
      Assigned Subjects<br>
      <span class="fs-3"><%= totalSubjects %></span>
    </div>
  </div>

  <div class="col-md-6">
    <div class="stat-card bg-success text-white">
      <i class="fa fa-user-graduate"></i><br>
      Assigned Students<br>
      <span class="fs-3"><%= totalStudents %></span>
    </div>
  </div>

</div>

<!-- QUICK ACTIONS -->
<div class="card p-4">
<h5>Quick Actions</h5>

<div class="row g-3">
  <div class="col-md-4">
    <a href="markAttendance.jsp" class="btn btn-outline-primary w-100">
      <i class="fa fa-calendar-check"></i> Attendance
    </a>
  </div>
  <div class="col-md-4">
    <a href="enterMarks.jsp" class="btn btn-outline-success w-100">
      <i class="fa fa-pen"></i> Enter Marks
    </a>
  </div>
  <div class="col-md-4">
    <a href="ViewStudents.jsp" class="btn btn-outline-info w-100">
      <i class="fa fa-users"></i> Students
    </a>
  </div>
</div>

</div>

</div>

<script>
function toggleDark(){
  document.getElementById("body").classList.toggle("dark");
}
</script>

```

```
</body>
</html>
```

Teacher Profile

Teacher Profile

Ask Gemini

localhost:8080/StudentMangementProject/teacherProfile.jsp

Teacher Panel

Dashboard

My Profile

Attendance

Enter Marks

Students

Dark Mode

Logout

Teacher Profile



sonam kumari

Role: Teacher

Assigned subjects and courses are shown below

Assigned Subjects & Courses

#	Course	Subject
1	BSC.IT	PHP

TeacherProfile.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>

<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
    return;
}

String username = (String) session.getAttribute("username");
String role = (String) session.getAttribute("role");

if("teacher".equals(role)){
    response.sendRedirect("login.jsp");
    return;
}
%>

<!DOCTYPE html>
<html>
<head>
<title>Teacher Profile</title>

<!-- Bootstrap -->
<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet">

<!-- Font Awesome -->
<link href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.1/css/all.min.css" rel="stylesheet">

<style>
body{
    font-family:"Segoe UI",sans-serif;
    background:#f4f6fb;
}
.sidebar{
    width:260px;
    height:100vh;
    position:fixed;
    background:#1f2937;
    color:white;
    padding:20px;
}
```

```

.sidebar h4{
    text-align:center;
    margin-bottom:30px;
}
.sidebar a{
    display:block;
    color:#d1d5db;
    padding:12px;
    border-radius:8px;
    text-decoration:none;
    margin-bottom:6px;
}
.sidebar a:hover{
    background:#374151;
    color:white;
}
.main{
    margin-left:270px;
    padding:30px;
}
.profile-card{
    border-radius:16px;
    box-shadow:0 12px 30px rgba(0,0,0,0.15);
}
.dark{
    background:#020617;
    color:white;
}
.dark .card,
.dark table{
    background:#1f2937;
    color:white;
}
.dark .sidebar{
    background:#020617;
}
</style>
</head>

<body id="body">

<!-- SIDEBAR -->
<div class="sidebar">
    <h4><i class="fa fa-chalkboard-teacher"></i> Teacher Panel</h4>

    <a href="teacherDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
    <a href="teacherProfile.jsp"><i class="fa fa-user"></i> My Profile</a>
    <a href="markAttendance.jsp"><i class="fa fa-calendar-check"></i> Attendance</a>

```

```

<a href="enterMarks.jsp"><i class="fa fa-pen"></i> Enter Marks</a>
<a href="ViewStudents.jsp"><i class="fa fa-user-graduate"></i> Students</a>

<a href="#" onclick="toggleDark()">
  <i class="fa fa-moon"></i> Dark Mode
</a>

<form action="LogoutServlet" method="post">
  <button class="btn btn-danger w-100 mt-3">
    <i class="fa fa-sign-out-alt"></i> Logout
  </button>
</form>
</div>

<!-- MAIN -->
<div class="main">

<h3 class="mb-4"><i class="fa fa-user"></i> Teacher Profile</h3>

<!-- PROFILE INFO -->
<div class="card profile-card p-4 mb-4">
  <div class="row align-items-center">
    <div class="col-md-3 text-center">
      <i class="fa fa-user-circle fa-6x text-primary"></i>
    </div>
    <div class="col-md-9">
      <h4><%= username %></h4>
      <p class="mb-1"><b>Role:</b> Teacher</p>
      <p class="mb-0 text-muted">
        Assigned subjects and courses are shown below
      </p>
    </div>
  </div>
</div>

<!-- ASSIGNED SUBJECTS -->
<div class="card p-4">
<h5 class="mb-3"><i class="fa fa-book"></i> Assigned Subjects & Courses</h5>

<table class="table table-bordered text-center">
<thead class="table-primary">
<tr>
  <th>#</th>
  <th>Course</th>
  <th>Subject</th>
</tr>
</thead>
<tbody>

```

```

<%
int count = 1;
try{
    Class.forName("com.mysql.cj.jdbc.Driver");
    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/admindb","root","2005");

    PreparedStatement ps = con.prepareStatement(
        "SELECT course_name, subject_name FROM subject_assign WHERE teacher_name=?");
    ps.setString(1, username);

    ResultSet rs = ps.executeQuery();

    boolean found = false;
    while(rs.next()){
        found = true;
    }
%>
<tr>
    <td><%= count++ %></td>
    <td><%= rs.getString("course_name") %></td>
    <td><%= rs.getString("subject_name") %></td>
</tr>
<%
    }

    if(!found){
%>
<tr>
    <td colspan="3">No subjects assigned</td>
</tr>
<%
    }

    con.close();
catch(Exception e){
    e.printStackTrace();
}
%>

</tbody>
</table>
</div>

</div>

<script>
function toggleDark(){

```

```

    document.getElementById("body").classList.toggle("dark");
}
</script>

</body>
</html>

```

Mark Attendance

The screenshot shows a web browser window with the URL `localhost:8080/StudentMangementProject/markAttendance.jsp`. The page has a dark blue sidebar on the left with the following menu items: Teacher, Dashboard, Attendance (highlighted), Enter Marks, Students, and Logout. The main content area is titled "Mark Attendance" and contains a form with two input fields: "Date" (with a date picker icon) and "Subject" (with a dropdown menu). Below the form is a table with three columns: Roll No, Student Name, and Status. The table contains five rows of data, all with a "Present" status. At the bottom of the form, there are two buttons: "Save Attendance" (green) and "Back" (grey).

Roll No	Student Name	Status
1	vishal kumar	Present
2	krrish soni	Present
3	vivek kumar	Present
4	Nitish kumar	Present
5	div kumari	Present

MarkAttendance.jsp:

```

<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>

<%@ page import="java.util.*" %>
<%@ page import="com.sms.dao.StudentDAO" %>
<%@ page import="com.sms.dao.AttendanceDAO" %>
<%@ page import="com.sms.model.Student" %>

<%
    StudentDAO studentDao = new StudentDAO();
    List<Student> students = studentDao.getAllStudents();

    AttendanceDAO attendanceDao = new AttendanceDAO();
    List<String> subjects = attendanceDao.getAllSubjects();
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Mark Attendance</title>

<link rel="stylesheet"
href="https://cdn.jsdelivr.net/npm/font-awesome@6.5.0/css/all.min.css"/>

<style>
*{ box-sizing:border-box; }
body{
    margin:0;
    font-family:"Segoe UI",Tahoma,sans-serif;
    background:#f4f6ff;
}

/* Layout */
.wrapper{
    display: flex;
    min-height: 100vh;
}

/* Sidebar */
.sidebar{
    width: 240px;
    background: #4338ca;
    color: white;

```

```

    padding:20px;
}
.sidebar h2{
    text-align:center;
    margin-bottom:30px;
}
.sidebar a{
    display:block;
    padding:12px 15px;
    color:white;
    text-decoration:none;
    border-radius:8px;
    margin-bottom:8px;
    transition:0.3s;
}
.sidebar a:hover,
.sidebar a.active{
    background:#3730a3;
}

/* Main */
.main{
    flex:1;
    padding:30px;
}

/* Card */
.card{
    background:white;
    padding:25px;
    border-radius:14px;
    box-shadow:0 10px 25px rgba(0,0,0,0.15);
}

/* Header */
.header{
    display:flex;
    justify-content:space-between;
    align-items:center;
    margin-bottom:20px;
}
.header h2{
    color:#4338ca;
}

/* Filters */
.filters{
    display:flex;

```

```

    gap:20px;
    flex-wrap:wrap;
    margin-bottom:20px;
}
.filters label{
    font-weight:600;
}
.filters input,
.filters select{
    padding:8px 12px;
    border-radius:8px;
    border:1px solid #ccc;
}

/* Table */
table{
    width:100%;
    border-collapse:collapse;
}
th,td{
    padding:12px;
    border-bottom:1px solid #ddd;
    text-align:center;
}
th{
    background:#4f46e5;
    color:white;
}
tr:hover{
    background:#eef1ff;
}

/* Status */
.status{
    padding:6px 10px;
    border-radius:6px;
}

/* Buttons */
.actions{
    margin-top:20px;
}
.btn{
    padding:10px 18px;
    border:none;
    border-radius:8px;
    cursor:pointer;
    font-size:15px;

```



```

}
.save-btn{
    background:#22c55e;
    color:white;
}
.back-btn{
    background:#6b7280;
    color:white;
    margin-left:10px;
}

/* Success */
.success{
    margin-top:15px;
    color:#16a34a;
    font-weight:600;
}
</style>
</head>

<body>

<div class="wrapper">

    <!-- Sidebar -->
    <div class="sidebar">
        <h2>👤 Teacher</h2>
        <a href="teacherDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
        <a href="markAttendance.jsp" class="active">
            <i class="fa fa-calendar-check"></i> Attendance
        </a>
        <a href="enterMarks.jsp"><i class="fa fa-pen"></i> Enter Marks</a>
        <a href="ViewStudents.jsp"><i class="fa fa-users"></i> Students</a>
        <a href="LogoutServlet"><i class="fa fa-sign-out-alt"></i> Logout</a>
    </div>

    <!-- Main -->
    <div class="main">
        <div class="card">

            <div class="header">
                <h2><i class="fa fa-calendar-check"></i> Mark Attendance</h2>
            </div>

            <form action="<%=request.getContextPath()%>/AttendanceServlet" method="post">

                <!-- Filters -->
                <div class="filters">

```

```

<div>
  <label>Date</label><br>
  <input type="date" name="date" required>
</div>

<div>
  <label>Subject</label><br>
  <select name="sub_name" required>
    <option value="">Select Subject</option>
    <% for(String sub : subjects){ %>
      <option value="<%= sub %>"><%= sub %></option>
    <% } %>
  </select>
</div>
</div>

<!-- Table -->
<table>
  <tr>
    <th>Roll No</th>
    <th>Student Name</th>
    <th>Status</th>
  </tr>

  <% for(Student s : students){ %>
  <tr>
    <td><%= s.getRollNo() %></td>
    <td><%= s.getStudentName() %></td>
    <td>
      <input type="hidden" name="rollNo[]" value="<%= s.getRollNo() %>">
      <input type="hidden" name="studentName[]" value="<%= s.getStudentName() %>">

      <select name="status[]" class="status">
        <option value="Present">Present</option>
        <option value="Absent">Absent</option>
        <option value="NA">NA</option>
      </select>
    </td>
  </tr>
  <% } %>
</table>

<!-- Buttons -->
<div class="actions">
  <button class="btn save-btn">
    <i class="fa fa-save"></i> Save Attendance
  </button>

```

```

        <button type="button" class="btn back-btn"
            onclick="location.href='teacherDashboard.jsp'">
            <i class="fa fa-arrow-left"></i> Back
        </button>
    </div>
</form>

<% if(request.getParameter("success")!=null){ %>
<div class="success">
    <i class="fa fa-check-circle"></i>
    Attendance saved successfully!
</div>
<% } %>

</div>
</div>

</div>

</body>
</html>

```

AttendanceServlet.java:

```

package com.sms.controller;

import java.io.IOException;
import java.sql.Date;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

import com.sms.dao.AttendanceDAO;
import com.sms.model.Attendance;

@WebServlet("/AttendanceServlet")
public class AttendanceServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest req, HttpServletResponse resp)
        throws ServletException, IOException {

```

```

String dateStr = req.getParameter("date");
String subName = req.getParameter("sub_name");

String[] rollNos = req.getParameterValues("rollNo[]");
String[] studentNames = req.getParameterValues("studentName[]");
String[] statuses = req.getParameterValues("status[]");

Date attendanceDate = Date.valueOf(dateStr);

AttendanceDAO dao = new AttendanceDAO();

for (int i = 0; i < rollNos.length; i++) {

    Attendance a = new Attendance();
    a.setRollNo(rollNos[i]);
    a.setStudentName(studentNames[i]);
    a.setStatus(statuses[i]);
    a.setAttendanceDate(attendanceDate);
    a.setSubName(subName);

    dao.saveAttendance(a);
}

resp.sendRedirect(req.getContextPath()
    + "/markAttendance.jsp?success=1");
}
}

```

AttendanceDao.java:

```

package com.sms.dao;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

import com.sms.model.Attendance;

public class AttendanceDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(

```

```

        "jdbc:mysql://localhost:3306/admindb",
        "root",
        "2005"
    );
}

// ♦ Get subjects for dropdown
public List<String> getAllSubjects() {

    List<String> subjects = new ArrayList<>();
    String sql = "SELECT sub_name FROM subjects";

    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(sql);
        ResultSet rs = ps.executeQuery()) {

        while (rs.next()) {
            subjects.add(rs.getString("sub_name"));
        }

    } catch (Exception e) {
        e.printStackTrace();
    }

    return subjects;
}

// ♦ Save attendance with subject & student name
public void saveAttendance(Attendance a) {

    String sql =
        "INSERT INTO attendance " +
        "(roll_no, attendance_date, status, studentname, sub_name) " +
        "VALUES (?, ?, ?, ?, ?)";

    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setString(1, a.getRollNo());
        ps.setDate(2, a.getAttendanceDate());
        ps.setString(3, a.getStatus());
        ps.setString(4, a.getStudentName());
        ps.setString(5, a.getSubName());

        ps.executeUpdate();

    } catch (Exception e) {
        e.printStackTrace();
    }
}

```

```
}  
}  
}
```

AttendanceModel.java:

```
package com.sms.model;  
  
import java.sql.Date;  
  
public class Attendance {  
  
    private String rollNo;  
    private String studentName;  
    private String subName;  
    private Date attendanceDate;  
    private String status;  
  
    public String getRollNo() {  
        return rollNo;  
    }  
    public void setRollNo(String rollNo) {  
        this.rollNo = rollNo;  
    }  
  
    public String getStudentName() {  
        return studentName;  
    }  
    public void setStudentName(String studentName) {  
        this.studentName = studentName;  
    }  
  
    public String getSubName() {  
        return subName;  
    }  
    public void setSubName(String subName) {  
        this.subName = subName;  
    }  
  
    public Date getAttendanceDate() {  
        return attendanceDate;  
    }  
    public void setAttendanceDate(Date attendanceDate) {  
        this.attendanceDate = attendanceDate;  
    }  
}
```

```

public String getStatus() {
    return status;
}
public void setStatus(String status) {
    this.status = status;
}
}

```

View listed students

Roll No	Name	Email	Contact	Course
1	vishal kumar	vishal@gmail.com	7870315835	BSC.IT
2	krrish soni	krrish@gmail.com	7079786840	BSC.IT
3	vivek kumar	Vivek@gmail.com	7077786849	BSC.IT
4	Nitish kumar	nitish@gmail.com	7079786878	BSC.IT
5	div kumari	div@gmail.com	7079786876	BSC.IT

View Students.jsp:

```

<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>

```

```

}
%>

<%@ page import="java.util.*" %>
<%@ page import="com.sms.dao.StudentDAO" %>
<%@ page import="com.sms.model.Student" %>

<%
    StudentDAO dao = new StudentDAO();
    List<Student> students = dao.getAllStudents();
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>View Students</title>

<link rel="stylesheet"
href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>

<style>
*{ box-sizing:border-box; }
body{
    margin:0;
    font-family:"Segoe UI",Tahoma,sans-serif;
    background:#f4f6ff;
}

/* Layout */
.wrapper{
    display:flex;
    min-height:100vh;
}

/* Sidebar */
.sidebar{
    width:240px;
    background:#4338ca;
    color:white;
    padding:20px;
}
.sidebar h2{
    text-align:center;
    margin-bottom:30px;
}
.sidebar a{
    display:block;

```



```

padding:12px 15px;
color:white;
text-decoration:none;
border-radius:8px;
margin-bottom:8px;
transition:0.3s;
}
.sidebar a:hover,
.sidebar a.active{
background:#3730a3;
}

/* Main content */
.main{
flex:1;
padding:30px;
}

/* Card */
.card{
background:white;
padding:25px;
border-radius:14px;
box-shadow:0 10px 25px rgba(0,0,0,0.15);
}

/* Header */
.header{
display:flex;
justify-content:space-between;
align-items:center;
margin-bottom:20px;
}
.header h2{
color:#4338ca;
}

/* Buttons */
.btn{
padding:10px 16px;
border:none;
border-radius:8px;
cursor:pointer;
font-size:14px;
}
.back-btn{
background:#6b7280;
color:white;

```

```

    margin-bottom:15px;
}

/* Table */
table{
    width:100%;
    border-collapse:collapse;
    margin-top:10px;
}
th,td{
    padding:12px;
    border-bottom:1px solid #ddd;
    text-align:center;
}
th{
    background:#4f46e5;
    color:white;
}
tr:hover{
    background:#eefiff;
}
</style>
</head>

<body>

<div class="wrapper">

    <!-- Sidebar -->
    <div class="sidebar">
        <h2>👤 Teacher</h2>
        <a href="teacherDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
        <a href="markAttendance.jsp"><i class="fa fa-calendar-check"></i> Attendance</a>
        <a href="enterMarks.jsp"><i class="fa fa-pen"></i> Enter Marks</a>
        <a href="ViewStudents.jsp" class="active"><i class="fa fa-users"></i> Students</a>
        <a href="LogoutServlet"><i class="fa fa-sign-out-alt"></i> Logout</a>
    </div>

    <!-- Main -->
    <div class="main">
        <div class="card">

            <div class="header">
                <h2><i class="fa fa-users"></i> View Students</h2>
                <button class="btn back-btn" onclick="goBack()">
                    <i class="fa fa-arrow-left"></i> Back
                </button>
            </div>

```

```

<table>
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th>Email</th>
    <th>Contact</th>
    <th>Course</th>
  </tr>

  <%
  if(students == null || students.isEmpty()){
  %>
  <tr>
    <td colspan="5">No students found</td>
  </tr>
  <%
  } else {
    for(Student s : students){
  %>
  <tr>
    <td><%= s.getRollNo() %></td>
    <td><%= s.getStudentName() %></td>
    <td><%= s.getEmail() %></td>
    <td><%= s.getContact() %></td>
    <td><%= s.getCourse() %></td>
  </tr>
  <%
  }
  }
  %>
</table>

```

```

</div>
</div>

```

```

</div>

```

```

<script>
function goBack(){
  window.location.href = "teacherDashboard.jsp";
}
</script>

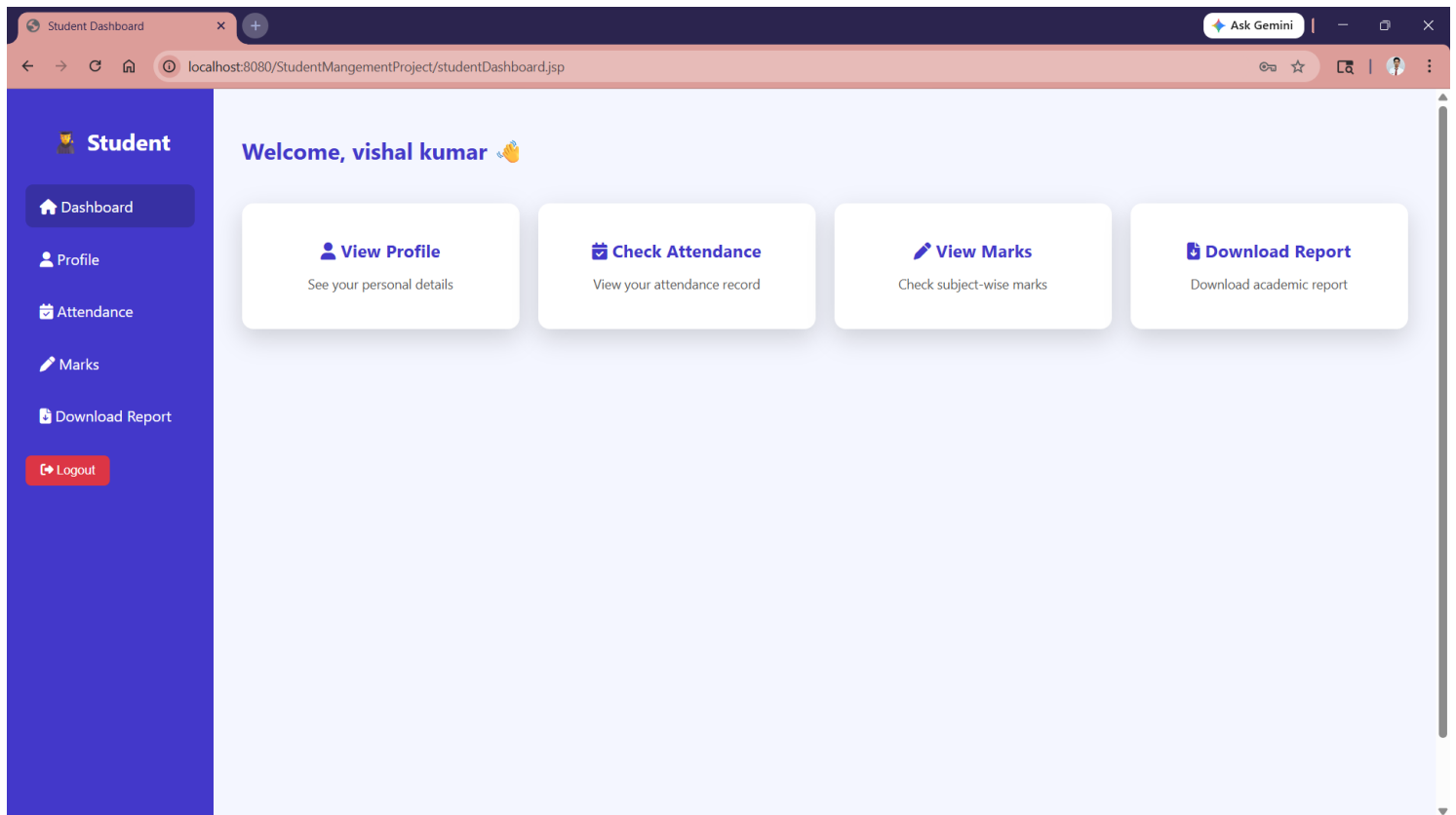
```

```

</body>
</html>

```

Student Dashboard



Student Dashboard.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>

<%
String username = (String) session.getAttribute("username");
```

```

String role = (String) session.getAttribute("role");

// Allow only logged-in students
if (username == null || role == null || !role.equals("student")) {
    response.sendRedirect("login.jsp");
    return;
}
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Student Dashboard</title>

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>

<style>
*{ box-sizing:border-box; }
body{
    margin:0;
    font-family:"Segoe UI",Tahoma,sans-serif;
    background:#f4f6ff;
    min-height:100vh;
}

/* Layout */
.wrapper{
    display:flex;
    min-height:100vh;
}

/* Sidebar */
.sidebar{
    width:220px;
    background:#4338ca;
    color:white;
    padding:20px;
}
.sidebar h2{
    text-align:center;
    margin-bottom:30px;
}
.sidebar a{
    display:block;
    padding:12px 15px;
    color:white;
    text-decoration:none;

```

```

    border-radius:8px;
    margin-bottom:10px;
    transition:0.3s;
}
.sidebar a:hover,
.sidebar a.active{
    background:#3730a3;
}

/* Main content */
.main{
    flex:1;
    padding:30px;
}

/* Header */
.header{
    display:flex;
    justify-content:space-between;
    align-items:center;
    margin-bottom:20px;
}
.header h2{
    color:#4338ca;
}

/* Cards */
.card-grid{
    display:grid;
    grid-template-columns:repeat(auto-fit,minmax(220px,1fr));
    gap:20px;
}
.card{
    background:white;
    padding:25px;
    border-radius:12px;
    box-shadow:0 10px 25px rgba(0,0,0,0.15);
    text-align:center;
    cursor:pointer;
    transition:transform 0.3s, box-shadow 0.3s;
}
.card:hover{
    transform:translateY(-5px);
    box-shadow:0 15px 30px rgba(0,0,0,0.3);
}
.card h3{
    margin:12px 0;
    color:#4338ca;
}

```

```

}
.card p{
    color:#555;
    font-size:14px;
}

/* Logout Button */
.logout-btn{
    background:#dc3545;
    color:white;
    border:none;
    padding:8px 16px;
    border-radius:6px;
    cursor:pointer;
}
.logout-btn:hover{
    background:#b02a37;
}

/* Footer */
footer{
    text-align:center;
    padding:15px;
    color:#555;
    font-size:14px;
    margin-top:30px;
}
</style>
</head>

<body>

<div class="wrapper">

<!-- Sidebar -->
<div class="sidebar">
    <h2> 🎓 Student</h2>
    <a href="studentDashboard.jsp" class="active"><i class="fa fa-home"></i> Dashboard</a>
    <a href="ProfileStudent"><i class="fa fa-user"></i> Profile</a>
    <a href="ProfileAttendance"><i class="fa fa-calendar-check"></i> Attendance</a>
    <a href="ProfileMarks"><i class="fa fa-pen"></i> Marks</a>
    <a href="ReportStudent"><i class="fa fa-file-download"></i> Download Report</a>
    <form action="LogoutServlet" method="post" style="margin-top:20px;">
        <button type="submit" class="logout-btn"><i class="fa fa-sign-out-alt"></i> Logout</button>
    </form>
</div>

<!-- Main -->

```

```

<div class="main">
  <div class="header">
    <h2>Welcome, <%= username %> 🙋 </h2>
  </div>

  <div class="card-grid">
    <div class="card" onclick="location.href='ProfileStudent'">
      <h3><i class="fa fa-user"></i> View Profile</h3>
      <p>See your personal details</p>
    </div>

    <div class="card" onclick="location.href='ProfileAttendance'">
      <h3><i class="fa fa-calendar-check"></i> Check Attendance</h3>
      <p>View your attendance record</p>
    </div>

    <div class="card" onclick="location.href='ProfileMarks'">
      <h3><i class="fa fa-pen"></i> View Marks</h3>
      <p>Check subject-wise marks</p>
    </div>

    <div class="card" onclick="location.href='ReportStudent'">
      <h3><i class="fa fa-file-download"></i> Download Report</h3>
      <p>Download academic report</p>
    </div>
  </div>

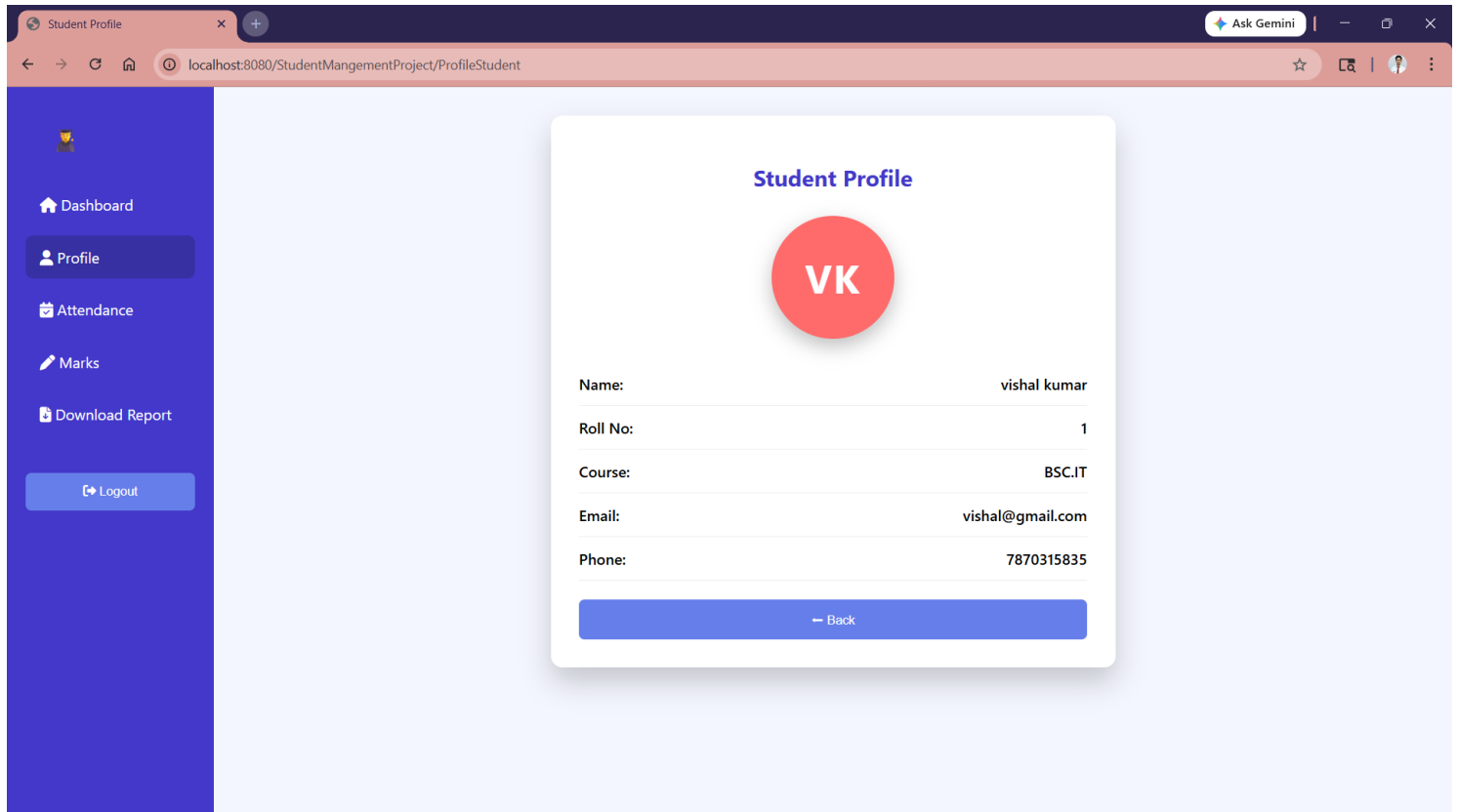
</div>

<footer>
  © 2026 Student Management System
</footer>

</body>
</html>

```


Student profile



Student Profile.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ page import="com.sms.model.ProfileStudent" %>

<%
    ProfileStudent s = (ProfileStudent) request.getAttribute("profileStudent");
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
```

```
<title>Student Profile</title>
```

```
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>
```

```
<style>
```

```
* { box-sizing: border-box; }
```

```
body{
```

```
    margin:0;
```

```
    font-family:"Segoe UI",Tahoma,sans-serif;
```

```
    background:#f4f6ff;
```

```
}
```

```
/* Wrapper */
```

```
.wrapper{
```

```
    display:flex;
```

```
    min-height:100vh;
```

```
}
```

```
/* Sidebar */
```

```
.sidebar{
```

```
    width:220px;
```

```
    background:#4338ca;
```

```
    color:white;
```

```
    padding:20px;
```

```
}
```

```
.sidebar h2{
```

```
    text-align:center;
```

```
    margin-bottom:30px;
```

```
}
```

```
.sidebar a{
```

```
    display:block;
```

```
    padding:12px 15px;
```

```
    color:white;
```

```
    text-decoration:none;
```

```
    border-radius:8px;
```

```
    margin-bottom:10px;
```

```
    transition:0.3s;
```

```
}
```

```
.sidebar a:hover,
```

```
.sidebar a.active{
```

```
    background:#3730a3;
```

```
}
```

```
/* Main content */
```

```
.main{
```

```
    flex:1;
```

```
    padding:30px;
```

```
}
```

```

/* Profile Card */
.container{
  max-width:600px;
  margin:auto;
  background:#fff;
  padding:30px;
  border-radius:12px;
  box-shadow:0 15px 30px rgba(0,0,0,0.2);
}

h2{
  text-align:center;
  color:#4338ca;
  margin-bottom:25px;
}

/* Avatar */
.profile-pic{
  display:flex;
  justify-content:center;
  margin-bottom:25px;
}

.avatar{
  width:130px;
  height:130px;
  border-radius:50%;
  color:#fff;
  display:flex;
  align-items:center;
  justify-content:center;
  font-size:42px;
  font-weight:bold;
  letter-spacing:2px;
  box-shadow:0 8px 20px rgba(0,0,0,0.25);
}

/* Profile details */
.profile-row{
  display:flex;
  justify-content:space-between;
  padding:12px 0;
  border-bottom:1px solid #eee;
}

.profile-row span{
  font-weight:600;
}

```

```

/* Back Button */
.back-btn{
    margin-top:20px;
    width:100%;
    padding:12px;
    border:none;
    background:#667eea;
    color:#fff;
    border-radius:6px;
    cursor:pointer;
}
.back-btn:hover{
    background:#4338ca;
}
</style>
</head>
<body>

<div class="wrapper">

    <!-- Sidebar -->
    <div class="sidebar">
        <h2> 🎓 Student</h2>
        <a href="studentDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
        <a href="ProfileStudent" class="active"><i class="fa fa-user"></i> Profile</a>
        <a href="ProfileAttendance"><i class="fa fa-calendar-check"></i> Attendance</a>
        <a href="ProfileMarks"><i class="fa fa-pen"></i> Marks</a>
        <a href="ReportStudent"><i class="fa fa-file-download"></i> Download Report</a>
        <form action="LogoutServlet" method="post" style="margin-top:20px;">
            <button type="submit" class="back-btn"><i class="fa fa-sign-out-alt"></i> Logout</button>
        </form>
    </div>

    <!-- Main content -->
    <div class="main">
        <div class="container">
            <h2>Student Profile</h2>

            <% if (s != null) { %>
            <!-- Avatar -->
            <div class="profile-pic">
                <div class="avatar" id="avatar"></div>
            </div>

            <!-- Profile Details -->
            <div class="profile-row">
                <span>Name:</span>
                <span id="name"><%= s.getStudentname() %></span>
            </div>
        </div>
    </div>

```

```

</div>
<div class="profile-row">
  <span>Roll No:</span>
  <span><%= s.getRollno() %></span>
</div>
<div class="profile-row">
  <span>Course:</span>
  <span><%= s.getCourse() %></span>
</div>
<div class="profile-row">
  <span>Email:</span>
  <span><%= s.getEmail() %></span>
</div>
<div class="profile-row">
  <span>Phone:</span>
  <span><%= s.getContact() %></span>
</div>
<% } else { %>
  <p style="text-align:center;color:red;">No student record found.</p>
<% } %>

  <button class="back-btn" onclick="goToDashboard()">⬅ Back</button>
</div>
</div>
</div>

<script>
function goToDashboard() {
  window.location.href = "studentDashboard.jsp";
}

/* Generate initials dynamically */
const nameText = document.getElementById("name")?.innerText || "";
const initials = nameText
  .split(" ")
  .map(word => word.charAt(0))
  .join("")
  .toUpperCase();
const avatar = document.getElementById("avatar");
avatar.innerText = initials || "ST";

/* Random background colors */
const colors = [
  "#667eea", "#764ba2", "#ff6b6b", "#1dd1a1",
  "#54a0ff", "#f368e0", "#ff9f43", "#48dbfb",
  "#576574", "#10ac84"
];
avatar.style.background = colors[Math.floor(Math.random()*colors.length)];

```

</script>

</body>

</html>

Profile student Servlet.java

```
package com.sms.controller;

import java.io.IOException;
import javax.servlet.RequestDispatcher;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;

import com.sms.dao.ProfileStudentDAO;
import com.sms.model.ProfileStudent;

@WebServlet("/ProfileStudent")
public class ProfileStudentServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        // session validation
        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.jsp");
            return;
        }

        String username = session.getAttribute("username").toString();

        ProfileStudentDAO dao = new ProfileStudentDAO();
        ProfileStudent student = dao.getStudentByUsername(username);

        request.setAttribute("profileStudent", student);
        RequestDispatcher rd =
            request.getRequestDispatcher("profileStudent.jsp");
        rd.forward(request, response);
    }
}
```

```
}
```

Profile Student Dao.java:

```
package com.sms.dao;

import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;

import com.sms.model.ProfileStudent;

public class ProfileStudentDAO {

    private static final String URL =
        "jdbc:mysql://localhost:3306/admindb";
    private static final String USER = "root";
    private static final String PASS = "2005";

    public ProfileStudent getStudentByUsername(String username) {

        ProfileStudent student = null;

        String sql =
            "SELECT rollno, studentname, email, contact, course " +
            "FROM mngstudent WHERE studentname = ?";

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(URL, USER, PASS);

            PreparedStatement ps = con.prepareStatement(sql);
            ps.setString(1, username);

            ResultSet rs = ps.executeQuery();

            if (rs.next()) {
                student = new ProfileStudent();
                student.setRollno(rs.getInt("rollno"));
                student.setStudentname(rs.getString("studentname"));
                student.setEmail(rs.getString("email"));
                student.setContact(rs.getString("contact"));
                student.setCourse(rs.getString("course"));
            }

            rs.close();
```

```

        ps.close();
        con.close();

    } catch (Exception e) {
        e.printStackTrace();
    }

    return student;
}
}

```

Profile Student model.java:

```

package com.sms.model;

public class ProfileStudent {

    private int rollno;
    private String studentname;
    private String email;
    private String contact;
    private String course;

    public int getRollno() {
        return rollno;
    }

    public void setRollno(int rollno) {
        this.rollno = rollno;
    }

    public String getStudentname() {
        return studentname;
    }

    public void setStudentname(String studentname) {
        this.studentname = studentname;
    }

    public String getEmail() {
        return email;
    }

    public void setEmail(String email) {
        this.email = email;
    }
}

```



```

public String getContact() {
    return contact;
}

public void setContact(String contact) {
    this.contact = contact;
}

public String getCourse() {
    return course;
}

public void setCourse(String course) {
    this.course = course;
}
}

```

Check Attendance

The screenshot displays a web application titled "Attendance Profile" in the browser tab. The URL is localhost:8080/StudentMangementProject/ProfileAttendance. The sidebar on the left contains a user profile icon and navigation links: Dashboard, Profile, Attendance (highlighted), Marks, Download Report, and a Logout button. The main content area features a white card titled "Attendance Record" with a calendar icon. Inside the card is a table with the following data:

Date	Subject	Status
2026-03-28	SQL Server	Present
2026-01-14	Cyber Security	Present

Below the table is a blue button labeled "← Back to Dashboard".

Check Attendance.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>
<%@ page import="java.util.*" %>
<%@ page import="com.sms.model.ProfileAttendance" %>

<%
    List<ProfileAttendance> list =
        (List<ProfileAttendance>) request.getAttribute("attendanceList");
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Attendance Profile</title>

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>

<style>
*{ box-sizing:border-box; }
body{ margin:0; font-family:"Segoe UI",Tahoma,sans-serif; background:#f4f6ff; }

/* Wrapper */
.wrapper{
    display:flex;
    min-height:100vh;
}

/* Sidebar */
.sidebar{
    width:220px;
    background:#4338ca;
    color:white;
    padding:20px;
}
.sidebar h2{
    text-align:center;
    margin-bottom:30px;
}
```

```

.sidebar a{
  display:block;
  padding:12px 15px;
  color:white;
  text-decoration:none;
  border-radius:8px;
  margin-bottom:10px;
  transition:0.3s;
}
.sidebar a:hover,
.sidebar a.active{
  background:#3730a3;
}

/* Main content */
.main{
  flex:1;
  padding:30px;
}

/* Container */
.container{
  background:#fff;
  padding:25px;
  border-radius:12px;
  box-shadow:0 10px 25px rgba(0,0,0,0.15);
  max-width:800px;
  margin:auto;
}

/* Heading */
h2{
  text-align:center;
  color:#4338ca;
  margin-bottom:20px;
}

/* Table styling */
table{
  width:100%;
  border-collapse:collapse;
  margin-top:15px;
}
th, td{
  padding:12px;
  text-align:center;
  border-bottom:1px solid #eee;
}

```

```

th{
    background:#667eea;
    color:white;
}
tr:nth-child(even){
    background:#f8f9ff;
}

/* Status colors */
.status-present{ color:green; font-weight:bold; }
.status-absent{ color:red; font-weight:bold; }
.status-na{ color:#999; font-weight:bold; }

/* Back button */
.back-btn{
    margin-top:20px;
    width:100%;
    padding:12px;
    border:none;
    background:#667eea;
    color:#fff;
    border-radius:6px;
    cursor:pointer;
    font-size:16px;
}
.back-btn:hover{
    background:#4338ca;
}
</style>
</head>
<body>

<div class="wrapper">
    <!-- Sidebar -->
    <div class="sidebar">
        <h2> 🧑 Student</h2>
        <a href="studentDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
        <a href="ProfileStudent"><i class="fa fa-user"></i> Profile</a>
        <a href="ProfileAttendance" class="active"><i class="fa fa-calendar-check"></i> Attendance</a>
        <a href="ProfileMarks"><i class="fa fa-pen"></i> Marks</a>
        <a href="ReportStudent"><i class="fa fa-file-download"></i> Download Report</a>
        <form action="LogoutServlet" method="post" style="margin-top:20px;">
            <button type="submit" class="back-btn"><i class="fa fa-sign-out-alt"></i> Logout</button>
        </form>
    </div>

    <!-- Main content -->
    <div class="main">

```

```

<div class="container">
  <h2> 📅 Attendance Record</h2>

  <% if (list != null && !list.isEmpty()) { %>
    <table>
      <tr>
        <th>Date</th>
        <th>Subject</th>
        <th>Status</th>
      </tr>
      <% for(ProfileAttendance a : list){ %>
        <tr>
          <td><%= a.getAttendanceDate() %></td>
          <td><%= a.getSubName() %></td>
          <td class="status-<%= a.getStatus().toLowerCase() %>">
            <%= a.getStatus() %>
          </td>
        </tr>
      <% } %>
    </table>
    <% } else { %>
    <p style="text-align:center;color:red; margin-top:20px;">
      No attendance records found.
    </p>
    <% } %>

    <button class="back-btn" onclick="goBack()">⬅ Back to Dashboard</button>
  </div>
</div>
</div>

<script>
function goBack() {
  window.location.href='studentDashboard.jsp';
}
</script>

</body>
</html>

```

Profile AttendanceDao.java:

```

package com.sms.dao;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

```

```

import com.sms.model.ProfileAttendance;

public class ProfileAttendanceDAO {

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/admindb",
            "root",
            "2005"
        );
    }

    public List<ProfileAttendance> getAttendanceByStudent(String studentName) {

        List<ProfileAttendance> list = new ArrayList<>();

        String sql =
            "SELECT attendance_date, sub_name, status " +
            "FROM attendance WHERE studentname = ? " +
            "ORDER BY attendance_date DESC";

        try (Connection con = getConnection();
            PreparedStatement ps = con.prepareStatement(sql)) {

            ps.setString(1, studentName);
            ResultSet rs = ps.executeQuery();

            while (rs.next()) {
                ProfileAttendance a = new ProfileAttendance();
                a.setAttendanceDate(rs.getDate("attendance_date"));
                a.setSubName(rs.getString("sub_name"));
                a.setStatus(rs.getString("status"));
                list.add(a);
            }

        } catch (Exception e) {
            e.printStackTrace();
        }

        return list;
    }
}

```

Profile Attendance Servlet.java:

```

package com.sms.controller;

import java.io.IOException;
import java.util.List;

import javax.servlet.*;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

import com.sms.dao.ProfileAttendanceDAO;
import com.sms.model.ProfileAttendance;

@WebServlet("/ProfileAttendance")
public class ProfileAttendanceServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.jsp");
            return;
        }

        String studentName = session.getAttribute("username").toString();

        ProfileAttendanceDAO dao = new ProfileAttendanceDAO();
        List<ProfileAttendance> attendanceList =
            dao.getAttendanceByStudent(studentName);

        request.setAttribute("attendanceList", attendanceList);
        request.getRequestDispatcher("profileAttendance.jsp")
            .forward(request, response);
    }
}

```

Profile attendance model.java:

```

package com.sms.model;

import java.sql.Date;

public class ProfileAttendance {

    private Date attendanceDate;

```

```

private String subName;
private String status;

public Date getAttendanceDate() {
    return attendanceDate;
}
public void setAttendanceDate(Date attendanceDate) {
    this.attendanceDate = attendanceDate;
}

public String getSubName() {
    return subName;
}
public void setSubName(String subName) {
    this.subName = subName;
}

public String getStatus() {
    return status;
}
public void setStatus(String status) {
    this.status = status;
}
}

```

Student marks

The screenshot displays a web application interface for managing student marks. The browser address bar shows the URL `localhost:8080/StudentMangementProject/ProfileMarks`. The application has a dark blue sidebar on the left with the following navigation options: Dashboard, Profile, Attendance, Marks (highlighted with a pencil icon), Download Report, and Logout. The main content area features a white card titled "Marks Record" containing a table with the following data:

Subject	Marks
Cyber Security	80
SQL Server	87

Below the table, there is a blue button labeled "← Back to Dashboard".

StudentMarks.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>
<%@ page import="java.util.*" %>
<%@ page import="com.sms.model.ProfileMarks" %>

<%
    List<ProfileMarks> list =
        (List<ProfileMarks>) request.getAttribute("marksList");
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Marks Profile</title>

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>

<style>
*{ box-sizing:border-box; }
body{ margin:0; font-family:"Segoe UI",Tahoma,sans-serif; background:#f4f6ff; }

/* Wrapper */
.wrapper{
    display: flex;
    min-height: 100vh;
}

/* Sidebar */
.sidebar{
    width: 220px;
    background: #4338ca;
    color: white;
    padding: 20px;
}
.sidebar h2{
    text-align: center;
    margin-bottom: 30px;
}
```

```

.sidebar a{
  display:block;
  padding:12px 15px;
  color:white;
  text-decoration:none;
  border-radius:8px;
  margin-bottom:10px;
  transition:0.3s;
}
.sidebar a:hover,
.sidebar a.active{
  background:#3730a3;
}

/* Main content */
.main{
  flex:1;
  padding:30px;
}

/* Container */
.container{
  background:#fff;
  padding:25px;
  border-radius:12px;
  box-shadow:0 10px 25px rgba(0,0,0,0.15);
  max-width:700px;
  margin:auto;
}

/* Heading */
h2{
  text-align:center;
  color:#4338ca;
  margin-bottom:20px;
}

/* Table styling */
table{
  width:100%;
  border-collapse:collapse;
  margin-top:15px;
}
th, td{
  padding:12px;
  text-align:center;
  border-bottom:1px solid #eee;
}

```

```

th{
    background:#667eea;
    color:white;
}
tr:nth-child(even){
    background:#f8f9ff;
}

/* Status */
.pass{ color:green; font-weight:bold; }
.fail{ color:red; font-weight:bold; }

/* Back button */
.back-btn{
    margin-top:20px;
    width:100%;
    padding:12px;
    border:none;
    background:#667eea;
    color:#fff;
    border-radius:6px;
    cursor:pointer;
    font-size:16px;
}
.back-btn:hover{
    background:#4338ca;
}
</style>
</head>
<body>

<div class="wrapper">
    <!-- Sidebar -->
    <div class="sidebar">
        <h2> 🎓 Student</h2>
        <a href="studentDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
        <a href="ProfileStudent"><i class="fa fa-user"></i> Profile</a>
        <a href="ProfileAttendance"><i class="fa fa-calendar-check"></i> Attendance</a>
        <a href="ProfileMarks" class="active"><i class="fa fa-pen"></i> Marks</a>
        <a href="ReportStudent"><i class="fa fa-file-download"></i> Download Report</a>
        <form action="LogoutServlet" method="post" style="margin-top:20px;">
            <button type="submit" class="back-btn"><i class="fa fa-sign-out-alt"></i> Logout</button>
        </form>
    </div>

    <!-- Main content -->
    <div class="main">
        <div class="container">

```

```

<h2>  Marks Record</h2>

<% if (list != null && !list.isEmpty()) { %>
<table>
  <tr>
    <th>Subject</th>
    <th>Marks</th>
  </tr>
  <% for(ProfileMarks m : list){ %>
  <tr>
    <td><%= m.getSubjectName() %></td>
    <td class="<%= m.getMarks() >= 35 ? "pass" : "fail" %>">
      <%= m.getMarks() %>
    </td>
  </tr>
  <% } %>
</table>
<% } else { %>
<p style="text-align:center;color:red; margin-top:20px;">
  No marks records found.
</p>
<% } %>

  <button class="back-btn" onclick="goBack()">⬅ Back to Dashboard</button>
</div>
</div>
</div>

<script>
function goBack(){
  window.location.href='studentDashboard.jsp';
}
</script>

</body>
</html>

```

Profile marks Servlet.java:

```

package com.sms.controller;

import java.io.IOException;
import java.util.List;

import javax.servlet.*;

```

```

import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

import com.sms.dao.ProfileMarksDAO;
import com.sms.model.ProfileMarks;

@WebServlet("/ProfileMarks")
public class ProfileMarksServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doGet(HttpServletRequest request,
        HttpServletResponse response)
        throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session == null || session.getAttribute("username") == null) {
            response.sendRedirect("login.jsp");
            return;
        }

        String studentName =
            session.getAttribute("username").toString();

        ProfileMarksDAO dao = new ProfileMarksDAO();
        List<ProfileMarks> marksList =
            dao.getMarksByStudent(studentName);

        request.setAttribute("marksList", marksList);
        request.getRequestDispatcher("profileMarks.jsp")
            .forward(request, response);
    }
}

```

Profile marks Dao.java:

```

package com.sms.dao;

import java.sql.*;
import java.util.ArrayList;
import java.util.List;

import com.sms.model.ProfileMarks;

public class ProfileMarksDAO {

```

```

private Connection getConnection() throws Exception {
    Class.forName("com.mysql.cj.jdbc.Driver");
    return DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/admindb",
        "root",
        "2005"
    );
}

public List<ProfileMarks> getMarksByStudent(String studentName) {

    List<ProfileMarks> list = new ArrayList<>();

    String sql =
        "SELECT subject_name, marks " +
        "FROM marks WHERE student_name = ?";

    try (Connection con = getConnection();
        PreparedStatement ps = con.prepareStatement(sql)) {

        ps.setString(1, studentName);
        ResultSet rs = ps.executeQuery();

        while (rs.next()) {
            ProfileMarks m = new ProfileMarks();
            m.setSubjectName(rs.getString("subject_name"));
            m.setMarks(rs.getInt("marks"));
            list.add(m);
        }

    } catch (Exception e) {
        e.printStackTrace();
    }

    return list;
}

```

Profile mark model.java:

```

package com.sms.model;

public class ProfileMarks {

    private String subjectName;
    private int marks;

```

```

public String getSubjectName() {
    return subjectName;
}
public void setSubjectName(String subjectName) {
    this.subjectName = subjectName;
}

public int getMarks() {
    return marks;
}
public void setMarks(int marks) {
    this.marks = marks;
}
}

```

Student Report

The screenshot displays a web application interface for a 'Student Academic Report'. The browser address bar shows 'localhost:8080/StudentMangementProject/ReportStudent'. The application has a dark blue sidebar with navigation links: Dashboard, Profile, Attendance, Marks, Report, and Logout. The main content area is titled 'Student Academic Report' and contains the following sections:

- Student Information:** A form showing details for 'vishal kumar' with Roll No. 1, Course BSC.IT, and Email vishal@gmail.com.
- Attendance (100.00%):** A table with columns Date, Subject, and Status.

Date	Subject	Status
2026-01-14	Cyber Security	Present
2026-03-28	SQL Server	Present
- Marks (Grade: B):** A table with columns Subject and Marks.

Subject	Marks
Cyber Security	80
SQL Server	87

At the bottom, there are two buttons: 'Download PDF' (green) and 'Back to Dashboard' (blue).

Student Report.jsp:

```

<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%
if(session.getAttribute("username")==null){
    response.sendRedirect("login.jsp");
}
%>
<%@ page import="java.util.*" %>
<%@ page import="com.sms.model.ReportStudent" %>
<%@ page import="com.sms.model.ReportAttendance" %>
<%@ page import="com.sms.model.ReportMarks" %>

<%
    ReportStudent student =
        (ReportStudent) request.getAttribute("student");

    List<ReportAttendance> attendanceList =
        (List<ReportAttendance>) request.getAttribute("attendanceList");

    List<ReportMarks> marksList =
        (List<ReportMarks>) request.getAttribute("marksList");

    int presentCount = 0, totalCount = 0;
    if(attendanceList != null){
        totalCount = attendanceList.size();
        for(ReportAttendance a : attendanceList){
            if("Present".equalsIgnoreCase(a.getStatus())) presentCount++;
        }
    }
    double attendancePercent = totalCount == 0 ? 0 : (presentCount * 100.0 / totalCount);

    int totalMarks = 0, subjectCount = 0;
    if(marksList != null){
        for(ReportMarks m : marksList){ totalMarks += m.getMarks(); subjectCount++; }
    }
    double avgMarks = subjectCount == 0 ? 0 : (totalMarks * 1.0 / subjectCount);
    String grade;
    if(avgMarks >= 85) grade="A";
    else if(avgMarks >= 70) grade="B";
    else if(avgMarks >= 50) grade="C";
    else if(avgMarks >= 35) grade="D";
    else grade="Fail";
%>

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Student Report</title>

```



```

<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf/2.5.1/jspdf.umd.min.js"></script>
<script src="https://cdnjs.cloudflare.com/ajax/libs/jspdf-autotable/3.5.29/jspdf.plugin.autotable.min.js"></script>

<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/css/all.min.css"/>

<style>
*{ box-sizing:border-box; margin:0; padding:0; font-family:"Segoe UI",Tahoma,sans-serif; }
body{ background:#f4f6ff; color:#333; }

/* Wrapper */
.wrapper{ display:flex; min-height:100vh; }

/* Sidebar */
.sidebar{
  width:220px;
  background:#4338ca;
  color:white;
  padding:20px;
}
.sidebar h2{ text-align:center; margin-bottom:30px; }
.sidebar a{
  display:block;
  padding:12px 15px;
  color:white;
  text-decoration:none;
  border-radius:8px;
  margin-bottom:10px;
  transition:0.3s;
}
.sidebar a:hover, .sidebar a.active{ background:#3730a3; }

/* Main content */
.main{ flex:1; padding:30px; }

/* Container sections */
.container{
  background:#fff;
  padding:25px;
  border-radius:12px;
  box-shadow:0 10px 25px rgba(0,0,0,0.15);
  margin-bottom:30px;
}
h2,h3{ text-align:center; color:#4338ca; margin-bottom:20px; }
.profile-row{ display:flex; justify-content:space-between; padding:10px 0; border-bottom:1px solid #eee; font-weight:600; }

/* Tables */

```

```
table{ width:100%; border-collapse:collapse; margin-top:15px; }
th,td{ border:1px solid #ddd; padding:12px; text-align:center; }
th{ background:#667eea; color:white; }
tr:nth-child(even){ background:#f8f9ff; }
```

```
/* Buttons */
.btn{ margin-top:20px; width:100%; padding:12px; border:none; border-radius:6px; cursor:pointer; font-size:15px; }
.pdf-btn{ background:#28a745; color:white; }
.pdf-btn:hover{ background:#218838; }
.back-btn{ background:#007bff; color:white; }
.back-btn:hover{ background:#4338ca; }
</style>
</head>
```

```
<body>
<div class="wrapper">
  <!-- Sidebar -->
  <div class="sidebar">
    <h2> 🧑 Student</h2>
    <a href="studentDashboard.jsp"><i class="fa fa-home"></i> Dashboard</a>
    <a href="ProfileStudent"><i class="fa fa-user"></i> Profile</a>
    <a href="ProfileAttendance"><i class="fa fa-calendar-check"></i> Attendance</a>
    <a href="ProfileMarks"><i class="fa fa-pen"></i> Marks</a>
    <a href="ReportStudent" class="active"><i class="fa fa-file-alt"></i> Report</a>
    <form action="LogoutServlet" method="post" style="margin-top:20px;">
      <button type="submit" class="back-btn"><i class="fa fa-sign-out-alt"></i> Logout</button>
    </form>
  </div>
```

```
  <!-- Main content -->
  <div class="main">

    <!-- Student Details -->
    <div class="container">
      <h2> 🎓 Student Academic Report</h2>
      <% if(student != null){ %>
        <div class="profile-row"><span>Name:</span><span id="sname"><%= student.getStudentname()
%></span></div>
        <div class="profile-row"><span>Roll No:</span><span id="roll"><%= student.getRollno() %></span></div>
        <div class="profile-row"><span>Course:</span><span><%= student.getCourse() %></span></div>
        <div class="profile-row"><span>Email:</span><span><%= student.getEmail() %></span></div>
      <% } %>
    </div>

    <!-- Attendance -->
    <div class="container">
      <h3> 📊 Attendance ( <%= String.format("%.2f", attendancePercent) %>% )</h3>
      <table id="attendanceTable">
```

```

<tr>
  <th>Date</th>
  <th>Subject</th>
  <th>Status</th>
</tr>
<% if(attendanceList != null && !attendanceList.isEmpty()){
  for(ReportAttendance a : attendanceList){ %>
    <tr>
      <td><%= a.getAttendanceDate() %></td>
      <td><%= a.getSubName() %></td>
      <td><%= a.getStatus() %></td>
    </tr>
  <% } } else { %>
    <tr><td colspan="3">No attendance records</td></tr>
  <% } %>
</table>
</div>

```

```

<!-- Marks -->
<div class="container">
  <h3>📊 Marks ( Grade: <%= grade %> )</h3>
  <table id="marksTable">
    <tr>
      <th>Subject</th>
      <th>Marks</th>
    </tr>
    <% if(marksList != null && !marksList.isEmpty()){
      for(ReportMarks m : marksList){ %>
        <tr>
          <td><%= m.getSubjectName() %></td>
          <td><%= m.getMarks() %></td>
        </tr>
      <% } } else { %>
        <tr><td colspan="2">No marks records</td></tr>
      <% } %>
    </table>
  </div>

```

```

<button class="btn pdf-btn" onclick="downloadPDF()">📄 Download PDF</button>
<button class="btn back-btn" onclick="location.href='studentDashboard.jsp'">⬅ Back to Dashboard</button>

```

```

</div>
</div>

```

```

<script>
function downloadPDF(){
  const { jsPDF } = window.jspdf;
  const doc = new jsPDF();

```

```

const name = document.getElementById("sname").innerText;
const roll = document.getElementById("roll").innerText;

doc.setFontSize(16);
doc.text("Student Academic Report",14,15);
doc.setFontSize(11);
doc.text("Name: " + name,14,25);
doc.text("Roll No: " + roll,14,32);

doc.autoTable({ startY:40, html:"#attendanceTable" });
doc.autoTable({ startY:doc.lastAutoTable.finalY+10, html:"#marksTable" });

doc.save(roll + "_" + name + "_Report.pdf");
}
</script>

</body>
</html>

```

studentReportServlet.java:

```

package com.sms.controller;

import java.io.IOException;
import java.sql.*;
import java.util.*;

import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.*;

import com.sms.model.*;

@WebServlet("/ReportStudent")
public class ReportStudentServlet extends HttpServlet {

    /**
     *
     */
    private static final long serialVersionUID = 1L;

    private Connection getConnection() throws Exception {
        Class.forName("com.mysql.cj.jdbc.Driver");
        return DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/admindb",
            "root",

```

```

        "2005"
    );
}

```

```

protected void doGet(HttpServletRequest req, HttpServletResponse resp)
    throws ServletException, IOException {

```

```

    HttpSession session = req.getSession();
    String username = (String) session.getAttribute("username");

```

```

    if (username == null) {
        resp.sendRedirect("login.jsp");
        return;
    }

```

```

    try (Connection con = getConnection()) {

```

```

        /* ===== STUDENT ===== */
        ReportStudent student = null;
        PreparedStatement ps1 = con.prepareStatement(
            "SELECT * FROM mngstudent WHERE studentname=?"
        );
        ps1.setString(1, username);
        ResultSet rs1 = ps1.executeQuery();

```

```

        if (rs1.next()) {
            student = new ReportStudent();
            student.setRollno(rs1.getString("rollno"));
            student.setStudentname(rs1.getString("studentname"));
            student.setCourse(rs1.getString("course"));
            student.setEmail(rs1.getString("email"));
        }

```

```

        /* ===== ATTENDANCE ===== */
        List<ReportAttendance> attendanceList = new ArrayList<>();
        PreparedStatement ps2 = con.prepareStatement(
            "SELECT attendance_date, sub_name, status FROM attendance WHERE studentname=?"
        );
        ps2.setString(1, username);
        ResultSet rs2 = ps2.executeQuery();

```

```

        while (rs2.next()) {
            ReportAttendance a = new ReportAttendance();
            a.setAttendanceDate(rs2.getDate("attendance_date"));
            a.setSubName(rs2.getString("sub_name"));
            a.setStatus(rs2.getString("status"));
            attendanceList.add(a);
        }

```

```

/* ===== MARKS ===== */
List<ReportMarks> marksList = new ArrayList<>();
PreparedStatement ps3 = con.prepareStatement(
    "SELECT subject_name, marks FROM marks WHERE student_name=?"
);
ps3.setString(1, username);
ResultSet rs3 = ps3.executeQuery();

while (rs3.next()) {
    ReportMarks m = new ReportMarks();
    m.setSubjectName(rs3.getString("subject_name"));
    m.setMarks(rs3.getInt("marks"));
    marksList.add(m);
}

req.setAttribute("student", student);
req.setAttribute("attendanceList", attendanceList);
req.setAttribute("marksList", marksList);

req.getRequestDispatcher("reportStudent.jsp").forward(req, resp);

} catch (Exception e) {
    e.printStackTrace();
    resp.sendRedirect("studentDashboard.jsp");
}
}
}

```

BIBLIOGRAPHY

Books

- *Java: The Complete Reference* – Herbert Schildt
- *Head First Servlets and JSP* – Bryan Basham, Kathy Sierra
- *Database System Concepts* – Abraham Silberschatz

Web Resources

- Official Java Documentation – <https://docs.oracle.com>
- MySQL Documentation – <https://dev.mysql.com/doc>

- Apache Tomcat Documentation – <https://tomcat.apache.org>
 - W3Schools (HTML, CSS, JavaScript) – <https://www.w3schools.com>
-

Tools & Technologies

- Java (Servlets & JSP)
- MySQL Database
- Apache Tomcat Server
- Eclipse / IntelliJ IDE